BAYESIAN NONPARAMETRIC GLOTTAL INVERSE FILTERING

by

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Abstract

A reliable and accurate white-box model of speech is still something of a holy grail in phonetics, forensics, and clinical medicine. The difficulty is that fitting such models to microphone recordings is a blind identification problem, known to practitioners as glottal inverse filtering, which requires inferring both source signal and filter simultaneously from a single waveform. Because the problem is severely underdetermined, we must lean hard on prior information to constrain the set of mathematically possible solutions: they must adhere to what we already know to be plausible or true. I will show how nonparametric Gaussian processes let us bake such prior knowledge directly into the model by (1) deriving a nonstationary kernel that can model the glottal cycle, (2) specializing that kernel into a surrogate model learned from simulated source signals via interdomain features, and (3) proposing a new and general framework for specifying arbitrary prior information about the filter; and how this improves reliability and accuracy at inference time.

Lay summary

In the past decade, we have made enormous strides in recognizing and synthesizing speech using large "black-box" models trained on massive datasets. Today, these models have learnt to speak and understand major world languages such as English, Chinese, and Spanish with impressive accuracy. Remarkably, they remain largely agnostic to the physical mechanisms underlying speech, relying instead on large-scale pattern recognition and vast quantities of data.

In contrast, "white-box" models grounded in accumulated scientific knowledge do focus explicitly on these physical mechanisms. While they are not remotely competitive with black-box models in speech recognition or synthesis, they do offer the valuable ability to extract biologically meaningful information about the speaker from the signal itself.

This is important. A growing body of evidence shows that quantitative features of voice and speech can act as digital biomarkers of neurological and clinical conditions. In Parkinson's disease, for example, changes in speech and language have been reported to precede diagnosis by as much as a decade, and acoustic measures such as voice quality, articulation rate, and prosody have shown promise for early detection and monitoring. Similarly, alterations in specific acoustic parameters have been linked to mood disorders, suggesting that speech analysis could complement clinical assessment of depression.

Nevertheless, white-box models often lack the flexibility of black-box models needed to deal with real-world speech. I therefore propose a "grey-box" approach that attempts to combine the best of both worlds. By integrating physical insight with data-driven modeling, I derive a class of grey-box models that allows for far more accurate analysis and decomposition of speech into separate contributions from the vocal folds and the vocal tract. These models are also capable of indicating how certain they are of their measurements, while running close to real-time speed on modern hardware.

This was achieved by combining new theoretical insights in glottal flow and vocal tract models with customized machine learning models. I hope that this work can contribute to a more reliable and holistic use of speech analysis in clinical practice, forensic work, and the scientific study of vocalization as a whole.

Preface

Write your preface here. This must include a description of your role in the work as presented, and the role of any collaborators if applicable. Additionally, include a statement on the use of Generative AI tools in producing this work.

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Acknowledgments

Write your acknowledgments here. This section is optional, feel free to remove it if you do not wish to include it. Nonetheless, this is the place to thank professional colleagues and people who have given you the most support during the course of your work.

1 Overview

We introduce Bayesian nonparametric glottal inverse filtering (BNGIF) and provide a high-level overview of the main concepts involved, followed by a summary of the contributions made in this thesis.

1.1 Introduction

In this thesis we present a new approach to the problem of *glottal inverse filtering* (GIF) called *Bayesian nonparametric glottal inverse filtering* (BNGIF). At the heart of this new approach lies a Bayesian nonparametric *generative model of voiced speech* that reflects general constraints on the voice production process. Though conceived for GIF, the generative model can be conditioned on both raw speech data and high level speech features, which makes it useful for other related tasks such as augmenting or denoising speech data.

Speech can be thought of as the output of a linear system in which an upstream source signal (for example, the rapid opening and closing of the vocal folds) is modulated by a downstream filter (for example, the oral cavity and lips). This linear approximation to speech production, called source-filter theory (Fant, 1960), still underlies most of what is known today as acoustic phonetics (Maurer, 2016), and both source and filter are comparably well understood.

The GIF problem refers to source-filter separation of speech; that is, simultaneously inferring both the glottal source signal and (some of) the vocal tract filter characteristics from the speech signal alone (Figure 1). In the BNGIF approach we systematically take advantage of known constraints on the voice production process to home in on scientifically plausible solutions to the GIF problem. That is the main theme of this thesis.

GIF is an important and difficult problem in acoustic phonetics (Auvinen et al., 2014; Kadiri et al., 2021) and various approaches have been developed over the last 60 years. Paraphrasing Alku (2011), applications mainly come in three broad categories: (i) fundamental research about voice production (for example, quantifying voice quality), (ii) medical applications (such as analysis of pathological voices), and (iii) speech technology applications (for example, voice modification). To this we would add (iv) recent forensic and biometric applications such as evidence-based speaker identification (Bonastre et al., 2015).

Our approach to the GIF problem is mostly relevant to the first category (i) above because we try to model the process of scientific inference – make assumptions, model the data, fit the data, evaluate the model, reiterate – as a whole rather than 'just' modeling the data alone (MacKay, 2005, Ch. 28). It has been shown repeatedly that the unique way to do this is

FIGURE HERE

Figure 1 | **Overview of source-filter separation.** The speech waveform (left) is modeled as a glottal source signal (top right) passing through a series of vocal tract filters (bottom left). The task is to find such plausible source-filter pairs (colored samples on the right) that together produce a given speech waveform.

the probabilistic or *Bayesian* framework (Knuth & Skilling, 2012). Within this mathematical framework our generative model of voiced speech ultimately expresses nothing more than our assumptions about voice production in a logically coherent way. Because we insist on modeling scientific inference itself the accuracy of those assumptions will translate directly to the accuracy of our measurements, providing a much-needed means to calibrate uncertainty for use in (i) fundamental research about voice production (Kent & Vorperian, 2018). This is the main benefit of and motivation for using the Bayesian framework for BNGIF, despite its considerable computational expense compared to standard GIF algorithms which are based on traditional signal processing methods.

Another important benefit is the standardized form of Bayesian generative models: posterior inference is mechanical once the prior and likelihood are specified, which allows for a wide variety of generic algorithms to fit the data in true plug-and-play fashion (Murphy, 2022). This standardized form also ensures that generative models are easily modifiable and extendable for more specific applications such as (ii-iv) mentioned above.[1]

Another distinguishing feature of BNGIF is the *nonparametric* nature of our generative model of speech. More precisely, the model contains both nonparametric and parametric components. Figure Figure 1 shows these two neatly divided components on the right hand side: the glottal source signal (top) is modeled nonparametrically with a Gaussian process (Rasmussen & Williams, 2006), while the vocal tract filter (bottom) is expanded into a familiar rational transfer function that is parameterized by an overall gain factor and a vector of poles and zeros.

Nonparametric in this context does not refer to 'distribution-free' methods used in descriptive statistics or nonparametric spectral analysis (Stoica & Moses, 2005); rather it means that the glottal source signal is modeled directly through an infinite-dimensional distribution over functions whose qualitative behavior is specified by a just few global properties like roughness and wiggliness. This is convenient because this is exactly the kind of broad information we possess about the glottal source signal, unlike the vocal tract filter which in source-filter theory is naturally specified in terms of resonances of the vocal tract (Stevens, 2000).

1.2 Problem statement

The central problem of this thesis is boxed in Figure 2.

[1] For example it is expected that our model of human voice production can be useful for mammal vocal communication research (Taylor & Reby, 2010) by simply tweaking some of its prior assumptions. An example of extending the model to downstream tasks is forensic phonetics, since "formant features […] are widely accepted features used in forensic acoustic-phonetic speaker verification" (Becker et al., 2008, p. 1) and "the Bayesian decision framework […] is now the standard paradigm for forensic speaker comparison" (Bonastre et al., 2015, p. 1). This Bayesian decision framework depends on likelihood ratios (Harrison, 2013) that can be calculated with the same computational tools used in this thesis.

Glottal inverse filtering (GIF)

Assume that the source-filter model of speech production (Fant, 1960) holds such that the speech signal at time $t \in \mathbb{R}$ is described by convolving three real-valued functions:

$$s(t) = u(t) * h(t) * r(t). \tag{1}$$

From left to right $s(t)$ is the speech pressure waveform, $u(t)$ is the glottal flow waveform, $h(t)$ is the impulse response of the vocal tract and $r(t)$ is the impulse response of the radiation characteristic, assumed known. Source-filter theory asserts that $s(t)$ can be thought of as the result of the source signal $u(t)$ passing through the filters $h(t)$ and $r(t)$.

The data consists of N noisy samples $\mathbf{d} = d_{1:N}$ of the speech pressure waveform $s(t)$ at times $\mathbf{t} = t_{1:N}$. We assume additive errors $\mathbf{e} = e_{1:N}$ such that

$$\mathbf{d} = \mathbf{s} + \mathbf{e} \Longrightarrow \quad d_n = s(t_n) + e_n \quad (n = 1 \cdots N). \tag{2}$$

The error components e_n are modeled as white noise with expected power σ_n^2 which means that $e_n \sim \text{Normal}(0, \sigma_n^2)$ for $n = 1 \cdots N$.

The data $(\mathbf{t}, \mathbf{d})$ is possibly augmented with noisy statistics of $s(t)$ such as a list of estimated glottal closing instants (GCIs).

The GIF problem is then to *find plausible estimates of the source* $u(t)$ *and the filter* $h(t)$ that are consistent with the given data.

Figure 2 | **The GIF problem** which we attack in this thesis.

Source-filter theory as expressed by equation (1) is the most important assumption in this thesis because it provides a definite handle on the GIF problem. Even just stating the problem would be difficult without it. In particular, note the four functions (or waveforms) in (1);: each of these play an important role in BNGIF. Before reviewing them in Section 1.3, we first consider how the *plausible estimates* mentioned in the statement above translate within the Bayesian framework adopted in this thesis.

Like most inverse problems, the main difficulty with GIF lies in the fact that it is vastly underdetermined: even in the noiseless case ($\mathbf{d} = \mathbf{s}$), many solutions (that is, pairs of functions $u(t)$ and $h(t)$) are compatible with given data $\mathbf{d}$ (Landau, 1983). It thus appears that next to the data $\mathbf{d}$ itself we need additional criteria to choose between possible solutions of (1);, and this is where the idea of regularization comes in.

Appendix A discusses how in the Bayesian framework regularization is imposed by assigning priors to the unknown functions $u(t)$ and $h(t)$ in such a way that these priors express known constraints on the voice production process. The main idea is that a probability in the Bayesian framework describes a state of knowledge, not necessarily related to an obm(s)erved frequency in the real world, so we can conveniently use prior probabilities to express constraints on the voice production process.

Even with that regularization in place, we will see that the problem is still underdetermined in the sense that we cannot in general expect to find unique solutions: the best we can hope for are *plausible estimates in the form of posterior samples of* $u(t)$ *and* $h(t)$. These are plausible in the sense that they are consistent with both the speech data $\mathbf{d}$ and whatever prior information about voice production we have expressed through the priors for $u(t)$ and $h(t)$.

1.3 Source-filter theory

FIGURE HERE

Figure 3 | Schematical representation of the source-filter model (1) with u, h, r, s and their Fourier transforms.

Source-filter theory[2] models the speech production process as a standard *linear time-invariant* (LTI) system (Antsaklis & Michel, 2006) which is essentially a linear approximation to the underlying nonlinear dynamics of voice production that is roughly valid up to frequencies of 4 or 5 kHz (Doval et al., 2006). The LTI system (1) used in this thesis is illustrated in Figure 3. We proceed with a brief sketch of the important concepts involved.

The *input* to the system is the source $u(t)$. The system has two *transfer functions* which determine how the amplitudes and phase of the frequencies present in the input $\tilde{u}(f)$ are modified or filtered by the system to produce the *output*. That frequency decomposition

$$\tilde{u}(f) = \mathcal{F}[u(t)](f) = \int_{-\infty}^{\infty} u(t) \exp\{-i2\pi f t\} \,\mathrm{d}t \tag{3}$$

is obtained by the Fourier transform $\mathcal{F}$ of $u(t)$. Note that the system's transfer function is assumed to be independent from its input: source-filter theory assumes source and filter to be decoupled.

The vocal tract transfer function $\tilde{h}(f)$ describes the resonances and antiresonances of the vocal tract (comprising the throat, mouth and nasal cavity), while the radiation characteristic $\tilde{r}(f)$ describes the lip radiation effect. These transfer functions can be described equivalently in the time domain by the *impulse responses*

$$\begin{aligned} h(t) = \mathcal{F}^{-1}\left[\tilde{h}(f)\right](t) &= \int_{-\infty}^{\infty} \tilde{h}(f) \exp\{i2\pi f t\} \,\mathrm{d}x \\ r(t) = \mathcal{F}^{-1}[\tilde{r}(f)](t) \quad &= \int_{-\infty}^{\infty} \tilde{r}(f) \exp\{i2\pi f t\} \,\mathrm{d}x \end{aligned} \tag{4}$$

[2]Currently the dominant paradigm in acoustic phonetics (Maurer, 2016), the source-filter theory of voice production is commonly attributed to (Fant, 1960) but appears to date back to the early 19th century (Chen, 2016). Its strength lies in its extremely useful and compact description of the acoustics of phonation, summarized by (Fant, 1960, p. 15) as "the speech wave is the response of the vocal tract filter systems to one or more sound sources." A modern perspective on source-filter theory is given by Svec et al. (2021).

using the inverse Fourier transform $\mathcal{F}^{-1}$. Unlike $u(t)$ and $h(t)$, $r(t)$ is assumed fixed and known: the lip radiation effect is modeled in the GIF literature as a standard 6 dB/octave high pass filter (Stevens, 2000, p. 128) which has transfer function $\tilde{r}(f) = i2\pi x$. This corresponds to differentiation in the time domain as (4) yields $r(t) = \delta'(t)$.[3]

The *output* of the system is the speech pressure signal $s(t)$

$$\begin{aligned} s(t) &= u(t) * h(t) * r(t) \qquad &&\text{(time domain)} \\ \implies \quad \tilde{s}(f) &= \tilde{u}(f) \quad \tilde{h}(f) \quad \tilde{r}(f) \qquad &&\text{(frequency domain)} \end{aligned} \tag{5}$$

assumed to be measured in the free field (that is, sufficiently far from the speaker's head) (Stevens, 2000). Filtering in LTI systems happens through multiplication in the frequency domain (5), which is equivalent to convolution in the time domain (1).

The meaning of these equations is best understood in the frequency domain. During speech production, the source drives the air in the vocal tract such that it vibrates with some initial distribution over frequencies $\tilde{u}(f)$. Excited, the vocal tract filter $\tilde{h}(f)$ reacts: frequencies in $\tilde{u}(f)$ close to its resonance frequencies are emphasized,[4] while frequencies in $\tilde{u}(f)$ close to its antiresonance frequencies are (often brutally) suppressed. The result of this spectral imprinting is then radiated from the lips into the free field. This radiation effect is contained within $\tilde{r}(f)$: high frequencies are emphasized, essentially because the lips act like small (passive) tweeters (Schroeder, 1999). We are able to *hear* the spectral imprint of the speaker's vocal tract on the initial energy distribution (Klatt, 1986), because it is what we use to tell the difference between different speech sounds, and ultimately to understand the speaker's message.

Since $r(t)$ is assumed known, we can further simplify (1) by substituting $r(t) = \delta'(t)$:

$$s(t) = u(t) * h(t) * \delta'(t) = [u(t) * \delta'(t)] * h(t) = u(t) * [h(t) * \delta'(t)]. \tag{6}$$

We can differentiate either $u(t)$ or $h(t)$ in (1) because convolution is associative; we choose to differentiate $u(t)$ as is customary in the literature (Doval et al., 2006). This finally yields

$$\boxed{s(t) = u'(t) * h(t)} \tag{7}$$

where $u'(t)$ is the *glottal flow derivative* (DGF) which now acts as the source driving the single filter in (7). The original source $u(t)$ can be recovered through integration:

[3]Convolution of a test function $f(f)$ with the derivative of the Dirac delta function is equivalent to differentiation: $f(t) * \delta'(t) = f'(t)$. This can be justified by distribution theory (Lighthill, 1958).

[4]The vocal tract filter is *passive*, meaning that $|\tilde{h}(f)| < 1$ for all frequencies x (Flanagan, 1965). The resonances of the vocal tract do not actively amplify frequencies; frequencies close to the resonance frequencies are just less dampened relative to neighbouring frequency bands.

$$u(t) = \int_{-\infty}^{t} u'(\tau)\,\mathrm{d}\tau. \tag{8}$$

Equation (7) is the basis for the nonparametric generative model of $s(t)$ which we use to separate speech data [noisy samples of $s(t)$] into source $u'(t)$ [equivalently $u(t)$ via (8)] and filter $h(t)$ [equivalently $\tilde{h}(f)$ via (4)].

With the general picture laid out, we now discuss the $u(t)$, $h(t)$ and $s(t)$ waveforms in further detail, and present a high-level overview of the proposed priors for each of them. Section 1.2 argued that effective regularization of the GIF problem requires realistic priors on $u(t)$ and $h(t)$ and we can show some of their realistic properties already by just sampling them. Note that these priors are nothing but generative models of $u(t)$, $h(t)$ and $s(t)$ – they are 'priors' because we can generate samples from these models according to a probability density that represents our prior information about the speech production process.

1.3.1 The glottal flow waveform $u(t)$

In source-filter theory the glottal flow $u(t)$ is the source component which provides the initial acoustic energy that is subsequently modified by the 'downstream' filter component (comprised of the vocal tract and the lips). This definition is quite general and the actual physical mechanism underlying $u(t)$ depends on the type of phonation employed.

In the GIF literature the type of phonation assumed is *voiced speech*, during which the vocal folds vibrate at the fundamental frequency F_0 while modulating a stream of air expelled from the lungs. The $u(t)$ waveform measures the flow rate (volume velocity, in ml/sec) of this airflow which looks like a quasiperiodic string of pulses, each pulse representing a "puff of air" passing through the glottis (Schroeder, 1999). The quasiperiodicity of $u(t)$ directly transfers to the voiced speech signal $s(t)$ because the vocal articulators move much more slowly than the vibrating vocal folds.

The most widely used model for the DGF $u'(t)$ in voiced speech is the Liljencrants-Fant model of Fant et al. (1985). We use this in Chapter 4 as the basis for a parametric prior for $u(t)$ denoted formally as

$$u(t) \sim \pi_{\mathrm{LF}(u(t))} \tag{9}$$

This parametric prior serves as the basis for a more expressive and nonparametric Gaussian process prior derived in Chapter 5:

$$u(t) \sim \pi_{\mathrm{GP}(u(t))} \tag{10}$$

The GP underlying this prior has a Matérn kernel that is calibrated to π_{LF} using Bayesian transfer learning (Xuan et al., 2021).

FIGURE HERE

Figure 4 | DGF and GF samples from π_{LF} and π_{GP}.

Samples from π_{LF} and π_{GP} are shown in Figure 4. Due to its nonparametric nature the latter is capable of representing a wide variety of $u(t)$ curves both in terms of roughness and overall pulse shape. Therefore, the nonparametric π_{GP} prior can be expected to yield a more realistic model of the glottal flow in real voiced speech compared to the parametric π_{LF} prior.

1.3.2 The impulse response of the vocal tract $h(t)$

Source-filter theory asserts that the physical resonances and antiresonances of the vocal tract show up as local maxima and minima in the envelope of the power spectrum $|\tilde{h}(f)|^2$ of the vocal tract transfer function. These extrema are known as *formants* and *antiformants*, respectively, and their amplitudes, frequencies and bandwidths give a compact description of the resonant and damping ('antiresonant') behavior of the vocal tract. During speech production, we manipulate these (anti)formant frequencies on the fly by skillfully changing the vocal tract configuration (such as rounding the lips, moving the tongue or closing the mouth) in just the right way as to produce the desired vowel or consonant. Humans really are master musicians of the vocal instrument.

The standard LTI assumption and canonical approach in source-filter theory is that $\tilde{h}(f)$ is a *rational transfer function* whose *poles* and *zeros* are identified with the aforementioned formants and antiformants, respectively (Stevens, 2000). We adopt this assumption in this thesis but without the one-to-one correspondence between (formants and pole pairs) and (antiformants and zero pairs). Instead, a single formant may be modeled by one or more poles while some 'spectrum shaping' poles need not correspond to formants at all; and similarly for antiformants and zeros. For this reason we prefer to think of $\tilde{h}(f)$ as a parametric rational *expansion* of the true underlying vocal tract transfer function. The parameters controlling this Padé expansion are its order, an overall gain factor and a vector of poles and zeros. All of these can be inferred from speech data.

A (strictly proper) rational transfer function $\tilde{h}(f)$ in the frequency domain implies through (4) that the impulse response $h(t)$ in the time domain consists of a real-valued superposition of exponentially decaying sinusoids. This real-valued superposition is more convenient mathematically for our purposes than a complex rational function, so in Chapter 3 we express our two priors for $h(t)$ in the time domain.[5]

FIGURE HERE

Figure 5 | Samples from $h(t)$ prior in time and spectral domain.

The first one is a parametric prior over pole-zero expansions of the true underlying vocal tract transfer function of a given order:

$$h(t) \sim \pi_{\mathrm{PZ}(h(t))} \tag{11}$$

[5]This might seem counterintuitive at first, but it is actually a common approach in Bayesian spectrum analysis (Bretthorst, 1988; Turner & Sahani, 2014).

Likewise, the second prior parametrizes all-pole expansions, which, lacking zeros, are a special case of pole-zero expansions.

$$h(t) \sim \pi_{\text{AP}(h(t))} \tag{12}$$

Samples from π_{PZ} and π_{AP} both in the time domain and the frequency domain are shown in Figure Figure 5. The qualitative differences between their power spectra in Figure Figure 5 are due to the fact that pole-zero expansions are more expressive than all-pole expansions. That is, samples from π_{AP} are more constrained because they are essentially samples from π_{PZ} for which the zeros are set up to cancel out exactly.

Without zeros π_{AP} cannot easily model antiformants of the vocal tract, which for example occur with nasal consonants [now sing]. This is often used as an argument against the use of all-pole transfer functions in speech processing (Mehta et al., 2012). Its samples also have an expected spectral tilt that is too steep to be realistic. In contrast π_{PZ} can model a more flexible class of spectra with a wide range of (anti)resonant behavior and possible spectral tilts, and is to be preferred to π_{AP} on these grounds, but at the cost of a roughly doubled amount of parameters.

Nevertheless, in practice all-pole expansions such as the ones represented by π_{AP} are generally thought to be sufficiently expressive to represent the vocal tract filter (Flanagan, 1965; Fulop, 2011) and have been computationally the preferred choice since the very beginning of digital acoustic phonetics (Atal & Hanauer, 1971). Next to theoretical grounds (Titze, 2000) this is mainly because they admit an efficient representation known as linear prediction (Markel & Gray, 1976), which is the traditional workhorse of acoustic analysis.

1.3.3 The speech pressure waveform $s(t)$

FIGURE HERE

Figure 6 | Samples from π_{VS}.

The speech pressure waveform

$$s(t) = u'(t) * h(t) \tag{13}$$

is the output of the LTI system that describes speech production according to source-filter theory. The priors for $u(t)$ and $h(t)$ introduced in ((10), (11), (12)) induce a prior for $s(t)$ through (7), which is derived analytically in Chapter 2. This prior is the aforementioned nonparametric generative model of voiced speech that underlies BNGIF:

$$s(t) \sim \pi_{\text{VS}(s(t))} \tag{14}$$

Samples from π_{VS} are shown in Figure Figure 6. The quasiperiodicity exhibited by these waveforms is necessary for speech with a voiced sound and is induced by the quasiperiodicity of samples from π_{GP} as can be seen in the right panel of Figure Figure 4.

The π_{VS} prior is nonparametric because it is simply another GP whose behavior is governed by a learnable kernel that describes the intricate covariance structure of voiced speech. Accordingly, the samples shown in Figure Figure 6 are nothing but draws from a high-dimensional multivariate normal distribution. BNGIF infers the source $u'(t)$ [equivalently $u(t)$ via (8)] and filter $h(t)$ [equivalently $\tilde{h}(f)$ via (4)] from (7) by learning the kernel that best describes the given speech data. This is a challenging but ultimately standard GP inference task that can be done in a variety of ways (Frigola, 2015; Simpson et al., 2020).

1.4 Contributions

The relatively long history of research into GIF naturally produced several lines along which the problem may be approached, but, following two recent reviews (Drugman et al., 2019a; Kadiri et al., 2021), it is possible to separate these approaches loosely into two broad classes: inverse filtering and joint source-filter optimization.[6] Before reviewing the specific contributions made by BNGIF to the GIF literature, and what they are hoped to accomplish at what cost, we first situate the approach taken in this thesis within these two classes.

Inverse filtering Miller (1959) founded the basis for the first and larger class of GIF methods, which holds most of the popular GIF methods used today such as closed phase analysis (Wong et al., 1979) and iterative adaptive inverse filtering (Alku, 1992). The inverse filtering approach aims to recover the glottal source signal $u(t)$ by extracting from the speech data $\mathbf{d}$ an estimate of the vocal tract filter $\hat{\tilde{h}}(f)$ and then digitally filtering $\mathbf{d}$ by $1/\hat{\tilde{h}}(f)$, the inverse of that (hopefully nonzero) estimate. These methods are based on efficient digital signal processing algorithms. Most of them rely on LP analysis and differ mainly in how the all-pole vocal tract transfer function $\tilde{h}(f)$ is estimated (Kadiri et al., 2021). No explicit parametric form of $u(t)$ is assumed because $u(t)$ is recovered by applying the digital filter $1/\hat{\tilde{h}}(f)$ to a vector of speech samples $\mathbf{d}$.

Joint source-filter optimization Milenkovic (1986) proposed a different approach for the second class of GIF methods, a logical consequence of the increased computing power available in the 1980s. With joint source-filter optimization methods the fit of a parametric model to the speech data $\mathbf{d}$ is optimized to estimate both source $u(t)$ and filter $\tilde{h}(f)$ simultaneously (as opposed to consecutively as in the first class). These methods involve the nonconvex optimization of an objective function to determine the parameters $\boldsymbol{\theta}$ of some assumed functional form of the DGF $u'(t) = f(t; \boldsymbol{\theta})$, typically an approximation of the LF model (Alzamendi & Schlotthauer, 2017; Degottex, 2010; Fu & Murphy, 2006; Schleusing, 2012). This optimization is computationally costly compared to the inverse filtering methods in the first class, and a variety of tricks are used to reduce runtime and improve robustness, such as two-stage optimization (Fu & Murphy, 2006) and regressing $\boldsymbol{\theta}$ to a single parameter (Degottex, 2010).

[6]For the sake of the argument we ignore a here smaller third class which uses mixed-phase decomposition methods (Degottex, 2010; Drugman et al., 2019a; Kadiri et al., 2021) such as complex cepstrum decomposition (CCD) (Drugman et al., 2011).

Although we are not aware of any study that compares the performance of these two approaches systematically, it is possible to make a few general statements. Inverse filtering methods are generally efficient and nonparametric, while joint source-filter optimization methods have the distinct advantage of cleanly separating $u(t)$ and $\tilde{h}(f)$ from the outset. In doing so they avoid having to derive $u(t)$ directly from a filter estimate $\hat{\tilde{h}}(f)$ that inevitably contains pronounced effects from the former (Auvinen et al., 2014), to which Schroeder (1999, p. 94) refers as a "suspiciously circular sounding 'bootstrap' method."

So, where does that leave BNGIF? The main motivation for and innovation of BNGIF is that it mixes something of both classes: it is a joint source-filter optimization method capable of the flexibility of inverse filtering methods through its nonparametric model for $u(t)$. In addition, the Bayesian framework adopted by BNGIF handles uncertainty in a straightforward way through sampling the posterior, which can be seen as generalization of the nonconvex optimization that characterizes (and plagues) the class of joint source-filter optimization methods it belongs to. In contrast, inverse filtering methods pay for their efficiency by having no principled way of dealing with uncertainty, other than operationalising it as input variability.

1.4.1 Contributions and merits of BNGIF

Nonparametric glottal flow model BNGIF utilizes a nonparametric glottal flow model. That is, the unknown function $u(t)$ is modeled nonparametrically with the GP prior (10);, calibrated to the widely used LF model for $u'(t)$. As explained above, this is intended to strike a balance between the nonparametric flexibility of inverse filtering methods and the parametric control of source-filter optimization methods.

Conversely, BNGIF also avoids two particular shortcomings of currently used GIF methods. Inverse filtering methods typically model (a part of) $u(t)$ with only a few poles, either implicitly (such as the CP method mentioned above) or explicitly (such as IAIF, also mentioned above), which has since long been known to be "incapable of representing realistic physiologic glottal volume velocity or area waveforms" (Deller, 1983, p. 62). Source-filter optimization methods, on the other hand, use analytical models for the glottal flow derivative $u'(t)$ (Doval et al., 2006) which are convenient but necessarily "limited in their ability to capture the behavior of the glottal source in natural speech" (Kadiri et al., 2021, p. 1926).

Uncertainty quantification It is commonly agreed that uncertainty analysis and quantification is important for fundamental scientific research and forensic or biometric applications of acoustic phonetics, for example when characterizing quasi-vowels of great apes (Ekström et al., 2023) or calculating the evidence for a proposition in the context of forensic speaker comparison (Bonastre et al., 2015). However, in the specific case of GIF, "a difficult blind separation problem since neither the vocal tract response nor the glottal flow contribution are actually observable" (Degottex et al., 2014, p. 962), we would argue that uncertainty quantification takes on a nearly essential quality, if only as a rough measure of the GIF algorithm's trust in its own output.

As the reader probably anticipates by now, the natural framework for uncertainty quantification is Bayesian probability theory. Section 1.1 touched upon the reason for this, which is plain and simple: Bayesian theory prescribes how to reason consistently and quantitatively under uncertainty (that is, how to conduct inference mathematically), and it does so uniquely, as first demonstrated by (Cox, 1946). Uncertainty analysis is automatically incorporated in a Bayesian approach by sampling the posterior for the given problem, and that is of course what BNGIF does,[7] as explained in Section 1.2 and illustrated in Figure 1. Uncertainty quantification of various GIF-derived statistics such as the open quotient (OQ) and first formant frequency (F_1) then reduces to calculating the empirical mean $\pm$ standard deviation of these statistics directly from posterior samples of $u(t)$ and $\tilde{h}(f)$.

Despite their appeal for uncertainty quantification, Bayesian approaches to GIF are rare in the literature, and next to BNGIF we know of only three other probabilistic GIF methods: (Auvinen et al., 2014; Rao & Ghosh, 2018; Wang et al., 2016). (Bleyer et al., 2017) confirm that the first Bayesian inversion approach to GIF is described in (Auvinen et al., 2014), who call it MCMC-GIF. Surprisingly, these methods all produce point estimates of the glottal flow and vocal tract filter and do not seem to use posterior samples for uncertainty analysis.

Adaptive GCI estimation Contrary to the sliding windows approach used by conventional LP analysis for formant estimation, GIF methods are *pitch-synchronous* in nature and require knowledge of the glottal closure instants (GCIs) in order to work properly (Drugman et al., 2019b). This means that the user has to run an independent GCI detection algorithm first before GIF methods can be used.[8] A prominent (and according to (Chien et al., 2017), unique) exception to this is the aforementioned IAIF algorithm (Alku, 1992), which implements heuristic GCI estimation as part of its iterative structure.

On the other hand, BNGIF infers GCIs alongside and simultaneously with the source $u(t)$ and filter $\tilde{h}(f)$, which allows newly discovered information (during inference) about either one of these to constrain the others. This adaptive strategy helps to 'bootstrap' the GCIs from the speech signal, since GCIs are instrumental in recovering $u(t)$, but ultimately need to be derived from (and are defined in terms of) the latter. Another adaptive aspect of BNGIF's built-in GCI estimation is that noisy GCI estimates may *optionally* be supplied by the user a priori; BNGIF will then use this information to speed up inference, and yield refined estimates as part of its output. These adaptive abilities are quite unique in the GIF literature as far as we are aware.

Modeling inertia and nonstationarity The inertia of the vocal apparatus and articulators imposes constraints on the speed at which the speech signal $s(t)$ can be produced and altered. It is often the case that $s(t)$ appears relatively 'stable' or stationary during short time intervals of about 10 to 50 msec, an observation which underlies the ubiquitous utility and adoption

[7]Or rather: tries to do, as sampling the posterior is intractable for most real-world problems, and efficient approximations must be used instead.

[8]Note that robust GCI detection "is a very difficult task in practice" (Alzamendi & Schlotthauer, 2017, p. 14) for which many algorithms have been proposed (Drugman et al., 2019b, Sec. 3.3).

of the short-time Fourier transform (STFT) in acoustic phonetics (Little, 2011). In the case of voiced speech, the providence of GIF, these inertial constraints produce correlations across *dozens* of pitch periods . The source $u(t)$ and filter $\tilde{h}(f)$ constituents of voiced speech are both inherently nonstationary yet strongly correlated at the level of neighbouring pitch periods. We can actually *hear* this fact: pitch and timbre are perceptual renditions of these statistical regularities (Schnupp et al., 2011).

BNGIF can model nonstationary voiced speech which exhibits these inertial correlations, and takes advantage of the latter to speed up inference (since any correlation means more predictability, which often leads to faster exploration of plausible hypotheses). Figures Figure 4 and Figure 6 illustrate the local inertia and nonstationarity of $u(t)$ and $s(t)$ respectively. But BNGIF also takes into account 'filter correlations'; that is, the constrained variation of $\tilde{h}(f)$ from pitch period to pitch period. A pleasant consequence of this approach is probabilistic (anti)formant tracking, which we discuss just below.

The approach of BNGIF contrasts with many other GIF methods, which work one pitch period at a time and do not model inertial correlations between the pitch periods explicitly. For example, none of the algorithms considered in a recent review by (Chien et al., 2017) of the most important and representative inverse filtering methods use inter-pitch period correlations, and we are not aware of any joint source-filter optimization methods exploiting these correlations either.

Probabilistic (anti)formant tracking Formant tracking refers to estimating from speech data the (smooth) trajectories along which formants move over time, which in turn reflect the movement of the vocal articulators during phonation. Because of their efficiency as low-dimensional 'information bearers', formant tracks are highly important in acoustic phonetics and many algorithms exist for measuring them, mostly based on LP analysis (Van Soom & de Boer, 2020). These algorithms usually operate on a sliding window approach, estimating formants within each window and then using a post-processing step to join these point estimates into smooth trajectories (Whalen et al., 2022). A prominent example of this strategy is the widely used program (Boersma, 2001), whose default tracking function implements the Viterbi algorithm (Viterbi, 1967) to post-process the formant candidates into smooth trajectories.

In the case of BNGIF, formant tracking happens during inference and is achieved by exploiting the inertial correlations of the filter $\tilde{h}(f)$, as discussed above. That is, pole pairs are subjected to inter-pitch period correlations, and this takes care of 'trajectory smoothing' automatically while avoiding an ad-hoc post-processing step.

The same is true for the zero pairs: antiformant tracking is implicitly enabled in the case of a pole-zero expansion of the rational transfer function $\tilde{h}(f)$. We are not aware of other GIF methods with this ability, since GIF methods almost invariably assume all-pole transfer functions (Alku, 2011; Bleyer et al., 2017; Kadiri et al., 2021), even though spectral zeros occur during the open phase of the glottis due to coupling to the subglottal cavity (Ananthapadmanabha & Fant, 1982; Plumpe et al., 1999).

There are a handful of formant tracking algorithms, however, that implement the same principle – that is, smoothing of formant tracks during inference rather than as a post-processing step – of which the Kalman-based autoregressive moving average (Mehta et al., 2012) method is a relatively recent example. BNGIF and KARMA also have in common the ability to optionally use noisy a priori estimates of the formant trajectories to speed up inference.

Avoiding harmonic attraction Higher values of the fundamental frequency F_0 pose challenges for formant tracking algorithms based on LP analysis due to a phenomenon (Whalen et al., 2022) call "harmonic attraction." This refers to the fact that, as a result of spectral smoothing, the peaks of the estimated envelope obtained through LP are highly biased toward the harmonic partials of a speech spectrum (Yoshii & Goto, 2013), such that the estimated formant frequencies $\boldsymbol{F}$ tend to align with multiples of F_0. For higher values of F_0, this alignment causes the systematic correlations between F_0 and $\boldsymbol{F}$ known as F_0 bias. In the case that $F_0 \approx F_1$, known as $F_0 - F_1$ congruence (Kent & Vorperian, 2018), harmonic attraction can be especially challenging, since LP analysis, as a spectral smoothing device, lacks a conceptual model for separating the two – in contrast to, for example, the auditory processing regions of our brains (Schnupp et al., 2011).

Next to formant tracking, harmonic attraction equally affects GIF methods based on inverse filtering since the F_0-biased formant estimates result in insufficient cancelation of the true vocal tract resonances, which in turn distorts inference of the glottal flow $u(t)$ (Auvinen et al., 2014). This problem is well known, and there is extensive literature on the degradation of inverse filtering methods under increased values of F_0. In fact, 'increased values of F_0' often means 'not a low-pitched male voice' (Airaksinen et al., 2014), meaning that harmonic attraction renders a large majority of cases potentially problematic for inverse filtering methods! We also note that the $F_0 - F_1$ congruence mentioned above is especially problematic for GIF in the case of close rounded vowels, which remains "a major factor that limits the applicability of inverse filtering algorithms to accurate glottal flow estimation from continuous speech" (Chien et al., 2017, p. 17).

BNGIF is a joint-source filter optimization method and thereby automatically avoids harmonic attraction and the problems associated with $F_0 - F_1$ congruence. In addition, the probabilistic approach of BNGIF (as opposed to most GIF methods) is known to improve performance and reduced sensitivity to prior estimation of GCIs (Auvinen et al., 2014; Bleyer et al., 2017).

2 Infinite kernel linear prediction

The inference engine at the heart of BNGIF is a nonparametric Bayesian extension of linear prediction due to Yoshii & Goto (2013). The paper, presented over a decade ago at ICASSP 2013, has attracted fewer citations than its conceptual depth warrants — which, for our purposes, is fortunate, because it provides exactly the principled Bayesian foundation we need to replace the DSP heuristics of traditional glottal inverse filtering with a framework of testable probabilistic hypotheses. This chapter develops the model in full.

We proceed in three steps that mirror the paper's own structure: a classical probabilistic formulation of LP, a kernelized generalization in which the excitation is replaced by a Gaussian process, and the full Bayesian nonparametric limit in which the kernel is itself inferred from the data. To these we add two contributions of our own: a Woodbury-Cholesky reformulation of the inference equations that avoids all explicit $M \times M$ matrix inversions, and two reparameterizations of the prior hyperparameters into quantities with direct physical interpretations.

2.1 Linear prediction

Let $\mathbf{x} = (x_1, ..., x_M)^\top \in \mathbb{R}^M$ be M consecutive speech samples drawn from a short analysis frame. The autoregressive (AR) model of order P asserts that each sample is a linear combination of the P most recent ones plus an excitation residual:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m. \tag{15}$$

The vector $\mathbf{a} = (a_1, ..., a_P)^\top \in \mathbb{R}^P$ collects the *predictor coefficients* and $\boldsymbol{\varepsilon} = (\varepsilon_1, ..., \varepsilon_M)^\top$ is the *excitation*. In source-filter terms these play distinct roles: $\boldsymbol{\varepsilon}$ represents the glottal excitation and $\mathbf{a}$ encodes the resonance structure of the vocal tract.

Two matrices appear repeatedly. The *AR operator* $\boldsymbol{\Psi}(\mathbf{a}) \in \mathbb{R}^{M \times M}$ is approximately lower-triangular with unit diagonal and sub-diagonal entries given by $-a_1, ..., -a_P$; it satisfies $\boldsymbol{\varepsilon} = \boldsymbol{\Psi}\mathbf{x}$, or equivalently $\mathbf{x} = \boldsymbol{\Psi}^{-1}\boldsymbol{\varepsilon}$. The *lag matrix* $\mathbf{X} \in \mathbb{R}^{M \times P}$ has m-th row $(x_{m-1}, ..., x_{m-P})$ and is the design matrix for the normal equations.

The classical probabilistic LP model assumes white Gaussian excitation,

$$\boldsymbol{\varepsilon} \sim \mathtt{Normal}(\mathbf{0}, \nu\mathbf{I}), \tag{16}$$

which yields the likelihood

$$\mathbf{x} \sim \mathtt{Normal}(\mathbf{0}, \nu\boldsymbol{\Psi}^{-1}\boldsymbol{\Psi}^{-\top}). \tag{17}$$

Maximum-likelihood estimation under (17) yields the familiar normal equation $\mathbf{X}^\top\mathbf{X}\mathbf{a} = \mathbf{X}^\top\mathbf{x}$, which is what every standard LPC implementation solves.

The model works as designed for speech with broadband, noise-like excitation. It fails for voiced speech. During voiced phonation the excitation is periodic, not white, and fitting (17) to a pitched signal forces the estimated envelope to track the harmonic partials of the spectrum: the poles of the all-pole filter chase the pitch harmonics rather than the formants. This is the central defect that motivates everything that follows.

2.2 Kernel linear prediction

The fix is to replace the white-noise excitation model with a richer one. We model the excitation as a continuous function $\varepsilon(t)$ over time, approximated by a weighted sum of J basis functions plus a broadband residual:

$$\varepsilon(t) = \varphi(t)^\top \mathbf{w} + \eta(t), \tag{18}$$

where $\varphi(t) = \left(\varphi_1(t), ..., \varphi_{J(t)}\right)^\top$ and $\eta(t)$ is the residual. Sampling at times $t_1, ..., t_M$ and writing $\mathbf{\Phi} \in \mathbb{R}^{M \times J}$ for the design matrix with rows $\varphi(t_m)^\top$ gives $\boldsymbol{\varepsilon} = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\eta}$ in vector form.

With independent Gaussian priors

$$\mathbf{w} \sim \texttt{Normal}(\mathbf{0}, \nu_w \mathbf{I}), \quad \boldsymbol{\eta} \sim \texttt{Normal}(\mathbf{0}, \nu_e \mathbf{I}), \tag{19}$$

marginalizing over $\mathbf{w}$ yields

$$\boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \nu_w \mathbf{K} + \nu_e \mathbf{I}), \quad \mathbf{K} = \mathbf{\Phi}\mathbf{\Phi}^\top. \tag{20}$$

The matrix $\mathbf{K}$ is a *kernel matrix* with entries $K_{mm'} = \varphi(t_m)^\top \varphi(t_{m'})$, and the key observation of Kameoka et al. (2010) is that any positive semidefinite matrix can serve as $\mathbf{K}$ directly via the kernel trick — $K_{mm'} = k(t_m, t_{m'})$ for some kernel function k — without constructing the basis functions at all.

Substituting (20) into $\mathbf{x} = \mathbf{\Psi}^{-1}\boldsymbol{\varepsilon}$ gives the kernelized LP likelihood,

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}(\nu_w \mathbf{K} + \nu_e \mathbf{I})\mathbf{\Psi}^{-\top}\right). \tag{21}$$

Classical LP (17) is the special case $\mathbf{K} = \mathbf{I}$ with $\nu = \nu_w + \nu_e$. A periodic kernel with period T concentrates excitation power at harmonics and thereby *decorrelates* the envelope estimate from the pitch structure — exactly the fix we needed.

Excitation semantics It is worth being precise about how the structured and broadband components of $\boldsymbol{\varepsilon}$ map onto the GIF picture. We write the excitation as $\varepsilon(t) = s(t) + \eta(t)$, where $s(t) = \varphi(t)^\top \mathbf{w}$ is the structured GP component and $\eta(t)$ is broadband noise, and we distinguish this from $u'(t)$, the true physical DGF. In this thesis $u'(t)$ is the theoretical DGF in the sense of source-filter theory, and inferred $\varepsilon(t)$ is our practical DGF estimate. During voiced regions these are close but not equal, and we always report $\varepsilon(t)$ rather than $s(t)$ alone as our output.

This is deliberate. Even ordinary LPC — the $s = 0$ limit where all excitation comes from $\eta(t)$ — already recovers recognizable DGF estimates from long enough analysis frames, because $\eta(t)$ is free to inject the sharp broadband transients needed to excite all formant resonances simultaneously. Those sharp transients also model the broadband energy injections associated with glottal closure instants in the vocal apparatus. Extending from LPC to IKLP enriches what $s(t)$ can model but leaves $\eta(t)$ active and free to do this useful work — which is precisely what we want.

2.3 Infinite kernel linear prediction

2.3.1 From a fixed kernel to an infinite hypothesis bank

The kernelized model (21) requires designing $\mathbf{K}$ to match the excitation structure, but for voiced speech the fundamental period T is itself unknown and must be inferred. The natural Bayesian response is multiple kernel learning: prepare a bank of I candidate kernels $\mathbf{K}_1, ..., \mathbf{K}_I$ and define the excitation kernel as their weighted sum,

$$\mathbf{K} = \sum_{i=1}^{I} \theta_i \mathbf{K}_i, \quad \theta_i \geq 0. \tag{22}$$

Each weight $\theta_i \geq 0$ quantifies the degree to which hypothesis i explains the excitation. In the original formulation of Yoshii & Goto (2013), the bank indexes candidate fundamental periods $T_1, ..., T_I$ on a grid, so the weights θ_i directly encode a posterior over F0 candidates.

This is an important but too-narrow framing. The index i is a generic *hypothesis axis*: it can index candidate periods, candidate glottal flow morphologies, candidate phonation modalities, or any structured excitation model that can be expressed as a kernel. We will exploit this generality systematically in later chapters, where the bank is populated with GP priors learned from synthetic DGF data. The resulting multiple-kernel likelihood is

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{I} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\mathbf{\Psi}^{-\top}\right). \tag{23}$$

MAP estimation of $\boldsymbol{\theta}$ under a regularizing prior does not yield truly sparse weights. Yoshii & Goto (2013) solve this by taking $I \to \infty$ and placing a *gamma process* (GaP) prior on the infinite weight sequence: for any finite truncation level I,

$$\theta_i \sim \texttt{Gamma}(\alpha / I, \alpha), \tag{24}$$

and as $I \to \infty$ the vector $\boldsymbol{\theta}$ converges to a draw from a GaP with concentration α. It is proven that the effective number of active elements — those with $\theta_i > \varepsilon$ for any $\varepsilon > 0$ — is almost surely finite, and for $I \gg \alpha$ only $\mathcal{O}(\alpha)$ weights are substantially positive. Sparsity is a theorem about the prior, not an approximation.

The full IKLP likelihood is

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{\infty} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\mathbf{\Psi}^{-\top}\right). \tag{25}$$

This single model simultaneously infers the spectral envelope through $\mathbf{a}$, the fundamental frequency through the dominant θ_i, and voiced/unvoiced status through the ratio $\mathbb{E}[\nu_w]/\mathbb{E}[\nu_w + \nu_e]$ — all at once, in a principled Bayesian framework, with none of the separate heuristic steps that characterize classical DSP approaches to GIF.[9]

2.4 Inference

2.4.1 Priors and variational family

To complete the model we place priors on the positive scale parameters and on $\mathbf{a}$. Following Yoshii & Goto (2013), the scales receive gamma priors,

$$\nu_w \sim \texttt{Gamma}(a_w, b_w), \quad \nu_e \sim \texttt{Gamma}(a_e, b_e). \tag{26}$$

For the AR coefficients we generalize the isotropic prior $\mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \lambda\mathbf{I})$ of the original paper to a full multivariate Gaussian,

$$\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_a, \mathbf{\Sigma}_a), \quad \mathbf{Q} := \mathbf{\Sigma}_a^{-1}, \tag{27}$$

with precision matrix $\mathbf{Q}$. The isotropic case is $\boldsymbol{\mu}_a = \mathbf{0}$, $\mathbf{Q} = \lambda^{-1}\mathbf{I}$. We need the full form in Chapter 3, where informative priors derived from stability theory and spectral constraints replace the isotropic default.

Exact posterior inference is intractable, so we use a mean-field variational family,

$$q(\boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e) = q(\mathbf{a})\, q(\nu_w)\, q(\nu_e) \prod_{i=1}^{I} q(\theta_i), \quad q(\mathbf{a}) = \delta(\mathbf{a} - \mathbf{a}^*), \tag{28}$$

with $\mathbf{a}$ treated as a MAP point estimate. Each remaining factor is updated by coordinate ascent on the ELBO $\mathcal{L}$:

$$\log p(\mathbf{x}) \geq \underbrace{\mathbb{E}[\log p(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e)]}_{\text{expected log-likelihood}} + \sum_z \mathbb{E}[\log p(z)] - \sum_z \mathbb{E}[\log q(z)] =: \mathcal{L}. \tag{29}$$

Compared to MCMC-based GIF approaches — where Auvinen et al. (2014) report runtimes on the order of 2.5 hours of single-CPU time per 25-millisecond analysis frame — variational inference keeps the Bayesian treatment while reducing inference to a sequence of closed-form updates. CAVI is convergence-guaranteed, parallelizable over batches of frames, and compatible with modern autodiff and GPU workflows even when the update steps are computed analytically.

[9] Setting $P = 0$ is also valid: the AR operator becomes $\mathbf{\Psi} = \mathbf{I}$ and IKLP reduces to a pure multi-kernel Bayesian inference engine over the excitation process with no envelope component. This is useful as a standalone pitch detector or phonation-type classifier.

2.4.2 Tractable lower bound via matrix inequalities

The expected log-likelihood involves $\mathbb{E}[\log|\mathbf{K}|]$ and $\mathbb{E}[\mathbf{x}^\top\mathbf{\Psi}^\top\mathbf{K}^{-1}\mathbf{\Psi}\mathbf{x}]$, where $\mathbf{K} = \nu_w \sum_i \theta_i \mathbf{K}_i + \nu_e \mathbf{I}$, and neither expectation is tractable in closed form. Two matrix-variate inequalities provide bounds.

The log-determinant $\log|\mathbf{V}|$ is concave, so a first-order Taylor expansion around any PSD matrix $\mathbf{\Omega}$ gives

$$\log|\mathbf{V}| \leq \log|\mathbf{\Omega}| + \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{V}) - M. \tag{30}$$

The quadratic form $\mathbf{z}^\top\mathbf{V}^{-1}\mathbf{z}$ is convex, and an inequality of Sawada et al. (2012) gives

$$\mathbf{z}^\top \left(\sum_i \mathbf{V}_i\right)^{-1} \mathbf{z} \leq \sum_i \mathbf{z}^\top \mathbf{\Upsilon}_i^\top \mathbf{V}_i^{-1} \mathbf{\Upsilon}_i \mathbf{z}, \tag{31}$$

where $\{\mathbf{\Upsilon}_i\}$ are auxiliary matrices summing to the identity.

Applying both inequalities to the expected log-likelihood and optimizing the auxiliary quantities in closed form gives the tractable lower bound $\mathcal{L}' \leq \mathcal{L}$, with optimal values

$$\mathbf{\Omega} = \mathbb{E}[\nu_w] \sum_i \mathbb{E}[\theta_i]\mathbf{K}_i + \mathbb{E}[\nu_e]\mathbf{I}, \tag{32}$$

$$\mathbf{\Upsilon}_i = \tilde{w}_i \mathbf{K}_i \mathcal{S}^{-1}, \quad \mathbf{\Upsilon}_0 = \tilde{\nu}_e \mathcal{S}^{-1}, \tag{33}$$

where the *quadratic operator* $\mathcal{S}$ and harmonic-mean weights are defined by

$$\mathcal{S} = \sum_i \tilde{w}_i \mathbf{K}_i + \tilde{\nu}_e \mathbf{I}, \quad \tilde{\nu}_e = 1/\mathbb{E}[\nu_e^{-1}], \quad \tilde{w}_i = 1/(\mathbb{E}[\nu_w^{-1}]\mathbb{E}[\theta_i^{-1}]). \tag{34}$$

Both $\mathbf{\Omega}$ and $\mathcal{S}$ have the form $\nu\mathbf{I} + \sum_i w_i \mathbf{K}_i$; they differ only in which moment or harmonic-mean expectations enter as weights. This shared algebraic structure is what the Woodbury reformulation of Section 2.5 exploits.

2.4.3 GIG posteriors and CAVI updates

The sufficient statistics of a gamma prior are $\log z$ and z; the bounds above introduce terms in z and $1/z$. The optimal variational posteriors for θ_i, ν_w, ν_e therefore all take the form of a *generalized inverse Gaussian*,

$$\mathtt{GIG}(z \mid \gamma, \rho, \tau) \propto z^{\gamma-1} \exp(-1/2(\rho z + \tau/z)), \quad z > 0, \tag{35}$$

with moments

$$\mathbb{E}[z] = \sqrt{\tau/\rho}\, K_{\gamma+1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \quad \mathbb{E}[z^{-1}] = \sqrt{\rho/\tau}\, K_{\gamma-1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \tag{36}$$

where K_γ is the modified Bessel function of the second kind.

With the LP residual $\mathbf{r} := \mathbf{\Psi}(\mathbf{a})\mathbf{x}$, the weighted residual $\mathbf{w} := \mathcal{S}^{-1}\mathbf{r}$, and the per-kernel quadratic statistics

$$\mathbf{u}_i := \mathbf{\Phi}_i^\top \mathbf{w}, \quad q_i := \|\mathbf{u}_i\|^2, \quad q_0 := \|\mathbf{w}\|^2, \tag{37}$$

the CAVI parameter updates take the compact form. For the kernel weights:

$$\gamma_{\theta_i} = \alpha / I, \quad \rho_{\theta_i} \leftarrow 2\alpha + \mathbb{E}[\nu_w]\ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_{\theta_i} \leftarrow (1/\mathbb{E}[\nu_w^{-1}])\left(1/\mathbb{E}[\theta_i^{-1}]^2\right) q_i. \tag{38}$$

For the structured scale:

$$\gamma_w = a_w, \quad \rho_w \leftarrow 2b_w + \sum_i \mathbb{E}[\theta_i]\ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_w \leftarrow \sum_i \left(1/\mathbb{E}[\nu_w^{-1}]^2\right)(1/\mathbb{E}[\theta_i^{-1}]) q_i \tag{39}$$

For the broadband scale:

$$\gamma_e = a_e, \quad \rho_e \leftarrow 2b_e + \mathrm{tr}(\mathbf{\Omega}^{-1}), \quad \tau_e \leftarrow \left(1/\mathbb{E}[\nu_e^{-1}]^2\right) q_0. \tag{40}$$

The AR update is MAP under the generalized prior (27), giving the normal equation

$$(\mathbf{X}^\top \mathcal{S}^{-1}\mathbf{X} + \mathbf{Q})\mathbf{a}^* = \mathbf{X}^\top \mathcal{S}^{-1}\mathbf{x} + \mathbf{Q}\boldsymbol{\mu}_a. \tag{41}$$

All updates are iterated until the ELBO converges or a maximum iteration count is reached.

The dominant costs per iteration are the I trace terms $\mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i)$ and the operator actions $\mathcal{S}^{-1}\mathbf{r}$ and $\mathcal{S}^{-1}\mathbf{X}$, all requiring solutions to linear systems in matrices of the form $\nu\mathbf{I} + \sum_i w_i \mathbf{K}_i$. The next section reduces all of these to triangular solves on a single small matrix.

2.5 Woodbury-Cholesky reformulation

Both $\mathbf{\Omega}$ and $\mathcal{S}$ are instances of the abstract operator

$$\mathbf{S}(\nu, \mathbf{w}) := \nu \mathbf{I}_M + \sum_{i=1}^{I} w_i \mathbf{K}_i \in \mathbb{R}^{M \times M}. \tag{42}$$

The direct approach materializes this $M \times M$ matrix, factors it by Cholesky ($\mathcal{O}(M^3)$), and solves systems against it ($\mathcal{O}(M^2)$ each). For $M = 2048$ samples at $I = 400$ kernels this is expensive; for large frames it is prohibitive.

Suppose each kernel admits a low-rank factorization $\mathbf{K}_i = \mathbf{\Phi}_i \mathbf{\Phi}_i^\top$ with $\mathbf{\Phi}_i \in \mathbb{R}^{M \times r_i}$. In the settings we care about this is exact: the periodic kernels of the original paper are parameterized by Fourier features, and the arc cosine GP priors of later chapters are defined by their feature matrices $\mathbf{\Phi}_i$. Form the weighted stacked factor

$$\tilde{\mathbf{\Phi}} := [\sqrt{w_1}\mathbf{\Phi}_1, ..., \sqrt{w_I}\mathbf{\Phi}_I] \in \mathbb{R}^{M \times L}, \quad L = \sum_i r_i, \tag{43}$$

so that $\mathbf{S}(\nu, \mathbf{w}) = \nu \mathbf{I}_M + \tilde{\mathbf{\Phi}}\tilde{\mathbf{\Phi}}^\top$. The Woodbury matrix identity gives its inverse as

$$\mathbf{S}^{-1}\mathbf{v} = \nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\mathbf{\Phi}}\mathbf{A}^{-1}\tilde{\mathbf{\Phi}}^{\top}\mathbf{v}, \quad \mathbf{A} := \mathbf{I}_L + \nu^{-1}\tilde{\mathbf{\Phi}}^{\top}\tilde{\mathbf{\Phi}} \in \mathbb{R}^{L\times L}. \tag{44}$$

The $L \times L$ *core matrix* $\mathbf{A}$ is the only object that needs to be factored. Its Cholesky decomposition $\mathbf{A} = \mathbf{R}^{\top}\mathbf{R}$ — computed once per operator instance per CAVI iteration — makes the following quantities available through triangular solves alone.

Operator solves To compute $\mathbf{S}^{-1}\mathbf{v}$: form $\mathbf{t} = \tilde{\mathbf{\Phi}}^{\top}\mathbf{v} \in \mathbb{R}^{L}$, solve the triangular systems $\mathbf{R}^{\top}\mathbf{y} = \mathbf{t}$ then $\mathbf{R}\mathbf{u} = \mathbf{y}$ to get $\mathbf{u} = \mathbf{A}^{-1}\mathbf{t}$, then return $\nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\mathbf{\Phi}}\mathbf{u}$. No explicit inverse is formed.

Log-determinant By the matrix determinant lemma, $|\mathbf{S}| = \nu^{M}\ |\mathbf{A}|$, so

$$\log|\mathbf{S}| = M\log\nu + 2\sum_{\ell=1}^{L}\log R_{\ell\ell}. \tag{45}$$

Trace of inverse

$$\operatorname{tr}(\mathbf{S}^{-1}) = (M-L)/\nu + \nu^{-1}\ \operatorname{tr}(\mathbf{A}^{-1}). \tag{46}$$

Per-kernel trace Define $\mathbf{B}_i := \tilde{\mathbf{\Phi}}^{\top}\mathbf{\Phi}_i \in \mathbb{R}^{L\times r_i}$. Then

$$\operatorname{tr}(\mathbf{S}^{-1}\mathbf{K}_i) = \nu^{-1}\ \|\mathbf{\Phi}_i\|_F^2 - \nu^{-2}\ \|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2, \tag{47}$$

where $\|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2$ is the squared Frobenius norm of the solution to $\mathbf{R}^{\top}\mathbf{Z} = \mathbf{B}_i$.

AR normal equation With $\mathcal{S}^{-1}$ available as a triangular-solve operator, the $P \times P$ Gram matrix $\mathbf{G} = \mathbf{X}^{\top}\mathcal{S}^{-1}\mathbf{X}$ is assembled column-by-column, and (41) is then a single $P \times P$ Cholesky solve of $\mathbf{G} + \mathbf{Q}$.

Every dense operation in the CAVI loop has been reduced to triangular solves on matrices of size at most $\max(L, P) \ll M$.

2.5.1 Complexity

Let M be the frame length, P the AR order, I the number of kernels, r the rank per kernel, $L = Ir$ the total factor dimension, and B the number of frames in a batch.

In the dense baseline, building $\mathbf{S}$ costs $\mathcal{O}(IM^2r)$, factoring it costs $\mathcal{O}(M^3)$, each solve costs $\mathcal{O}(M^2)$, and storing it requires $\mathcal{O}(M^2)$ memory.

In the Woodbury-Cholesky regime, the core matrix $\mathbf{A}$ is assembled in $\mathcal{O}(ML^2)$ and factored in $\mathcal{O}(L^3)$. Each operator solve costs $\mathcal{O}(ML + L^2)$, the log-determinant costs $\mathcal{O}(L)$ after factorization, the per-kernel traces cost $\mathcal{O}(L^2r)$ each, and memory drops to $\mathcal{O}(ML + L^2)$: the $M \times M$ covariance is never stored. The crossover is favorable whenever $L \ll M$, precisely the regime of interest.

For B frames the loop vectorizes linearly in B. Since no dense $M \times M$ matrix is ever formed, the formulation is insensitive to its conditioning. Running in 32-bit floating point gives a substantial throughput gain on accelerators while retaining numerical stability through the Cholesky factorization and a small diagonal jitter on $\mathbf{A}$.

2.6 Prior reparameterizations

The priors (24) and (26) are written in shape-rate form, natural for conjugate analysis but opaque for prior elicitation. We reparameterize both into quantities with direct physical interpretations.

2.6.1 Kernel sparsity: α-calibration

Under (24), define $T := \sum_i \theta_i$ and the normalized weights $p_i = \theta_i / T$. For any finite truncation I, the sum $T \sim \texttt{Gamma}(\alpha, \alpha)$ has $\mathbb{E}[T] = 1$ and $\mathrm{sd}(T) = \alpha^{-\frac{1}{2}}$, while the proportions follow a symmetric Dirichlet:

$$(p_1, ..., p_I) \sim \texttt{Dirichlet}(\alpha/I, ..., \alpha/I). \tag{48}$$

Small α concentrates mass toward the vertices of the probability simplex — one dominant hypothesis — while large α spreads it uniformly.

A natural summary of effective sparsity is the exponentiated Shannon entropy,

$$I_{\text{eff}} := \exp(\mathbb{E}[H(\mathbf{p})]), \quad H(\mathbf{p}) = -\sum_i p_i \log p_i, \tag{49}$$

which counts the number of equally probable hypotheses that would produce the same entropy. Since the Dirichlet expectation of H is known analytically,

$$\mathbb{E}[H(\mathbf{p})] = \psi(\alpha + 1) - \psi(1 + \alpha/I), \tag{50}$$

we obtain

$$I_{\text{eff}(\alpha;I)} = \exp(\psi(\alpha + 1) - \psi(1 + \alpha/I)). \tag{51}$$

As $I \to \infty$ with α fixed, I_{eff} saturates at the constant $\exp(\psi(\alpha + 1) + \gamma_{\text{EM}})$ where γ_{EM} is the Euler-Mascheroni constant. At $\alpha = 1$ and large I this gives $I_{\text{eff}} \to e \approx 2.72$, meaning roughly a handful of simultaneously active hypotheses on average. This matches what Yoshii & Goto (2013) observe empirically at $\alpha = 1$, $I = 400$.

To calibrate α to a desired effective count I^0_{eff}, one solves (50) for α; the solution $\alpha^*(I)$ approaches 1 rapidly for $I^0_{\text{eff}} = e$ as I grows. The local sensitivity of I_{eff} to α near this calibration is described by the log-elasticity

$$b(\alpha; I) := \frac{\partial \log I_{\text{eff}}}{\partial \log \alpha} = \alpha[\psi_1(\alpha + 1) - I^{-1}\psi_1(1 + \alpha/I)], \tag{52}$$

where ψ_1 is the trigamma function. As $I \to \infty$ this approaches $b \to \psi_1(2) = \pi^2/6 - 1 \approx 0.645$, giving the local approximation

$$I_{\text{eff}(\alpha;I)} \approx e(\alpha/\alpha^*(I))^{0.645}. \tag{53}$$

So α behaves like a concentration parameter whose effect on sparsity is governed by a fixed exponent near the default calibration.

2.6.2 Excitation power decomposition: the (p, s, κ)-reparameterization

When $b_w = b_e =: b$, the total excitation power $\nu_w + \nu_e \sim \texttt{Gamma}(a_w + a_e, b)$ and the structured fraction $\pi := \nu_w/(\nu_w + \nu_e) \sim \texttt{Beta}(a_w, a_e)$ are independent of each other. This motivates the reparameterization

$$a_w = \kappa p, \quad a_e = \kappa(1 - p), \quad b_w = b_e = \kappa/s, \tag{54}$$

under which the total power has prior $\nu_w + \nu_e \sim \texttt{Gamma}(\kappa, \kappa/s)$ with

$$\mathbb{E}[\nu_w + \nu_e] = s, \quad \mathrm{Var}(\nu_w + \nu_e) = s^2/\kappa, \tag{55}$$

and the structured fraction has prior $\pi \sim \texttt{Beta}(\kappa p, \kappa(1 - p))$ with

$$\mathbb{E}[\pi] = p, \quad \mathrm{Var}(\pi) = p(1 - p)/(\kappa + 1). \tag{56}$$

The parameter κ controls concentration only — it tightens both distributions simultaneously without moving their means. The parameters (p, s) directly control the prior expected values: s is calibrated to the expected dynamic range of the analysis frame, and p reflects prior beliefs about phonation type. A frame expected to be clearly voiced might receive $p = 0.8$; an unvoiced prior might use $p = 0.1$.

The default of Yoshii & Goto (2013), $(a_w, a_e, b_w, b_e) = (1, 1, 1, 1)$, corresponds to $(p, s, \kappa) = (1/2, 2, 2)$: equal prior probability of structured and broadband excitation, total power expected at 2, with moderate concentration. A unit-power neutral default is $(p, s, \kappa) = (1/2, 1, 1)$.

2.7 Summary

IKLP generalizes classical LP in three directions at once. The white excitation assumption is replaced by a GP prior whose kernel is a sparse mixture learned from the data. The single scalar noise variance is decomposed into structured and broadband components whose ratio tracks voiced/unvoiced status. And the AR prior is generalized to a full multivariate Gaussian, ready to accept the informative spectral priors of Chapter 3.

All of this fits inside a single CAVI loop with closed-form GIG updates and a convergence guarantee. The Woodbury-Cholesky reformulation of Section 2.5 removes the cubic bottleneck, reducing every inner-loop computation to triangular solves on matrices of size $\max(L, P) \ll M$. This makes the method both fast and composable: the kernel bank $\{\mathbf{\Phi}_i\}$ is the only interface between the inference engine and the prior on the excitation. Replacing the periodic kernels of the original paper with the arc cosine GP priors of later chapters requires no change to the inference architecture — only the features $\mathbf{\Phi}_i$ change.

The GIF problem of Chapter 1 requires finding plausible source-filter pairs consistent with a speech observation. IKLP, equipped with the right kernel bank, is exactly a hypothesis-testing engine for that problem.

3 Priors for autoregressive filters

The vocal tract filter is modeled throughout this thesis as an $\mathrm{AR}(P)$ process: a rational all-pole transfer function whose P coefficients $\mathbf{a} = (a_1, ..., a_P)^\top$ encode the resonant structure of the vocal tract. Estimating $\mathbf{a}$ from speech data is done traditionally by LPC methods, or in our case, by IKLP. Like any Bayesian model, IKLP requires a prior $p(\mathbf{a})$, and throughout their paper Yoshii & Goto (2013) simply assert

$$p(\mathbf{a}) = \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P), \quad \text{with a fixed } \lambda := 0.1. \tag{57}$$

This choice, which we will refer to as the *isotropic prior*, is clearly something more of a blunt regularizer than an informative prior. It's there to prevent the normal matrix in (41) from becoming singular, not to encode anything known a priori about the vocal tract.

While it gets the job done, two things about (57) are perhaps unsatisfying. First, the isotropic prior with $\lambda = 0.1$ implies a standard deviation of about 3.16 per coefficient. This is a very loose prior under which most induced AR filters will almost surely be unstable. Second, we actually know a great deal about vocal tract filters: their resonances are spaced roughly 500 Hz apart, their spectral tilt is on the order of $-6\ \mathrm{dB/octave}$, and, plain from the physics of speech production, the filter must be at least be passive and causal. Except for causality none of this is encoded in (57).

This chapter then develops two priors that attempt to do better in this regard. The first, the *Monahan prior*, is the closest Gaussian to the uniform distribution over stable AR coefficient vectors; at first sight, this sounds like exactly the thing to do, but we show that it doesn't work. The second, the *soft spectral shaping prior*, is a new construction that encodes a number of spectral features useful in speech – resonances and spectral rolloff – directly into the prior mean and covariance through a closed-form projection. Both priors remain Gaussian and are therefore compatible with IKLP's variational inference without modification, and we test their merit in Chapter 7.

3.1 The isotropic prior

3.1.1 Expected output power

To understand the inductive bias that $\lambda = 0.1$ represents, consider what an isotropic Gaussian prior

$$\mathtt{Isotropic}(\lambda) := \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \tag{58}$$

does to the expected output power of an $\mathrm{AR}(P)$ process driven by white noise with variance ν as in equations (15) and (16) from the previous chapter:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m, \quad \text{where} \quad \begin{cases} \mathbf{a} & \sim \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \\ \varepsilon_m & \sim \mathtt{Normal}(0, \nu) \end{cases} \tag{59}$$

Up to scale, then, the output power is determined by the frequency response

$$H(e^{i\omega}) = \frac{1}{A(e^{i\omega})} \quad \text{where} \quad A(z) = 1 - \sum_{p=1}^{P} a_p z^{-p}. \tag{60}$$

To second order, the Maclaurin expansion of $|H(e^{j\omega})|^2$ in a_p gives

$$|H(e^{j\omega})|^2 \approx 1 + 2\sum_{p=1}^{P} a_p \cos(p\omega) - \sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos((p-\ell)\omega) \tag{61}$$

$$+ 4\sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos(p\omega)\cos(\ell\omega) + \ldots \tag{62}$$

Taking the expectation over the zero-mean isotropic prior (57), the linear terms and the cross-terms $p \neq \ell$ vanish and the expected spectral power becomes

$$\mathbb{E}_{\mathbf{a}}[|H(e^{j\omega})|^2] \approx 1 - P\lambda + 4\lambda \sum_{p=1}^{P} \cos^2(p\omega). \tag{63}$$

Integrating over frequency and applying Parseval's identity gives the expected output variance of (59):

$$\mathbb{E}[\mathrm{var}(x_m)] \approx \nu(1 + P\lambda). \tag{64}$$

With $\nu := 1$, $P = 30$ and $\lambda = 0.1$ as in Yoshii & Goto (2013), this gives an expected standard deviation of $\sqrt{1+3} = 2$, which is indeed $O(1)$. Nevertheless, power scales linearly with P, which contradicts the prevalent idea that increasing P just prepares the model for more fine-structured data, not "louder" data. The lesson from (64) is thus: to keep power approximately controlled under an isotropic prior, one needs $\lambda \propto 1/P$. This rule is a known heuristic in signal processing literature (Rabiner et al., 1993, for example; Sørbye & Rue, 2017), here we have derived it from a probabilistic argument.

3.1.2 Stability

Beyond raw power, there is a more fundamental problem: `Isotropic`(λ) places most of its mass outside the stability region $\mathcal{A}_P \subset \mathbb{R}^P$, which is the set of coefficient vectors for which all roots of $A(z) = 1 - \sum_p a_p z^{-p}$ lie strictly inside the unit disk. Physically, *stability* means the filter's impulse response decays over time: the vocal tract is a passive resonator that dissipates energy, so its transfer function cannot have poles on or outside the unit circle. An unstable AR filter would predict an impulse response that grows without bound, which is physically impossible for a tube of flesh and air (the vocal tract).

For $P = 1$ the stability region is the interval $(-1, 1)$, and a unit Gaussian already places about 32% of its mass outside. For larger P the stability region is a complicated manifold and the fraction of unstable draws tends to grow rapidly – see Figure 7.

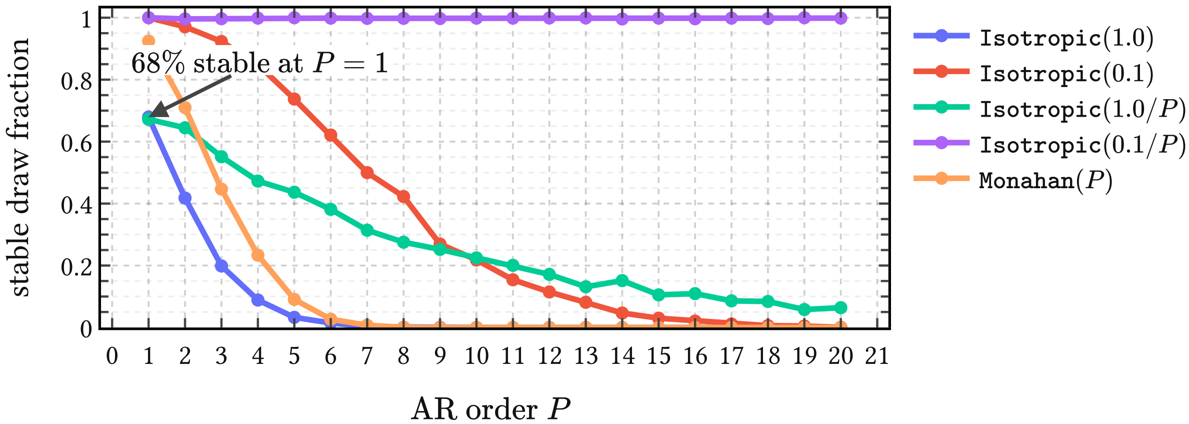


Figure 7 | **Probability that a prior draw lies in the stable AR region $\mathcal{A}_P$** for different values of λ with the isotropic prior (58). Stable prior mass collapses quickly under isotropic Gaussian AR priors, unless the $\lambda \propto 1/P$ rule is used with a $O(0.1)$ proportionality constant; the Monahan prior (68) performs underwhelmingly.

This figure also reveals that the proportional constant in the $\lambda \propto 1/P$ rule does seem to matter, because a Gaussian prior for $\mathbf{a}$ entangles expected power output and stability. At one extreme, $\lambda \to 0$ yields a filter than is trivially stable but will only produce signals with vanishing power. However, turning up expected power output to $O(1)$ by increasing λ then gradually decreases stability. The `Isotropic`$(0.1/P)$ prior is seen to strike a reasonable balance here. But can't we encode our preference for stability directly?

3.2 The Monahan prior

3.2.1 The Monahan map

A classical result from time series analysis (Monahan, 1984) is that the partial autocorrelation function (PACF) parameters $\boldsymbol{\varphi} = (\varphi_1, ..., \varphi_P)^\top$ provide a smooth bijection between the hypercube $(-1, 1)^P$ and the stability region $\mathcal{A}_P$:

$$T : (-1, 1)^P \to \mathcal{A}_P, \quad \mathbf{a} = T(\boldsymbol{\varphi}). \tag{65}$$

The map T is defined by the so-called Levinson recursion which we illustrate below in Figure 8, but whose details are unimportant for our purpose. Its main value is that drawing $\varphi_p \overset{\text{iid}}{\sim}$ `Uniform`$(-1, 1)$ produces only stable AR coefficient vectors. The push-forward of this uniform through T defines a density

$$g(\mathbf{a}) = 2^{-P} \ |\det \nabla_{\mathbf{a}} T^{-1}(\mathbf{a})|, \quad \mathbf{a} \in \mathcal{A}_P, \tag{66}$$

on the stability region, where 2^{-P} is just the density of the uniform base measure on $(-1, 1)^P$. This g is the natural "uniform over stable filters" measure: it assigns equal mass to equal volumes in PACF space, mapped into coefficient space by T.

Since tractable inference in AR models usually requires Gaussian priors, which (66) is not, the best we can do then is to approximate g by a Gaussian $p(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P)$ that envelops $\mathcal{A}_P$ in some optimal way.

3.2.2 Deriving the prior

We want the Gaussian to encompass $g(\mathbf{a})$, such that it is neither overconfident about any subregion of $\mathcal{A}_P$, nor wasteful in placing mass far outside it. We thus minimize $D_{\mathrm{KL}}\left(g \parallel \texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S})_P\right)$ over all Gaussians and solve for $\boldsymbol{\mu}_P$ and $\mathbf{S}_P$. This is a standard convex problem whose unique solution is the moment-matched Gaussian:

$$\boldsymbol{\mu}_P = \mathbb{E}_g[\mathbf{a}] = \mathbf{0}, \quad \mathbf{S}_P = \mathbb{E}_g[\mathbf{a}\mathbf{a}^\top]. \tag{67}$$

Surprisingly, the mean is identically zero by the sign-symmetry of the uniform PACF law and the fact that the Monahan map is odd in each coordinate. The Monahan prior is therefore

$$\texttt{Monahan}(P) := \texttt{Normal}(\mathbf{0}, \mathbf{S}_P), \tag{68}$$

where $\mathbf{S}_P$ captures the full geometry of the stability region in P dimensions within the approximation of a Gaussian family.

$\mathbf{S}_P$ has no closed form but is easy to estimate by Monte Carlo: draw N samples $\boldsymbol{\varphi}^{(n)} \overset{\text{iid}}{\sim} \texttt{Uniform}(-1,1)^P$, compute $\mathbf{a}^{(n)} = T\left(\boldsymbol{\varphi}^{(n)}\right)$, and average the outer products,

$$\hat{\mathbf{S}}_P = \frac{1}{N}\sum_{n=1}^{N} \mathbf{a}^{(n)}\mathbf{a}^{(n)\top} \to \mathbf{S}_P. \tag{69}$$

This converges at standard $\mathcal{O}\left(N^{-1/2}\right)$ rate and can be precomputed for any desired P. At first sight this all appears promising: the Monahan map guarantees stability by construction, and (68) provides a convenient, tractable prior for IKLP that one might hope captures the shape of $\mathcal{A}_P$ sufficiently well.

3.2.3 Why it does not work

Sampling from $\texttt{Monahan}(P)$, however, leads to a rapidly vanishing fraction of stable filters as P increases, as shown by the last curve in Figure 7. The issue is geometric: g is supported entirely on $\mathcal{A}_P$, which is a highly non-elliptical subset of $\mathbb{R}^P$, while any nondegenerate Gaussian has full support on $\mathbb{R}^P$. Figure 8 illustrates this in one and two dimensions.

What if we tuck $\texttt{Monahan}(P)$ inside the support of $g(\mathbf{a})$ and make sure every draw leads to a stable filter? This would be done by minimizing the reverse projection $D_{\mathrm{KL}}(\texttt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P) \parallel g)$. Unfortunately, it is unbounded for any nondegenerate Gaussian since g assigns zero density outside $\mathcal{A}_P$. There is thus no Gaussian approximation that is simultaneously tractable and faithful to the stability constraint: the stable set is simply too irregular to be well-approximated by an ellipsoid.

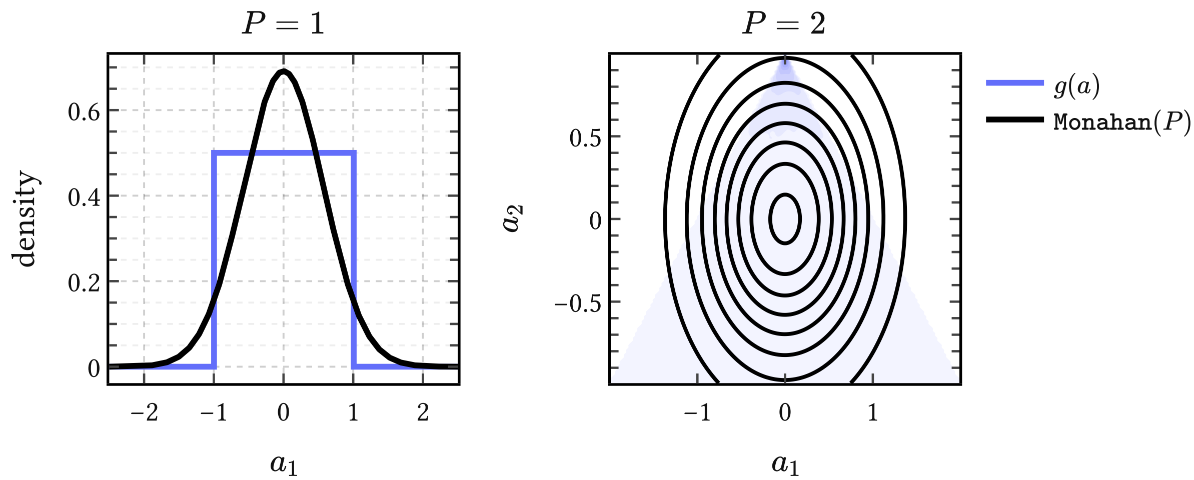


Figure 8 | **Geometry mismatch.** For $P = 1$ the Monahan map is just $a_1 = \varphi_1$ and $\mathcal{A}_1 = (-1, 1)$. For $P = 2$ it gives $(a_1, a_2) = (\varphi_1(1 - \varphi_2), \varphi_2)$, so uniform PACF draws induce a density $g(\mathbf{a})$ supported on the triangular stable region. The Monahan prior $\texttt{Monahan}(P)$ matches the first two moments of g, but it cannot reproduce the bounded, non-elliptical shape of that support faithfully, and mass leaks out of the region of stable filters.

3.3 Spectral shaping priors

Setting worries about power and stability aside for a moment, suppose we expect that the AR filter has a resonance near 500 Hz, or that its spectrum rolls off at a particular rate, or that it has a formant structure typical of a particular vowel. All of these are relevant for acoustic phonetics. How can we "fold" such beliefs into a the first and second moments of a simple Gaussian prior?

The spectral shaping priors that we introduce in this section tackle this problem in some generality. We present a recipe for encoding "spectral beliefs" into the prior mean of $p(\mathbf{a})$ through a single closed-form projection (82). This works by shifting the burden on the user to encode their desired *spectral features* as *roots of* $A(z)$, which are then imposed on average.

3.3.1 Spectral features as root constraints

Features of an AR filter's spectral envelope $|H(e^{i\omega})|^2 = 1/\ |A(e^{i\omega})|^2$ can be traced to the pole structure of $H(z)$, or, equivalently, roots of $A(z)$. That is, controlling where $A(z)$ has its roots is an operational handle on the spectral shape of the filter.

For example, a conjugate pair of roots close to the unit circle at angle ω_0 produces a resonance peak at that frequency; the closer the roots sit to the unit circle, the sharper the peak. A real root near $z = 1$ boosts low frequencies and therefore induces negative spectral tilt, which is exactly the speech-like case of interest. These and other examples of spectral features are collected in Table 1.

One interesting way to encode such spectral features is through *polynomial divisibility*. Saying "we want $A(z)$ to have a root at z_0" is the same as saying "the factor $(1 - z_0^{-1} z)$ divides

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho\cos(\omega_0)z + \rho^2 z^2$	2
Real pole at ρ	$1 - \rho z$	1
Negative spectral tilt $-6k$ dB/oct	$(1 - \rho z)^k$	k
Seasonal pole of period s	$1 - z^s$	s
Repeated resonance at (ρ, ω_0)	$\left(1 - 2\rho\cos(\omega_0)z + \rho^2 z^2\right)^k$	$2k$

Table 1 | **Spectral features and their polynomial encodings.** Each feature corresponds to a polynomial $Q(z)$ whose roots impose the desired pole structure on $|H(z)|^2 = 1/ \ |A(z)|^2$.

$A(z)$." A collection of desired roots defines a polynomial $Q(z)$, and the spectral constraint becomes

$$Q(z) \mid A(z), \tag{70}$$

which is read as "Q divides A". For instance, a resonance near frequency ω_0 with damping $1 - \rho$ corresponds to the conjugate pair $z = \rho e^{\pm i\omega_0}$, giving the quadratic

$$Q(z) = 1 - 2\rho\cos(\omega_0)z + \rho^2 z^2. \tag{71}$$

Generalizing, assume $K \geq 1$ features are desired simultaneously. The individual $\{Q_k(z)\}_{k=1}^K$ are combined into a single *constraint polynomial*

$$M(z) = \texttt{lcm}\{Q_1(z), ..., Q_K(z)\}. \tag{72}$$

Over $\mathbb{C}$ this amounts to collecting the union of desired roots with their maximum multiplicities, and the effective constraint degree is $L_{\text{eff}} = \deg M$. For example, a prior that expects both (1) a resonance and (2) a 18 dB/octave negative rolloff uses $M(z) = Q_1(z) \cdot Q_2(z)$ with $L_{\text{eff}} = 2 + 3 = 5$, where the degrees can be read off the last column of Table 1.

3.3.2 Divisibility is linear

The observation that makes (70) a tractable ansatz for encoding inductive bias towards given spectral features is that polynomial divisibility is a *linear* condition on the coefficients of A. $M(z) \mid A(z)$ if and only if the remainder of the polynomial division $A \bmod M$ vanishes, and this remainder depends linearly on $\mathbf{a}$.

More precisely, for a constraint polynomial $M(z)$ of degree L_{eff}, there exists a matrix $\mathbf{F} \in \mathbb{R}^{L_{\text{eff}} \times P}$ and a vector $\mathbf{c} \in \mathbb{R}^{L_{\text{eff}}}$ such that the remainder

$$\mathbf{f}(\mathbf{a}; M) \coloneqq \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0} \quad \Leftrightarrow \quad M \mid A. \tag{73}$$

The matrix $\mathbf{F}$ and offset $\mathbf{c}$ are determined by the coefficients of $M(z)$ and the monic leading term $a_0 = 1$ of $A(z)$. The set of AR(P) coefficient vectors compatible with the constraint is therefore an affine subspace of $\mathbb{R}^P$ of dimension $P - L_{\text{eff}}$.

Worked example To see this concretely, consider an AR(4) polynomial

$$A(z) = 1 - a_1 z - a_2 z^2 - a_3 z^3 - a_4 z^4 \tag{74}$$

and suppose we want to encode a resonance at (ρ, ω_0) via the quadratic $Q(z) = 1 - cz + dz^2$ where $c = 2\rho\cos(\omega_0)$ and $d = \rho^2$. Divisibility means there exists a quotient $R(z) = 1 - r_1 z - r_2 z^2$ such that $A(z) = Q(z)R(z)$. Expanding and matching the coefficients of z, z^2, z^3, z^4 gives

$$a_1 = c + r_1, \qquad a_2 = r_2 - cr_1 - d, \tag{75}$$

$$a_3 = dr_1 - cr_2, \quad a_4 = dr_2. \tag{76}$$

The last two equations pin down the quotient: $r_2 = \frac{a_4}{d}$ and $r_1 = \frac{a_3 + ca_4/d}{d}$. Substituting back into the first two gives two linear equations in $(a_1, ..., a_4)$ alone,

$$\underbrace{\begin{pmatrix} 1 & 0 & -\frac{1}{d} & -\frac{c}{d^2} \\ 0 & 1 & \frac{c}{d} & \frac{c^2}{d^2} - \frac{1}{d} \end{pmatrix}}_{\mathbf{F}} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} + \underbrace{\begin{pmatrix} -c \\ d \end{pmatrix}}_{\mathbf{c}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{77}$$

from which the matrix $\mathbf{F}$ and vector $\mathbf{c}$ in (73) are read off. The resonance constraint thus carves out a $(P - L_{\text{eff}}) = 2$-dimensional affine subspace of $\mathbb{R}^4$.

3.3.3 The soft spectral shaping prior

We can now state derive the soft spectral shaping prior for $\mathbf{a}$. Let $p_0(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ represent our initial state of knowledge of the AR coefficients and let $M(z)$ be the constraint polynomial (72) encoding the desired spectral features. For example, the assignments

$$p_0(\mathbf{a}) := \texttt{Isotropic}(0.1/P), \quad M(z) := Q(z), \tag{78}$$

where $Q(z)$ is the resonance defined in (71), express a state of knowledge where we expect the AR filter to have unit power output, be stable, and have a single resonance.

We now update $p_0(\mathbf{a})$ to $p^*(\mathbf{a})$ by imposing the new piece of information that the statistic $\mathbf{f}(\mathbf{a}; M)$ has zero expectation under the new state of knowledge expressed by p^* – that is,

$$\mathbb{E}_{p^*}[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{79}$$

Knuth & Skilling (2012) have shown that the unique variational potential for updating information is the Kullback-Leibler divergence, so our task is to

$$\text{find} \quad p^* = \operatorname*{argmin}_p D_{\text{KL}}(p \parallel p_0) \quad \text{given} \quad \mathbb{E}_p[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{80}$$

Since $p_0(\mathbf{a})$ is Gaussian and $\mathbf{f}(\mathbf{a}; M) = \mathbf{F}\mathbf{a} + \mathbf{c}$ is linear in $\mathbf{a}$, we can solve this in closed form:

$$p^*(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M), \tag{81}$$

which is the unique normalized density closest to p_0 under the constraint (79). The solution keeps $\mathbf{\Sigma}_0$ unchanged, so $\mathbf{\Sigma}_M = \mathbf{\Sigma}_0$, but it does shift the mean to

$$\boldsymbol{\mu}_M = \boldsymbol{\mu}_0 - \mathbf{\Sigma}_0 \mathbf{F}^\top (\mathbf{F}\mathbf{\Sigma}_0 \mathbf{F}^\top)^{-1} (\mathbf{F}\boldsymbol{\mu}_0 + \mathbf{c}). \tag{82}$$

Wrapping up, we write the resulting prior formally as

$$\texttt{SoftSpectral}(\boldsymbol{\mu}_0, \mathbf{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \mathbf{\Sigma}_0) \equiv p^*(\mathbf{a}), \tag{83}$$

and refer to it as the *soft spectral shaping prior.*

An example is shown in the left and middle panels of Figure 9, where we encode three resonances that have center frequency and bandwidth typical for the first three vowel formants. The prior spectrum induced by the base prior (left panel) has no discernable resonance structure. The middle panel shows the prior spectrum obtained after updating our state of information per (80). We can see the three resonances clearly imprinted, but what is more interesting is that the prior is noncommital with respect to their actual locations: the zoom on the F_1 region shows clearly that the actual peak locations "happen around" the mean: this is the maximum entropy principle at work. There is also some interesting tail structure which indicates that the prior is anticipating extra resonances at higher frequencies.

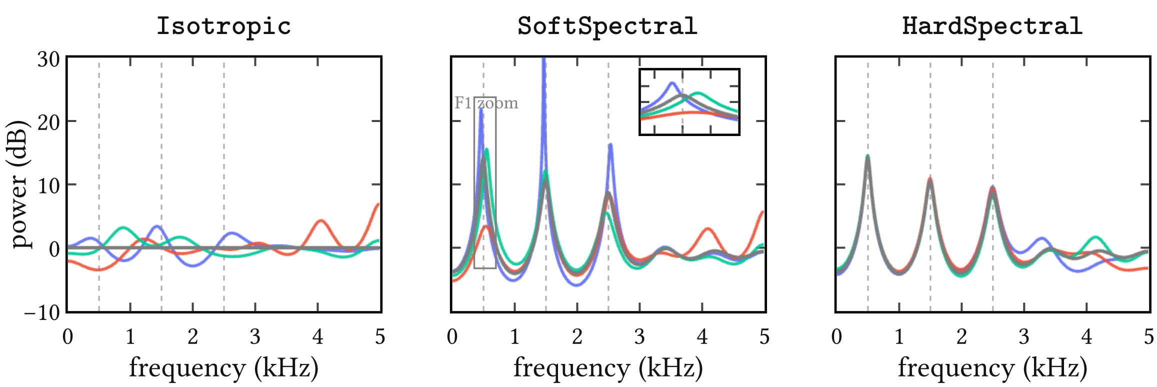


Figure 9 | **Example of imposing spectral features** on an isotropic base prior leading to the soft and hard spectral shaping priors from (83) and (86). We used a constraint polynomial $M(z) = Q_1(z) \cdot Q_2(z) \cdot Q_3(z)$ to encode the spectral features for this example. These are three resonances at $(F_1, F_2, F_3) = (500, 1500, 2500)$ Hz as indicated by the dashed vertical lines. Each panel shows the spectrum mean (grey) together with three samples (colors) of the associated prior. From left to right, these are (1) the base prior $p_0(\mathbf{a}) := \texttt{Isotropic}(0.1/P)$ with $P = 12$; (2) the soft spectral shaping prior $\texttt{SoftSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$ with an inset that zooms in on the F_1 region; and (3) the hard spectral shaping prior $\texttt{HardSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$.

Hard divisibility If one prefers to enforce divisibility $M \mid A$ exactly rather than in expectation – for instance when a particular formant location is known with certainty – this can derived to by solving a related D_{KL} problem. However, since we already know $p^*(\mathbf{a})$ to be Gaussian, we can just condition it on the observation

$$\mathbf{f}(\mathbf{a}; M) = \mathbf{0}. \tag{84}$$

Then the standard conditioning formula gives the hard version of (83): the same mean shift, but the covariance is additionally deflated along the constrained directions,

$$\mathbf{\Sigma}_M = \mathbf{\Sigma}_0 - \mathbf{\Sigma}_0 \mathbf{F}^\top \left(\mathbf{F} \mathbf{\Sigma}_0 \mathbf{F}^\top\right)^{-1} \mathbf{F} \mathbf{\Sigma}_0, \tag{85}$$

and we write the *hard spectral shaping prior* as

$$\texttt{HardSpectral}(\boldsymbol{\mu}_0, \mathbf{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \mathbf{\Sigma}_M). \tag{86}$$

Since we are rarely certain of spectral features in advance, this prior might be overconfident so less useful for applications.

An example is shown in the last panel of Figure 9. Compared to the soft spectral shaping prior spectra (middle panel), it is seen that the hard spectral shaping prior collapses the uncertainty around the first three resonances, which is allowed to increase again towards the tail end of the spectrum.

3.4 Modeling or expanding?

Jaynes (1991) recounts in a memorable passage the confusion of the renowned experimental physicist R. W. Wood, who in 1911 proposed optical experiments to determine whether the sidebands $\omega \pm \nu$ in the amplitude-modulated wave

$$(1 + m \cos \nu t) \cos \omega t = \cos \omega t + \frac{m}{2} \cos(\omega + \nu) t + \frac{m}{2} \cos(\omega - \nu) t \tag{87}$$

are “physically real frequencies actually present,” or whether “there is in reality only one frequency ω, but with varying amplitude.” Today it is clear that the above is a mathematical identity; the two descriptions are two ways of saying the same thing, and “no experiment could have any bearing on it,” as Jaynes wrote.

$\mathrm{AR}(P)$ filters raise similar questions. The polynomial $A(z)$ has P roots, and each conjugate pair near the unit circle produces a resonance peak in the spectral envelope $|H(e^{i\omega})|^2$. Now, the vocal tract genuinely has resonances – the formants $F_1, F_2, F_3, \ldots$ with their bandwidths $B_1, B_2, B_3, \ldots$ – and these are as physically real as anything in acoustics: they correspond to standing wave patterns in the cavities of the vocal tract and show up as broad peaks in the spectrum.

When P is chosen small, say $P \approx 2N_F$ where N_F is the number of formants in the analysis bandwidth, each conjugate pole pair tends to line up with one formant, and the correspondence is fairly direct. In this regime, $A(z)$ is acting as a *physical model* of the vocal tract, and it makes sense to constrain it: stability is a physical requirement (the vocal tract is a passive resonator), spectral tilt reflects the glottal source coupling, and the formant locations themselves are knowable a priori within a vowel class (Peterson & Barney, 1952).

But this is not the only way to read $A(z)$. In signal processing practice, P is routinely set much larger than $2N_F$ – the well-known rule of thumb $P \approx f_s/1000 + 2$ for a sampling rate f_s gives $P = 2$ at 10 kHz, which is generous enough that many poles will have no individual formant counterpart. These "extra" poles shape spectral tilt, model subglottal coupling, smooth over source harmonics, or simply approximate fine spectral structure that no small set of resonances could capture on its own.[10] In this regime, $A(z)$ is no longer a model but just a *rational expansion* of whatever the true transfer function happens to be – a Padé-type approximant whose individual poles are artifacts of the expansion order, not of the physics (Press et al., 1992).

This distinction underlies frequent confusions between poles and formants in acoustic phonetics (Titze et al., 2015). It also matters from the Bayesian point of view. If we are *modeling*, then encoding formant structure into the prior is encoding real physics and we should expect it to help inference. If we are *expanding*, then the same prior could easily turn into a straitjacket on what is essentially a flexible basis, and we should expect it to hurt inference. These two situations map cleanly on whether we assume informative or noninformative priors.

The same tension applies to the concept of stability itself. In the modeling regime, stability is a genuine physical constraint and enforcing it through the prior is well motivated. But in the expansion regime, a root landing slightly outside the unit circle need not signal physical instability – it may simply mean that the rational approximation of order P is locally imperfect, and penalizing it amounts to penalizing the approximation for being an approximation.

3.4.1 Modeling *and* expanding?

Where does the soft spectral shaping prior (83) stand on all this? In theory at least, rather well, because it operates at what one might call the *suggestion* level.

The constraint polynomial $M(z)$ encodes beliefs about the few poles that *are* likely to have physical counterparts – the formants F_1, F_2, F_3 with their bandwidths, perhaps a spectral tilt factor – and (82) shifts the prior mean to a coefficient vector $\boldsymbol{\mu}_M$ that is compatible with these features in expectation. But note that the covariance $\boldsymbol{\Sigma}_0$ is left untouched. Indeed, the remaining $(P - L_{\text{eff}})$ degrees of freedom, corresponding to the "expansion" poles that have no individual physical meaning, are entirely free to go wherever the data require. So we are simultaneously modeling *and* expanding.

[10] A single broad formant, for instance, can easily be shaped by two or three poles acting together rather than a single conjugate pair (Van Soom & de Boer, 2021). Conversely, a pole may sit far from the unit circle and contribute little more than a nearly flat spectral factor. The correspondence between poles and formants is thus many-to-many, not one-to-one, except approximately in the parsimonious regime.

Nevertheless, even this soft, suggestive encoding of spectral structure may not always help. The $\lambda \propto 1/P$ scaling rule derived in (64) keeps expected power controlled under an isotropic prior, and does so without imposing any spectral shape at all. Whether the soft spectral shaping prior (83) actually improves inference over this simple baseline – or whether the mean shift inadvertently constrains the expansion poles in ways that hurt the fit – is ultimately an empirical question that depends on the data, the phonation type, and the model order P.

3.5 Summary

The IKLP machinery from the previous chapter requires us to choose a prior $p(\mathbf{a})$ for the AR coefficients. In this chapter we discuss three families of such priors, each attempting to encode progressively more structure about the vocal tract filter.

We began with the *isotropic prior* and showed that it entangles two things one would rather control independently: expected output power, which requires $\lambda \propto 1/P$ to stay bounded, and stability, which tends to collapse with P. In practice, the choice $\lambda \coloneqq 0.1/P$ appears to be a good comprise, however.

The *Monahan prior* attempts to address the stability issue in a more principled way by approximating the uniform distribution over stable AR coefficient vectors, but the geometry of the stability region is too irregular to be faithfully captured by any Gaussian, and the attempt does not survive increasing P.

The *soft* and *hard spectral shaping priors* take a different approach entirely: rather than constraining where the coefficients may live, they encode what the resulting spectrum should roughly look like. These priors answer the question of how to anticipate complex speech-like spectra with something as simple as a Gaussian prior.

The proposed construction rests on the idea that an operational handle on "$A(z)$ should have certain roots" is given by the concept of polynomial divisibility. That is, we can encode the desired spectral features as roots of $A(z)$, and ask that they divide the latter "on average." This turns out to be a linear constraint on the coefficients $\mathbf{a}$, and we can solve the associated D_{KL} update in closed form. We present soft and hard flavors of the principle, where the former merely suggests spectral features and the latter insists on them.

This is, as far as we know, a new possibility to have it both ways in general AR modeling: *model* what you know, *expand* where you don't. The soft and hard spectral shaping priors have broad applicability beyond speech processing and can be used for modeling any time series, whether finance or EEG signals. Because they are derived from a well-posed maximum entropy problem, they form the bridge between work in the literature on *model-based* (informative) priors on the one hand, and *expansion-based* (uninformative) priors on the other (Cuevas et al., 2021; Gibson, 2018; Sørbye & Rue, 2017).

4 From parametric to nonparametric glottal flow models

Glottal flow models (GFMs) describe the source signal $u(t)$ as it drives the vocal tract during voiced speech. These models operate in the time domain because the delicate phase characteristics of the *glottal cycle* are an integral part of vocal communication.[11]

Decades of empirical work have over time produced a "model zoo" of GFMs: handcrafted waveforms of $u(t)$ with handfuls of carefully chosen parameters. Because these are *parametric* models, they all share the same basic trait: interpretable, but inevitably inflexible (Schleusing, 2012). Their parametric nature limits the range of time domain information they can encode. This motivates the construction of a family of *nonparametric* models, which are more expressive as they are able to grow "parametric capacity" depending on the data at hand.

With the many glottal flow models to follow, it is all too easy to lose connection to the actual physical phenomenon being studied. After a short look at the glottal cycle, we begin with a review of the classic Liljencrants-Fant model. Then we revisit the parametric piecewise polynomial models and generalize them to the nonparametric regime, where they are identified with the *arc cosine kernel* of Cho & Saul (2009). The latter is argued to combine the advantages of both worlds: the interpretability of parametric models with the flexibility of nonparametric ones. This then sets the stage for Chapter 5, which derives the *(quasi)periodic arc cosine kernel* as a learnable surrogate model of synthetic glottal flow data.

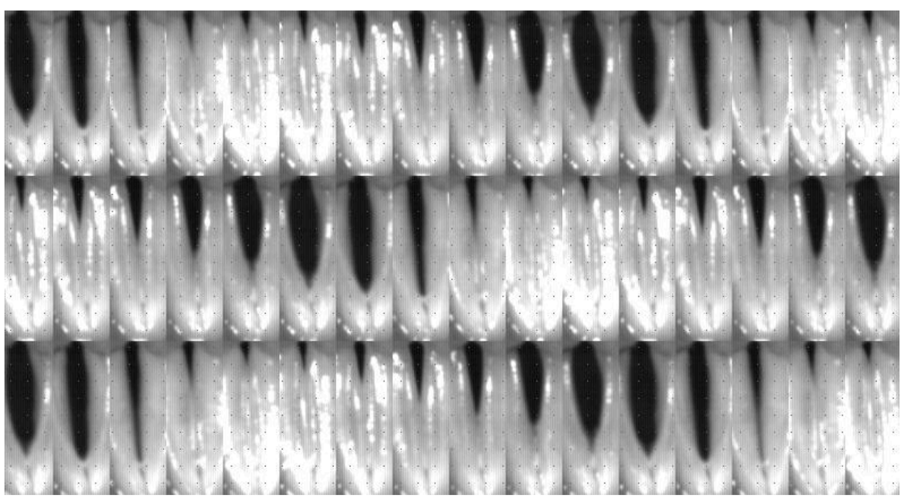


Figure 10 | **A look at the glottal cycle** via laryngeal stroboscopy of the vocal folds. Each frame here is a picture taken at intervals spanning different cycles to give the illusion of steady-state behavior. The glottis is the opening between the folds. After Chen (2016); Jopling (2009).

[11]For example, plosives: glottal stops are micro-events at the millisecond level that underlie semantic information two orders of magnitude higher up the scale. The timing of the vocal fold movement is so precisely controlled by our brain that the slightest deviations are studied as biomarkers for neurodegenerative diseases like Parkinson's (Ma et al., 2020).

The glottal cycle is the periodic opening and closing of the vocal folds, depicted in Figure 10. During one *pitch period*, a single cycle is completed, in this case spanning $T = 6$ ms. GFMs describe the rate of airflow $u(t)$ [typically expressed in cm³/s] through the opening between the vocal folds called the *glottis*, visible as a black slit in Figure 10. Importantly, it is not at maximum glottis aperture that the acoustic output is greatest; it is at the sudden *glottal closure instant* (GCI), as first shown by Miller (1959) in modern literature. At this point, the moving air column in the vocal tract is abruptly interrupted, and kinetic energy is converted efficiently into acoustic energy which then excites the vocal tract much like plucking a harp or clapping in a resonant room (Chen, 2016, Section 4.6). The sharpness of this transition governs much of the clarity and perceived strength of voiced speech (Fant, 1979). Thus, efficiency demands that the glottal cycle be strongly asymmetric: a slow buildup of flow during the open phase, followed by a rapid, almost impulsive closure to the closed phase.

4.1 The Liljencrants-Fant model

A model that has been very useful in describing how $u(t)$ varies during the glottal cycle is the *Liljencrants-Fant (LF) model* proposed by Fant et al. (1985). It has been the GFM of choice for many joint-inverse filtering approaches[12] but also been studied in its own right.[13]

The model states that $u'(t)$ in a single pitch period of length T consists of three parts, or phases. The *open phase* (O) is modeled as a rising and falling exponential sinusoid part, and transitions to the *closed phase* (C) via an exponential return during the *return phase* (R). The latter is thus part of (C). Mathematically, this is defined piecewise as

$$u'_{\mathrm{LF}}(t) = \begin{cases} -E_e e^{a(t-t_e)} \frac{\sin(\pi t/t_e)}{\sin(\pi t_e/t_p)} & 0 \leq t \leq t_e \quad \text{(O)} \\ -\frac{E_e}{\varepsilon T_a}\left(e^{-\varepsilon(t-t_e)} - e^{-\varepsilon(t_c-t_e)}\right) & t_e < t \leq t_c \quad \text{(C, R)} \\ 0 & t_c \leq t \leq T \quad \text{(C)} \end{cases} \tag{88}$$

where $\boldsymbol{\theta}_{\mathrm{LF}} = \{E_e, t_p, t_e, t_c, T_a, T\}$ are the six model parameters explained below in (91) and $\{a, \varepsilon\}$ are determined implicitly from the *closure constraint*

$$\int_0^T u'(t)\,\mathrm{d}t = 0 \quad \text{such that} \quad u(0) = u(T), \tag{89}$$

which holds for any GFM, and the continuity constraint

$$\lim_{\substack{t \to t_e \\ <}} u'(t) = \lim_{\substack{t \to t_e \\ >}} u'(t) \tag{90}$$

which is peculiar to GFMs with smooth return phases rather than hard, discontinuous GCI events (Veldhuis, 1998). Note that like most GFMs, the LF model is defined in $u'(t)$ not $u(t)$

[12] See **Joint source-filter optimization** (page 9).

[13] See Degottex (2010, Section 2.4) for a discussion. Research into the LF model falls mainly into study of its spectral characteristics (Doval et al., 2006) and (re)parametrization, notably Fant (1995); Perrotin et al. (2021).

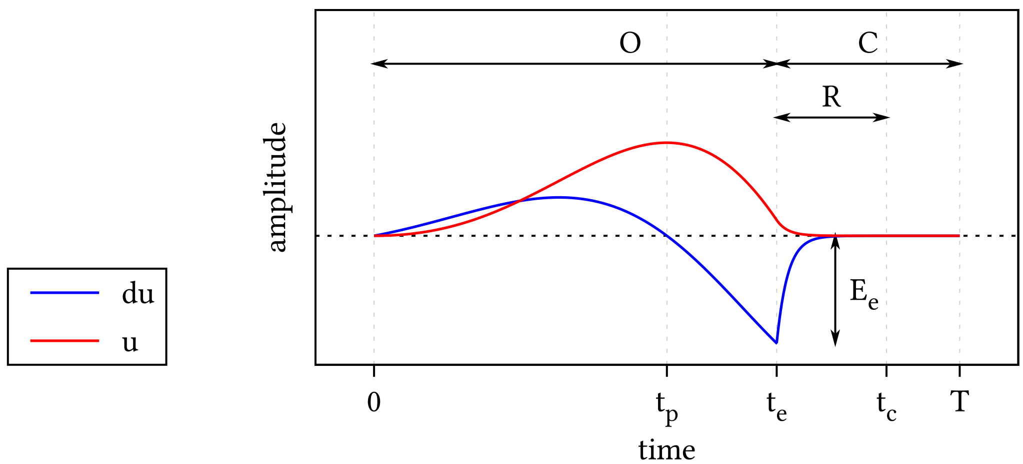


Figure 11 | **The Liljencrants-Fant model** for —— $u'_{\mathrm{LF}}(t)$ and —— $u_{\mathrm{LF}}(t)$ during a single period of length T. The open phase (O), closed phase (C) and return phase (R) are marked at the top and are determined by the changepoints $\{t_p, t_e, t_c, T\}$. The amplitude at peak excitation t_e is E_e. Not shown here is time constant of the return phase T_a, which during (R) determines the steepness of exponential decay towards zero in $u'(t)$.

due to the radiation effect (7), though $u(t)$ can be recovered easily via integration (8). Figure 11 shows the waveforms for $u'_{\mathrm{LF}}(t)$ and $u_{\mathrm{LF}}(t)$ for a typical setting of $\boldsymbol{\theta}_{\mathrm{LF}}$. When varying $\boldsymbol{\theta}_{\mathrm{LF}}$ the LF waveforms trace a family of broad to narrow asymmetric pulses with smooth to abrupt closure encoding positive ($u(t) \geq 0$) glottal flow.

Parameters of the LF model The LF model achieved broad adoption because it strikes a balance between simplicity and empirical accuracy. After several iterations, Fant et al. (1985) arrived at this parametrization:

$$\boldsymbol{\theta}_{\mathrm{LF}} = \begin{cases} E_e = \min_t u'(t) = u'(t_e) & \text{peak excitation amplitude} \\ t_p = \operatorname{argmax}_t u(t) & \text{instant of peak glottal flow} \\ t_e = \operatorname{argmin}_t u'(t) & \text{instant of peak excitation} \\ t_c & \text{instant of glottal closure (GCI)} \\ T_a & \text{time constant of the return phase} \\ T & \text{fundamental period} \end{cases} \tag{91}$$

which turned out to be effective in capturing the glottal cycle in both modal and more extreme phonations.[14] Note that, apart from E_e, which sets the vertical scale, these are all

[14]Actually, Fant et al. present LF as a "four-parameter model". They ignore E_e and set $T := t_c$ for convenience, thus arriving at a count of four not six as in (91). In their unifying theory of common GFMs, Doval et al. (2006) also collapse T to t_c. It is common too in applications (for example, O Cinneide, 2012). Of course, nothing is really lost because a closed phase of any length can be appended to any GFM by piecewise concatenation. We choose to keep t_c distinct from T explicitly because this becomes significant for short return phases and is experimentally observed to be important, especially in clinical scenarios (Kreiman et al., 2007; Rosenberg, 1971).

time domain parameters which determine *changepoints* in the glottal cycle in Figure 10. These tend to covary significantly and can in fact be regressed into a single convenient parameter R_d (Fant, 1995).

From the modeling perspective, the most important changepoints are t_e and t_c: when these lie close together and return is swift (T_a small and vanishing return phase), the glottal cycle is at peak efficiency and $\tilde{u}(t)$ has broadband spectral energy. This results in a clear voice and sharp features in the speech waveform data $\mathbf{d}$. Conversely, a smooth return (T_a medium to large) results in a longer return phase with t_c tending to T and the GCI becoming ill-defined. The result here is a more mellow voice and more smooth features in $\mathbf{d}$. Both modes are common in voiced speech (Maurer, 2016) and GIF algorithms must be able to handle both cases.

Implementation In practice, the LF model is unwieldy to compute (Gobl, 2017; Veldhuis, 1998). The closure and continuity constraints in (89) and (90) lack an analytical solution, meaning that for each evaluation of $u'(t)$ one must numerically solve for the auxiliary constants $\{a, \varepsilon\}$ by a bisection or root-finding routine. This seems like a high price to pay for an analytical model, and worse, this optimization is brittle. As noted by Perrotin et al. (2021), simplified LF models exist that are perceptually equivalent to LF and easier to work with.

Nevertheless, it remains the de facto standard in parametric source-filter modeling and we therefore use it here. Our code in Appendix C implements the LF equations in `JAX` (Bradbury et al., 2020), so the bisection routines are compiled to native code for a speedup and the entire model is differentiable and easily batcheable. The code also implements four LF parametrizations, including the previously mentioned R_d formulation of Fant (1995).

4.1.1 One of many: the glottal flow "model zoo"

Over the years, a large number of glottal flow models have been proposed in acoustic phonetics. A sample of these is shown in Figure 12, already an impressive lineup by 1986. Despite their visual differences, these models are, essentially, variations on a single theme: they all describe a glottal pulse that tracks the periodic opening and closing of the glottis.

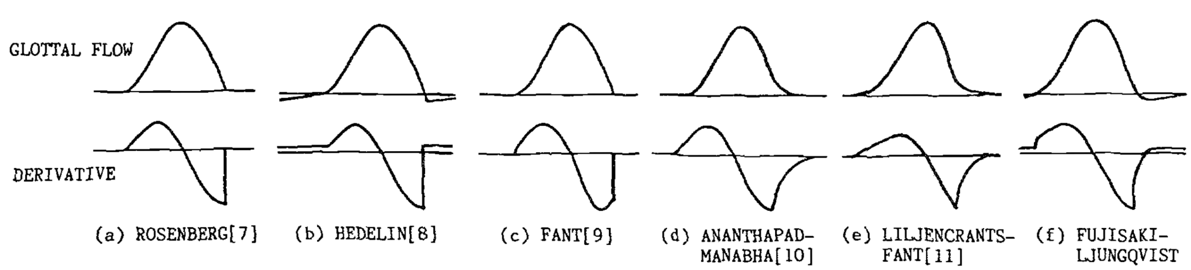


Figure 12 | **A lineup of GF models** back from 1986 (handdrawn?), together with their derivatives (DGFs). These models are visually very similar. From Fujisaki & Ljungqvist.

[15]These are LF, KLGLOTT88 (Klatt & Klatt, 1990), Rosenberg C (Rosenberg, 1971), R++ (Veldhuis, 1998).

Doval et al. (2006) demonstrated that the most common GFMs[15] share a common mathematical structure and can be unified under a single framework; the $\boldsymbol{\theta}_{\mathrm{LF}}$ parameters in (91) can in fact express all of them. These GFMs all model a glottal flow $u(t)$ that is positive or null, continuous, and typically differentiable except at the glottal closure instant (GCI). Within each period, $u(t)$ rises during the opening phase, falls during the closing phase, and returns to zero during the closed phase, possibly through a short return phase that smooths the closure. The derivative $u'(t)$ alternates positive, zero, and negative in the expected order and integrates to zero over one period, satisfying the closure constraint (89). It is continuous in case of a nonempty return phase and discontinuous otherwise (meaning a jump at t_c as in Figure 13), known as hard closure.

Allow partial closure? The strict closure constraint in (89) is of course just a convenient modeling choice and must be expected to be violated to varying degrees in reality, as real folds often leak, especially in breathy or pathological voices. On the one hand, it clearly is a defining feature[16] of the periodic nature of the glottal cycle. On the other, empirical data show the limits of this hard constraint. For example, Kreiman et al. (2007, p. 601) observe that

> in many cases in our data, returning flow derivatives to 0 at the end of the cycle conflicted with the need to match the experimental data and conflicted with the requirement for equal areas under positive and negative curves in the flow derivative. [...] These conflicts between model features, constraints, and the empirical data were handled by abandoning the equal area constraint.[17]

The approach we take in Chapter 5 is to learn a softened version of the closure constraint (89) from data rather than set it a priori. This means we learn both whether the flow typically returns to zero and, if not, to what extent this constraint is plausibly violated.

Allow negative flow? Most GFMs enforce a positive glottal flow and assert $u(t) \geq 0$. But there are exceptions such as Fujisaki & Ljungqvist (1986). They allow for transient negative flow which could represent a lowering of the vocal folds after GCI (presumably when air is sucked back into the lungs). As with the closure constraint, we will learn this effect from data in Chapter 5 and do not forbid $u(t) < 0$ a priori.

[16]The pioneering study on GIF by Miller (1959) had to deconvolve taped speech waveforms with analogue resistor networks and manual optimization. A review article on GIF methods notes the part played by the defining quality of the closure constraint in this search for $u(t)$: "the fine-tuning of the resistors of the inverse network was conducted by searching for such values that yielded zero flow after the instant of glottal closure" (Alku, 2011, p. 626).

[17]That "equal area constraint" is just another name for the closure constraint (89).

Degree d	Name	Reference
0	Triangular pulse model	Alku et al. (2002)
1	—	Titze (2000); Verdolini & Titze (1995)
2	KLGLOTT88	Klatt & Klatt (1990) Chang-Sheng Yang & Kasuya (1998)
3	R++	Veldhuis (1998)
3	FL model	Fujisaki & Ljungqvist (1986)
6	—	Childers & Hu (1994)

Table 2 | **Classic piecewise polynomial models** from the literature. Of these, KLGLOTT88 and R++ remain popular today (Doval et al., 2006).

4.2 Classic piecewise polynomial models

Polynomial GFMs occupy an interesting corner of the "model zoo." They fit into the same piecewise framework we have been using for the LF model, but with each part modeled as a polynomial of degree d. We list several GFMs proposed in the literature in Table 2.

This class of GFMs mainly appeal to simplicity and spectral strength. Piecewise polynomial models admit an analytical solution to the closure constraint while remaining computationally cheap. More importantly, the sharp transitions generated by low-order polynomials have bright spectra with slow decay, which is exactly what it takes to model hard GCIs (Childers & Hu, 1994) and score well on perceptual tests of synthetic speech (Rosenberg, 1971).

To get a feel for these models, we now turn to the simplest case: $d = 0$.

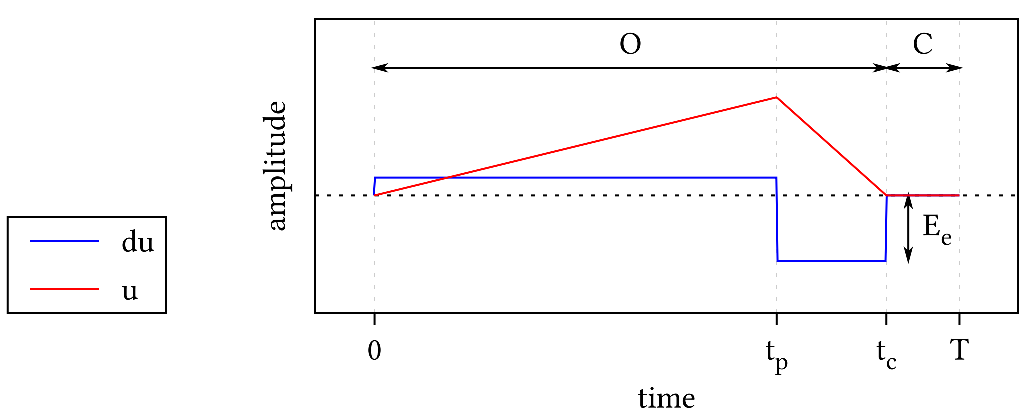


Figure 13 | **The triangular pulse model** for —— $u'_\Delta(t)$ and —— $u_\Delta(t)$ during a single period of length T. The parameters used in this model are identical to Figure 11, except for the lack of a return phase and that the 'instant' of maximum excitation t_e now spans the interval $[t_p, t_c]$.

4.2.1 The triangular pulse model

The simplest polynomial model is the *triangular pulse model* proposed by Alku et al. (2002). It takes $u(t)$ piecewise linear and $u'(t)$ piecewise constant, as shown in Figure 13. Mathematically, the triangular pulse model is given by:

$$u'_\Delta(t) = \begin{cases} +E_e\left(\frac{t_c}{t_p} - 1\right) & 0 \ < t \leq t_p & \text{(O)} \\ -E_e & t_p < t \leq t_c & \text{(O)} \\ 0 & t_c < t \leq T & \text{(C)} \end{cases} \tag{92}$$

The model has no return phase and only four parameters: $\boldsymbol{\theta}_\Delta = \{E_e, t_p, t_c, T\}$. Note that the amplitude of the first case in (92) is fixed because the closure constraint (89) removes one degree of freedom. Alku et al. (2002) point out that this enables a difficult-to-measure time domain quantity to be expressed as the ratio of two easy-to-measure quantities in the amplitude domain and exploit this fact to measure the open quotient more robustly.

4.3 Parametric piecewise polynomial models

The triangular pulse model (92) contains two jumps in the derivative $u'_\Delta(t)$, so we can write the latter more compactly as a linear combination of two Heaviside functions

$$u'_\Delta(t) = \begin{cases} a_1(t-b_1)^0_+ + a_2(t-b_2)^0_+ & 0 \ < t \leq t_c & \text{(O)} \\ 0 & t_c < t \leq T & \text{(C)} \end{cases} \tag{93}$$

where $(t-b)^0_+ = \texttt{Heaviside}(t-b)$ and

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = E_e \begin{pmatrix} \frac{t_c}{t_p} - 1 \\ -\frac{t_c}{t_p} \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 0 \\ t_p \end{pmatrix}. \tag{94}$$

During the open phase, we can interpret the rewritten triangular pulse model as a *parametric piecewise polynomial model* of *degree* $d = 0$ and *order* $H = 2$. It has parameters $\boldsymbol{\theta}_2 = \{\mathbf{a}, \mathbf{b}\}$ which describe $H = 2$ jumps with amplitudes $\mathbf{a}$ at locations $\mathbf{b}$. Note that in this formulation the amplitudes $\mathbf{a}$ differ from the piecewise amplitudes in (92) but are still linearly dependent given the changepoints $\mathbf{b}$ and t_c.

Continuing, nothing stops us from generalizing both H and d. The open phase part of (93) becomes:

$$u'_H(t) = \sum_{h=1}^{H} a_h(t-b_h)^d_+ \qquad 0 < t \leq t_c \qquad \text{(O)} \tag{95}$$

This is the general parametric piecewise polynomial model of degree d and order H. Again, its parameters $\boldsymbol{\theta}_H = \{\mathbf{a}, \mathbf{b}\}$ are the amplitudes $\mathbf{a} = (a_1, ..., a_H)^\top \in \mathbb{R}^H$ and the changepoints $\mathbf{b} = (b_1, ..., b_H)^\top \in \mathbb{R}^H$. These scale and shift H piecewise monomials of degree d:

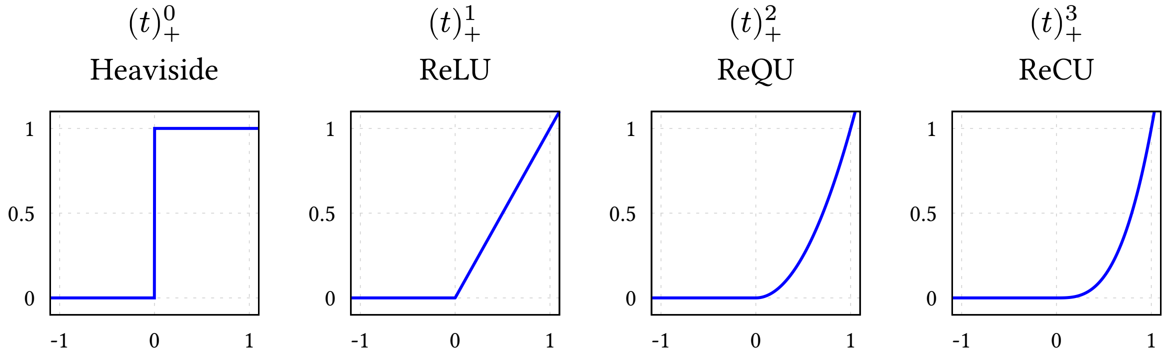


Figure 14 | **The RePU activation functions** —— $(t)_+^d$ for $d \in \{0, 1, 2, 3\}$.

$$(t-b)_+^d = (t-b)^d \ \texttt{Heaviside}(t-b) = \begin{cases} 0 & t < b \\ (t-b)^d & t \geq b \end{cases} \tag{96}$$

We recognize (96) as the family of activation functions called *rectified power units* (RePUs). These include Heaviside for $d = 0$ and the influential ReLU function for $d = 1$, both shown among higher degree variants in Figure 14. It may sound like a stretch to relate polynomial GFMs to neural nets, but this view of things will pay off greatly in Section 4.4 when we take the limit $H \to \infty$.

4.3.1 Regression view

GFMs ultimately play the role of regression models in the GIF problem. From this point of view, the general parametric model in (95) is just a standard *linear model*, and we will now develop it into a Bayesian regression model following MacKay (1998).

Suppose we observe N time samples $\boldsymbol{t} = \{t_n\}_{n=1}^N$ during one glottal cycle, and fix H basis functions of the form

$$\phi_h(t) = (t - b_h)_+^d, \tag{97}$$

where the changepoints $\mathbf{b}$ and t_c are treated as given. Collecting these into the design matrix $\mathbf{\Phi} \in \mathbb{R}^{N \times H}$ with

$$[\mathbf{\Phi}]_{nh} = \phi_h(t_n), \tag{98}$$

the general parametric model during the open phase becomes simply

$$\mathbf{u}' = \mathbf{\Phi}\mathbf{a}. \tag{99}$$

This is linear in the amplitudes $\mathbf{a}$ given $\mathbf{\Phi}$, so the model is rather constrained at this point. Indeed, $\mathbf{a}$ is currently the only source of variability and power in this model because we assume all other parameters fixed. Which prior then for $\mathbf{a}$ should we choose? If we want to maximize model support while adhering to the known additivity and power constraints, the optimal choice is the canonical Gaussian prior over amplitudes (Bretthorst, 1988):

$$p(\mathbf{a} \mid \mathbf{b}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{100}$$

Moving on to data space, the induced prior probability of $\mathbf{u}'$ is also Gaussian,

$$p(\mathbf{u}' \mid \mathbf{b}) = \int \delta(\mathbf{u}' - \mathbf{\Phi}\mathbf{a})\, p(\mathbf{a} \mid \mathbf{b})\, \mathrm{d}\mathbf{a} = \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top), \tag{101}$$

due to closure of the Gaussian under linear combinations, and

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{u}'] = \mathbf{\Phi}\, \mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{a}] = 0, \tag{102}$$

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\mathbf{u}'\mathbf{u}'^\top\right] = \mathbf{\Phi}\, \mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\mathbf{a}\mathbf{a}^\top\right] \mathbf{\Phi}^\top = \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top. \tag{103}$$

Because $\operatorname{rank}(\mathbf{\Phi}\mathbf{\Phi}^\top) \leq H$, prior samples of $\mathbf{u}'$ are intrinsically confined to an H-dimensional subspace of $\mathbb{R}^N$. Assuming $H < N$, we thus see that linear models like (99) are inherently low-rank. Their expressivity grows with H and saturates when H reaches N.

Closure constraint The closure constraint (89) becomes

$$\int_0^T u'_H(t)\, \mathrm{d}t = \sum_{h=1}^H a_h \int_0^{t_c} \phi_h(t)\, \mathrm{d}t = \sum_{h=1}^H a_h r_h = 0, \tag{104}$$

with $r_h = (t_c - b_h)^{d+1}/(d+1)$. Observe that the prior (100) already satisfies the closure constraint in expectation:

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\sum_{h=1}^H a_h r_h\right] = \sum_{h=1}^H \mathbb{E}_{\mathbf{a}|\mathbf{b}}[a_h]\, r_h = 0. \tag{105}$$

This motivates setting the prior mean of the $\mathbf{a}$ to zero, otherwise than symmetry of ignorance around $\mathbf{a} = \mathbf{0}$. We can also enforce it:[18]

$$\sum_{h=1}^H a_h r_h = 0 \Longrightarrow p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)) \tag{106}$$

where $\mathbf{r} = (r_1, ..., r_H)^\top$ and $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$. A convenient way to a sample from this updated prior makes use of the fact that this is a rank-one projection:

$$\mathbf{a} = (\mathbf{q}\mathbf{q}^\top)\mathbf{a} + (\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \mathbf{b}) \tag{107}$$

$$\Longrightarrow \mathbf{a}_\perp = \qquad (\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}). \tag{108}$$

How does this look in data space? By the same calculation (102), marginalizing over $\mathbf{a}$ yields

$$p(\mathbf{u}' \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{\Phi}^\top). \tag{109}$$

[18] The general solution is given by $\operatorname{argmin}_f D_{\mathrm{KL}}[f(\mathbf{a}) \parallel p(\mathbf{a} \mid \mathbf{b})]$ under the constraint (104), but the closure under linear conditioning of Gaussian distributions allow us to take a shortcut here.

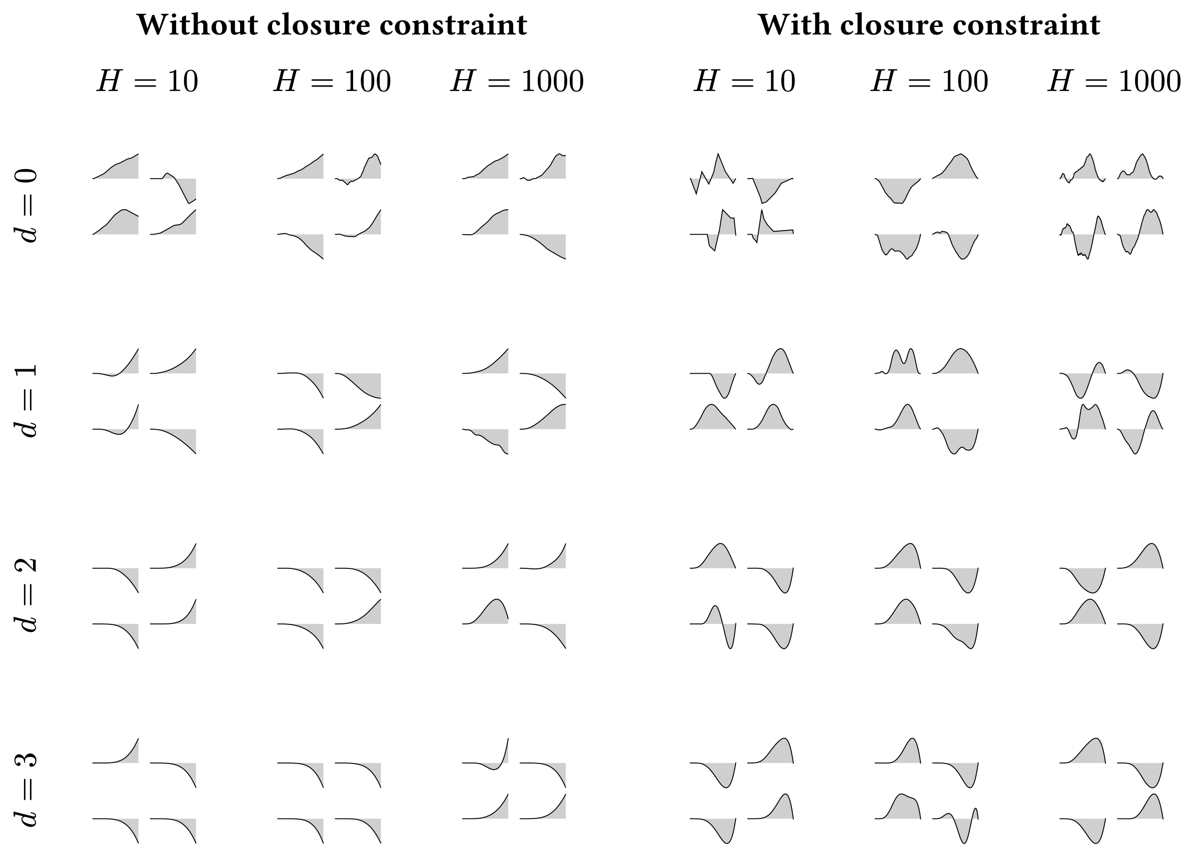


Table 3 | **Samples of $u(t)$** from the general parametric piecewise polynomial model (95) with or without the closure constraint (104). For each combination of degree d and order H, a total of H changepoints $b_h \sim \texttt{Uniform}(0, t_c)$ were drawn randomly, and four instances of $\mathbf{u}'$ conditioned on these were sampled either from (101) [**Without closure constraint**] or from (109) [**With closure constraint**]. Then $u(t) = \int_0^t u'(\tau)\,\mathrm{d}\tau$ was obtained by quadrature of $\mathbf{u}'$.

We can then obtain $\mathbf{u}$ from $\mathbf{u}'$ via integration using (8). Compared to samples from the isotropic prior (101), the anisotropic prior (109) induces glottal flows $u(t)$ that tend to look more pulse-like, as shown in Table 3. Indeed, moving to the bottom right corner, it is seen that increasing H and d tends to produce LF-like pulses out of the box!

Regression with (109) ensures that posterior mass vanishes at solutions $\mathbf{u}$ that violate the closure constraint. This illustrates how linear constraints on the amplitudes $\mathbf{a}$ in the linear model (99) can encode GFM properties at the cost of just a single rank-one downdate per constraint. Moving on from linear models to Gaussian processes, these linear constraints become linear functionals known as *interdomain features*, which may be used to impose structure directly in function space, without touching rank, but more challenging mathematically. In Chapter 5, we will use the interdomain approach to learn spectral features of nonparametric GFMs directly from data rather than hardcoding them as in (104).

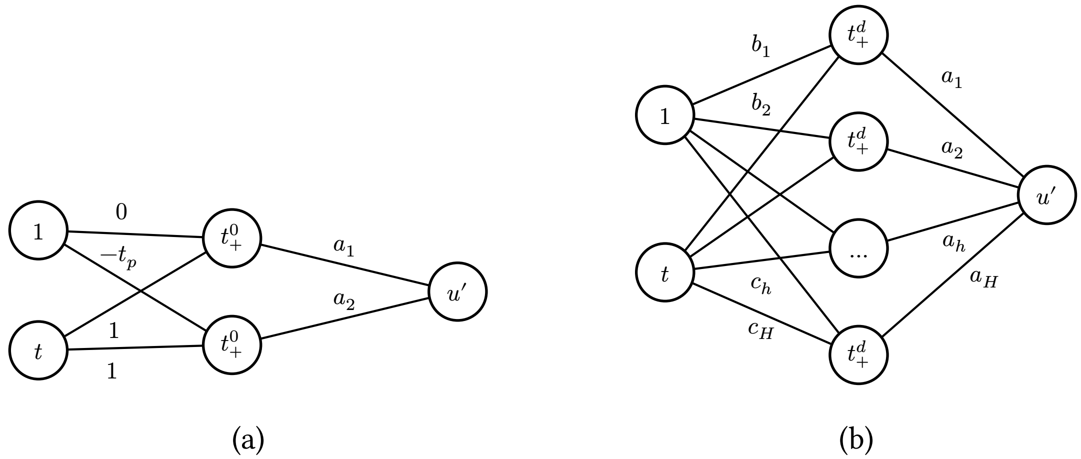


Figure 15 | **Neural network view.** (a) The triangular pulse model as a tiny neural network with a single hidden layer, linear readout and t_+^0 activation. (b) The general parametric polynomial model of degree d with order H with hidden weights $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$, output weights $\mathbf{a} \in \mathbb{R}^H$ and RePU activation function t_+^d.

Connection to classic polynomial models Observe that the analytical solution (106) to the closure constraint is valid for any degree $d \geq 0$ and order $H \geq 1$, so each classic polynomial model listed in Table 2 can be expressed in our linear model framework given suitable choices of $\mathbf{b}$ and t_c, without specialized derivations.

4.3.2 Neural network equivalence

Figure 15 illustrates that the triangular pulse model of Section 4.2.1 can be seen as a tiny neural network. Likewise, the general parametric piecewise polynomial model (95) can be recast as a single hidden layer of width H with a RePU activation and a linear readout:

$$u'_{\mathrm{NN}}(t) = \sum_{h=1}^{H} a_h (c_h t - b_h)_+^d. \tag{110}$$

Each unit computes $\phi_h(t) = \left(\mathbf{w}_h^\top \mathbf{x}_t\right)_+^d$ with hidden weights $\mathbf{w}_h = (c_h, -b_h)^\top$ and $\mathbf{x}_t = (1, t)^\top$, exactly like a neuron with a bias input. The $3H$ parameters of the neural net are thus $\mathbf{a}$ and $\mathbf{b}$ (as before) and the newly acquired degrees of freedom $\mathbf{c} = (c_1, ..., c_H)^\top$.

Seen from this point of view, a natural symmetry appears. We argued in Section 4.3.1 that in linear regression context the output weights $\mathbf{a}$ are best assigned an uninformative Gaussian prior, so

$$p(\mathbf{a}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{111}$$

But the neural network equivalence makes very clear that the hidden weights $\{\mathbf{b}, \mathbf{c}\}$ actually play similar roles: they enter linearly inside the activation, so the same argument applies, and we are encouraged to assign Gaussians again:

$$p(\mathbf{b}) = \texttt{Normal}(\mathbf{b} \mid \mathbf{0}, \sigma_b^2 \mathbf{I}), \quad p(\mathbf{c}) = \texttt{Normal}(\mathbf{c} \mid \mathbf{0}, \sigma_c^2 \mathbf{I}). \tag{112}$$

This step is not obvious from the viewpoint of any of the GFMs we've discussed before, where changepoints $\mathbf{b}$ were nonlinear hyperparameters fixed a priori; here they become "active" and "linear" degrees of freedom. Assuming t_c is known, the remaining hyperparameters to describe the open phase are thus the three scales $\boldsymbol{\theta}_{\text{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ controlling amplitude, typical changepoint location, and horizontal stretch, respectively.

Turning to data space again and having assigned priors to all parameters, we can now do a full marginalization unlike the conditional in (101). Updating the design matrix $\mathbf{\Phi}$ to

$$[\mathbf{\Phi}]_{nh} = \phi_h(t_n) = (c_h t_n - b_h)_+^d, \tag{113}$$

and assuming the isotropic priors (111) and (112) we have

$$p(\mathbf{u}') = \int \delta(\mathbf{u}' - \mathbf{\Phi}\mathbf{a})\, p(\mathbf{a})\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{a}\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{114}$$

$$= \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{115}$$

This is a Gaussian mixture with varying covariances (not means) as $\mathbf{\Phi}$ depends on the hidden weights $\{\mathbf{b}, \mathbf{c}\}$. A closed form for this integral exists only for $H \to \infty$ as $p(\mathbf{u}')$ converges to a Gaussian process, as we will see in Section 4.4.

Closure constraint The closure constraint (89) given $\{\mathbf{b}, \mathbf{c}\}$ remains a linear condition on $\mathbf{a}$, so we can define

$$p(\mathbf{a}, \mathbf{b}, \mathbf{c} \mid \texttt{closure}) = p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}, \mathbf{c})\, p(\mathbf{b})\, p(\mathbf{c}), \tag{116}$$

where $p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}, \mathbf{c})$ is identical to (106) but with $r_h = (c_h t_c - b_h)_+^{d+1} / c_h(d+1)$. This prior yields a mixture in data space that is similar to (114):

$$p(\mathbf{u}' \mid \texttt{closure}) = \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{117}$$

Here both $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$ and $\mathbf{\Phi}$ depend on $\{\mathbf{b}, \mathbf{c}\}$. This integral has no closed form for $H \to \infty$, but we can still impose the closure constraint on (114) indirectly in the spectral domain via interdomain features (Chapter 5).

Connection to LF and other GFMs The neural network view of the parametric polynomial model also clarifies expressivity. It is well known that a single RePU layer of sufficient width can approximate any continuous waveform on a bounded interval (Hornik, 1993). Therefore, with its analytical closure constraint (116) and the ability to represent sharp changepoint behavior, the parametric piecewise polynomial model in the form (110) effectively unifies the entire family of GFMs encountered in the literature, from triangular pulses to LF-type exponentials and sinusoids, given sufficiently large H.

4.4 Nonparametric piecewise polynomial models

The correspondence of piecewise polynomials GFMs with a single hidden layer shows that their expressivity depends mostly on the order H, as model order is equivalent to layer width in the neural network picture. Indeed, *parametric* linear models have a finite representation capacity determined primarily by rank because the modeled function can only explore the subspace spanned by the basisfunctions.[19] *Nonparametric* models grow capacity as the number of datapoints increases; in the case of Gaussian processes, posterior inference given N datapoints is equivalent to Bayesian linear regression with N linearly independent basisfunctions, hence full-rank (Rasmussen & Williams, 2006).

It thus makes sense to consider the limit $H \to \infty$ in our search for more expressive GFMs. Continuing where we left, we already saw in (114) that $p(\mathbf{u}')$ is a nontrivial Gaussian covariance mixture weighed by the isotropic priors defined in (112). This can be made explicit:

$$p(\mathbf{u}') = \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{118}$$

$$= \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \mathbf{Q})\, p(\mathbf{Q})\, \mathrm{d}\mathbf{Q}, \tag{119}$$

where we defined the push forward

$$p(\mathbf{Q}) = \int \delta(\mathbf{Q} - \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{120}$$

which essentially counts the number of $\{\mathbf{b}, \mathbf{c}\}$ configurations that give rise to a given $\mathbf{Q}$. Thus each random sample of $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ drawn from (112) induces in data space a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$,

$$\{\mathbf{b}, \mathbf{c}\} \mapsto \mathbf{Q} = \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}, \qquad \operatorname{rank}(\mathbf{Q}) \leq H \tag{121}$$

and hence a Gaussian component with weight $p(\mathbf{Q})$ in the mixture (118). Below we will show that by choosing $\sigma_a \propto 1/\sqrt{H}$ and letting $H \to \infty$, the density of states (120) degenerates to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}), \qquad \operatorname{rank}(\mathbf{K}) = N \text{ almost surely}, \tag{122}$$

for some $\mathbf{K} \in \mathbb{R}^{N \times N}$ derived below in (158), causing the mixture to collapse to a single Gaussian and thus completing the transition to an indexed Gaussian process with covariance matrix $\mathbf{K}$.

[19] More precisely: a higher rank naturally enlarges the subspace of attainable functions, but the prior over amplitudes still constrains where probability mass is placed within that subspace. Geometrically, the model explores an H-dimensional ellipsoid like (101) or (109) rather than the entire hyperplane. When data accumulate, the prior constraints are eventually overwhelmed and the model can move freely on that plane. Random kitchen sinks (Rahimi & Recht, 2008) take these two ideas to the extreme: forgo specialized modeling; just use H *random* basis functions with uninformative priors and rely on the fact that an entire H-dimensional hyperplane is reachable with high probability.

4.4.1 Mean and covariance description

While the mixture (118) has no closed form for finite H, progress can be made by computing its mean and covariance as before in (102):

$$\mathbb{E}[\mathbf{u}'] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\Big[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}[\mathbf{u}']\Big] = \mathbf{0}, \tag{123}$$

$$\mathbb{E}\Big[\mathbf{u}'\mathbf{u}'^{\top}\Big] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\Big[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}\Big[\mathbf{u}'\mathbf{u}'^{\top}\Big]\Big] = \sigma_a^2\,\mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{\Phi}\mathbf{\Phi}^{\top}], \tag{124}$$

The mean is zero everywhere due to our zero-mean prior $p(\mathbf{a})$. The covariance matrix, however, is nontrivial. To investigate, define

$$\mathbf{C} := \mathbb{E}\Big[\mathbf{u}'\mathbf{u}'^{\top}\Big] \in \mathbb{R}^{N\times N} \tag{125}$$

with elements

$$[\mathbf{C}]_{nm} = \sigma_a^2\,\mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\sum_{h=1}^{H}\phi_h(t_n)\phi_h(t_m)\right], \tag{126}$$

where $\phi_h(t) = (c_h t - b_h)_+^d$ for the neural network model $u'_{\mathrm{NN}}(t)$ was defined in (113). Since the $\phi_h(t)$ are statistically equivalent under i.i.d. priors like the ones we adopted in (112), define a single “mother wavelet” *without index h* as

$$\phi(t;b,c) = (ct-b)_+^d, \quad b \sim \texttt{Normal}(0,\sigma_b^2), \quad c \sim \texttt{Normal}(0,\sigma_c^2). \tag{127}$$

Then the elements of $\mathbf{\Phi}\mathbf{\Phi}^{\top}$ become sums of H i.i.d. random variables, and their expectation factorizes across h:

$$[\mathbf{C}]_{nm} = \sigma_a^2\sum_{h=1}^{H}\mathbb{E}_{\mathbf{b},\mathbf{c}}[\phi_h(t_n)\phi_h(t_m)] \tag{128}$$

$$= \sigma_a^2\sum_{h=1}^{H}\mathbb{E}_{b,c}[\phi(t_n;b,c)\phi(t_m;b,c)] \tag{129}$$

$$= \sigma_a^2\,H\,\mathbb{E}_{b,c}[\phi(t_n;b,c)\phi(t_m;b,c)] \tag{130}$$

$$= \sigma_a^2\,H\,\texttt{TACK}(t_n,t_m). \tag{131}$$

Here we anticipate the *temporal arc cosine kernel* of degree d, defined below in (146):

$$\texttt{TACK}(t,t') := \mathbb{E}_{b,c}[\phi(t;b,c)\phi(t';b,c)] \tag{132}$$

$$\equiv \int \phi(t;\mathbf{w})\phi(t';\mathbf{w})\ \texttt{Normal}(\mathbf{w}\mid\mathbf{0},\mathbf{\Sigma})\,\mathrm{d}\mathbf{w}, \tag{133}$$

where $\mathbf{\Sigma} := \begin{pmatrix}\sigma_b^2 & 0\\ 0 & \sigma_c^2\end{pmatrix}$ describes the covariance of the hidden weights $\mathbf{w} = (-b,c)^{\top}$. For legibility we suppress the dependence of TACK, written $\texttt{TACK}(t,t')$, on both the degree d and the covariance matrix $\mathbf{\Sigma}$.

To recapitulate, the first and second central moments of the mixture (118) are known for any H:

$$\mathbb{E}[\mathbf{u}'] = \mathbf{0}, \tag{134}$$

$$\mathbb{E}\left[\mathbf{u}'\mathbf{u}'^{\top}\right] \equiv \mathbf{C} = \sigma_a^2 \, H \, \texttt{TACK}(t, t'). \tag{135}$$

The third, fourth, … central moments are in general nonzero and difficult to compute. They will vanish, however, when $H \to \infty$ and we choose $\sigma_a \propto 1/\sqrt{H}$. Nevertheless, even before taking any limit, the first two moments of the prior already mirror something kernel-like. Note that this result hinges critically on the independence assumption for $\{b_h, c_h\}$ in (127). The closure-constrained prior of (117) would couple these parameters nonlinearly, destroying that independence and making the derivation intractable, which is why we temporarily set it aside.

4.4.2 The temporal arc cosine kernel

Of course, I wouldn't have had the courage to attempt this whole calculation had I not already known that a very similar limit has been evaluated succesfully in the neural-network-as-a-Gaussian-process literature. The marginalization we performed above is, in fact, a one-dimensional variant of the classic infinite-width limit of feedforward networks studied by Neal (1996); Williams (1998) and later generalized by Cho & Saul (2009).

In these works, the expectation over random weights $\mathbf{w}$ with $\texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$ priors gives rise to a family of kernels which describe an infinitely wide Bayesian network, and which differ only with respect to the activation function used and the imposed a priori covariance $\mathbf{\Sigma}$. This same reasoning will now allow us to identify our expected covariance (132) with a time-domain specialization of the *arc cosine kernel family* of degree d.

Table 4 organizes the two kernels we will introduce shortly. From here on we refer to these kernels in prose simply as ACK and TACK, and suppress their dependence on the listed parameters for legibility.

The arc cosine kernel (ACK) of degree d proposed by Cho & Saul (2009) is defined as the expected inner product between infinite-dimensional random RePU features of D-dimensional inputs $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^D$:

Name	Form	Parameters	Eq.
arc cosine kernel	$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \, \lVert \mathbf{x} \rVert^d \, \lVert \mathbf{x}' \rVert^d \, J_d(\theta)$	d	(137)
temporal arc cosine kernel	$\texttt{TACK}(t, t') = \frac{1}{2\pi} \, \lVert \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_t \rVert^d \, \lVert \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_{t'} \rVert^d \, J_d(\theta)$	$d, \mathbf{\Sigma}$	(146)

Table 4 | **Arc cosine kernels** introduced in this section. The ACK defined on arbitrary input dimension $D \geq 1$ was originally proposed by Cho & Saul (2009). The TACK is a simple application of the ACK for 1D time series.

$$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = 2 \int \left(\mathbf{w}^\top \mathbf{x}\right)_+^d \left(\mathbf{w}^\top \mathbf{x}'\right)_+^d \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{I}_D) \, \mathrm{d}\mathbf{w}, \tag{136}$$

where the hidden weights $\mathbf{w} \in R^D$, and $\mathbf{I}_D$ is the $D \times D$ identity matrix. This integral representation makes it clear that this is indeed a valid positive definite kernel in any input dimension $D \geq 1$. It admits a beautiful closed form:

$$\texttt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \, \|\mathbf{x}\|^d \, \|\mathbf{x}'\|^d \, J_d(\theta), \tag{137}$$

where

$$\theta = \arccos\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\|\mathbf{x}\| \, \|\mathbf{x}'\|}\right) \in [0, \pi] \tag{138}$$

is the positive angle between $\mathbf{x}$ and $\mathbf{x}'$, and $J_d(\theta)$ is given by Cho & Saul (2009, Eq. 4) as the generator expression

$$J_d(\theta) = (-1)^d (\sin\theta)^{2d+1} \left(\frac{1}{\sin\theta} \frac{\mathrm{d}}{\mathrm{d}\theta}\right)^d \left(\frac{\pi - \theta}{\sin\theta}\right). \tag{139}$$

The first few instances of which are

$$J_0(\theta) = \pi - \theta, \tag{140}$$

$$J_1(\theta) = \sin\theta + (\pi - \theta)\cos\theta, \tag{141}$$

$$J_2(\theta) = 3\sin\theta\cos\theta + (\pi - \theta)\left(1 + 2\cos^2\theta\right), \tag{142}$$

$$J_3(\theta) = 15\sin\theta - 11\sin^3\theta + (\pi - \theta)\left(9\cos\theta + 6\cos^3\theta\right). \tag{143}$$

Compared to the familiar Matérn kernels, the arc-cosine kernel is a rather atypical kernel, and probably only sparingly used in machine learning applications. It is, for one, blatantly nonstationary, and unlike stationary kernels such as the RBF, it can represent asymptotic behavior: its posterior mean need not revert to the prior mean outside the data regime, much like a neural network's response. The closed form (137) shows this structure explicitly: the integral representation factors into a polynomial term $\|x\|^d \, \|x'\|^d$, which controls the overall dynamic range and absorbs asymptotic variance, and an angular term $J_d(\theta)$, which captures the nonlinear thresholding of the RePU activation and thus allows for modeling the kind of changepoint structure we know is essential for GFMs.

The temporal arc cosine kernel (TACK) is the kernel which describes $\mathbf{C} \equiv \mathbb{E}\left[\mathbf{u}' \mathbf{u}'^\top\right]$ in (123). We can cast it as an affine shift of the bias augmented ACK on the input dimension $D = 2$. Define the auxiliary vectors

$$\mathbf{x}_t = \begin{pmatrix} 1 \\ t \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -b \\ c \end{pmatrix}. \tag{144}$$

and the covariance matrix for the hidden weights $\{b, c\}$

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_c^2 \end{pmatrix}. \tag{145}$$

Then, by substituting these into (137) and integrating with respect to the Gaussian weight prior $\texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$, we obtain the TACK:

$$\begin{aligned} \texttt{TACK}(t, t') &= \mathbb{E}_{b,c}[\phi(t; b, c)\phi(t'; b, c)] && (146)\\ &= \int (\mathbf{w}^\top \mathbf{x}_t)_+^d (\mathbf{w}^\top \mathbf{x}_{t'})_+^d \ \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})\, \mathrm{d}\mathbf{w} && (147)\\ &= \frac{1}{2}\, \texttt{ACK}\left(\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_t, \mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_{t'}\right) && (148)\\ &= \frac{1}{2\pi}\ \|\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_t\|^d\ \|\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_{t'}\|^d\ J_d(\theta), && (149) \end{aligned}$$

where the factor $1/2$ compensates for the convention used in Cho & Saul (2009). Here,

$$\theta = \arccos\left(\frac{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_{t'}}{\sqrt{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_t}\sqrt{\mathbf{x}_{t'}^\top \mathbf{\Sigma} \mathbf{x}_{t'}}}\right), \tag{150}$$

is the angle between $\mathbf{x}_t$ and $\mathbf{x}_{t'}$ in the geometry induced by $\mathbf{\Sigma}$. This anisotropic variant was also used in the context of D-dimensional feature extraction by Pandey & Dukkipati (2014).

Figure 16 makes the role of the degree parameter visible before we move on to the infinite-width limit. Across all three covariance settings, low degrees produce rougher draws with sharper local bends, whereas higher degrees regularize these bends into smoother open-phase trajectories. The covariance parameters then control where this structure is concentrated: smaller σ_b imply a more localized changepoint location and lead to sharp changepoint

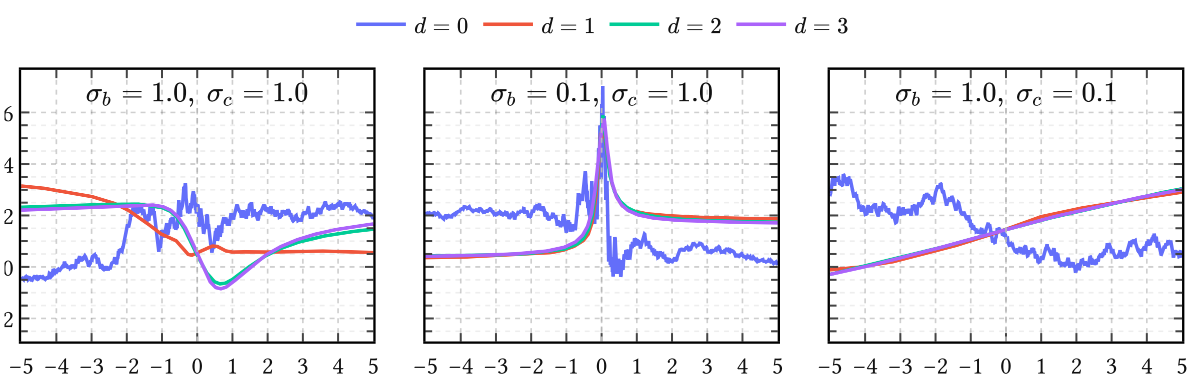


Figure 16 | **Prior samples from TACK for different degrees *d*.** Each column fixes a different covariance setting (σ_b, σ_c). Within each column the four curves correspond to different values of d. Increasing d progressively softens the hingepoint-like behavior, while decreasing σ_b relative to σ_c makes the prior more localized around the center.

behavior. Smaller values of σ_c just smear out time, which have the effect of longer timescales. Note that $d = 2$ and $d = 3$ are practically indistinguishable.

4.4.3 Taking the Gaussian-process limit

We turn again to the mixture (118). Recall from (121) that each random draw $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ from the prior (112) produces a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N\times N}$, which elementwise decomposes as a sum of H i.i.d. terms:

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m). \tag{151}$$

This sum tends to grow as $\mathcal{O}(H)$, since from (128)

$$\mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{Q}] = \mathbf{C} \propto H. \tag{152}$$

Such growth of the marginal variance of $p(\mathbf{u}')$ makes the $u'_{\text{NN}}(t)$ model (110) useless as a prior when we want strong inductive bias even for large H, which we do. Therefore, we couple σ_a and H by choosing $\sigma_a \propto 1/\sqrt{H}$, in accordance with the variance-preserving initialization principle (Glorot & Bengio, 2010). Thus, substituting $\sigma_a \to \sigma_a/\sqrt{H}$ in (151), we get

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \times \left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right]. \tag{153}$$

Observe that the quantity between brackets is a Monte Carlo estimate of the TACK (146). Therefore, by the strong law of large numbers we conclude that

$$\left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right] \longrightarrow \mathbb{E}_{b,c}[\phi(t;b,c)\phi(t';b,c)] \qquad \text{as } H \to \infty, \tag{154}$$

with deviations vanishing as $\mathcal{O}\left(1/\sqrt{H}\right)$. Therefore from (132)

$$[\mathbf{Q}]_{nm} \longrightarrow \sigma_a^2\, \texttt{TACK}(t_n, t_m) \qquad \text{as } H \to \infty. \tag{155}$$

Equivalently, the induced density $p(\mathbf{Q})$ of covariance matrices collapses to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}) \qquad \text{as } H \to \infty, \tag{156}$$

which in turn collapses the mixture (118) to

$$p(\mathbf{u}') \longrightarrow \texttt{Normal}(\mathbf{0}, \mathbf{K}) \qquad \text{as } H \to \infty. \tag{157}$$

Here the *kernel matrix* is the symmetric PSD matrix $\mathbf{K} \in \mathbb{R}^{N\times N} \propto \mathbf{C}$ given by

$$[\mathbf{K}]_{nm} = \sigma_a^2\, \texttt{TACK}(t_n, t_m). \tag{158}$$

In other words, the marginal $p(\mathbf{u}')$ converges entirely to a Gaussian and the first and second moments in (134) become *sufficient statistics* (Jaynes, 2003). All higher central moments vanish as $\mathcal{O}\left(1/\sqrt{H}\right)$ because the covariance fluctuations in (154) themselves decay at that rate. Since this argument holds for any finite collection of times $\boldsymbol{t} = \{t_n\}_{n=1}^N$, the limiting process is indeed officially a Gaussian process:

$$u'_{\mathrm{NN}}(t) \sim \texttt{GaussianProcess}\big(0, \sigma_a^2 \, \texttt{TACK}(t, t')\big). \tag{159}$$

Except for H, it has the same hyperparameters as the neural network model of Section 4.3.2: the scales $\boldsymbol{\theta}_{\mathrm{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ describing overall power, typical changepoint location and horizontal stretch, respectively; and the two higher-level hyperparameters $\{d, t_c\}$, which determine the polynomial degree and the instant of glottal closure, respectively.

Integrating out changepoints Note that t_c is the only changepoint parameter that survived the Gaussian-process limit. All other changepoints [each changepoint being defined by a pair of amplitude and location parameters $\{a_h, b_h\}$] have been analytically integrated out. That act of marginalization removed the parametric bottleneck carried by finite rank models like (88), (92), (95), (110) completely. This may come as a surprise, because we pointed out before that changepoint parameters as in (91) constitute the main "control points" of any GFM expressed in the time domain: what previously required a discrete arrangement of control points is now averaged into a single continuous covariance function capable of learning the "timing relationships [which] are very important for modelling the glottal flow signal" (Doval et al., 2006, p. 1).

More aspects of (159) are discussed in greater detail in Appendix D.

4.5 Summary

Great care was taken to argue that any *parametric* glottal flow model of the open phase of the glottal cycle can be expressed as a RePU network with a single hidden layer of width H, given H large enough.

Then we showed that in a Bayesian regression context with noninformative priors, the limit $H \to \infty$ corresponds to a *nonparametric* glottal flow model: a zero-mean GP with the temporal arc cosine kernel. This well-known limit, borrowed from classical GP literature, is applied here to unify the two dominant approaches to GIF. These are [parametric $\Leftrightarrow$ joint source-filter estimation] and [nonparametric $\Leftrightarrow$ inverse filtering], and we attempt here to combine the best of both worlds: parametric interpretability and nonparametric expressivity, respectively.

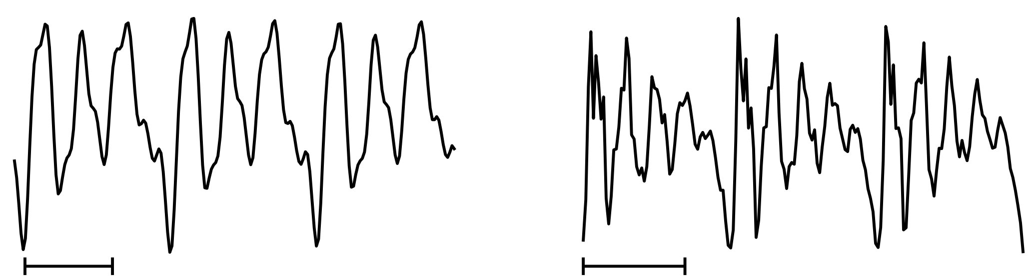
That is, the proposed probabilistic model has support for any parametric glottal flow model while retaining capacity to "let the data speak for itself" if need be. In the next chapters we will refine this initially tiny support gradually into a strong inductive bias by learning features from synthetic glottal flow simulations.

5 The periodic and quasiperiodic arc cosine kernel

Chapter 4 established parametric and nonparametric glottal flow models for a *single period* of the glottal cycle. In this chapter, our goal is to extend this basic building block to *multiple periods.* Recordings of voiced speech often visually exhibit self-similarity on timescales of $O(10\ \text{ms})$ because in normal speech the glottal cycle often reaches steady-state, and a clear pitch is perceived as a consequence.[20] An illustrative example for the case of steady-state vowels is given in Figure 17.

To model this steady-state source behavior we proceed in three steps. We begin with an idealized periodic kernel, the *periodic arc cosine kernel* (PACK), which is stationary and perfectly periodic. We then relax this to the *quasiperiodic arc cosine kernel* (QPACK), which retains the same within-cycle carrier while allowing slow amplitude variation and residual baseline structure across cycles. Finally we introduce *gated variants* (g-PACK and g-QPACK) in which the closed phase is incorporated explicitly. This last step breaks stationarity, but it turns out to be a useful one in practice.

The second half of the chapter turns from construction to learning. It is well known from functional data analysis that learning a single shared basis for a large corpus of training data requires aligning the data points, the process of which is called *registration.* (Ramsay & Silverman, 1997). We show how to do this accurately for the EGIFA data by aligning the



(a) (b)

Figure 17 | **Examples of steady-state vowels** during three pitch periods. Here (a) is perfectly periodic because it was synthesized with (88); and (b) is quasiperiodic because it is taken from a real recording.[21] The timescale is indicated by the horizontal bars ⊢—⊣ below the waveforms, each bar having a length of 5 msec.

[20]Results from clinical trials (Little et al., 2007) indicate that normal (non-pathological) voices are usually something called Type I (Titze, 1995); that is, phonation results in nearly periodic sounds, which supports the notion that uttered vowels in normal speech typically reach a steady-state before "moving on". Thus when a vowel is perceived with a clear and constant pitch, it is reasonable to assume that the vowel attained steady-state at some point (though perceptual effects forbid a one-to-one correspondence). In practice, steady-state vowels are identified simply by looking for such quasiperiodic strings, which typically consist of about 3 to 5 pitch periods in everyday speech (Rabiner & Schafer, 2007).

[21]Taken from a recorded steady-state instance of the vowel /æ/ at a fundamental frequency of 138 Hz. Source: `bdl/arctic\_a0017.wav:0.51-0.54sec` (Kominek & Black, 2004).

examples to GCI (and optional GOI) events. This also allows us to extract voiced groups from the data, which then serve as high-quality training data to *calibrate* the global kernel hyperparameters. These hyperparameters determine the shared basis seen by the whole corpus, and we learn and evaluate them directly from the registered data.

Three algorithms do the practical work of the chapter. PRISM (Algorithm 1) calibrates a single global basis on the registered corpus; LBGID (Algorithm 4) detects the GCI and GOI events needed for registration; and t-PRISM (Algorithm 2) robustly filters raw period tracks into clean voiced groups before calibration. For the sake of brevity their full details and development have been deferred to the algorithm chapters; here we just accept them as given tools and only motivate their usage.

5.1 The periodic arc cosine kernel

The TACK or temporal arc cosine kernel derived in (146) provides a nonstationary covariance structure for the open phase of a *single* glottal cycle. That covariance reflects the change-point geometry inherited from piecewise polynomial GFMs, and it does so in a principled, nonparametric manner. But a single pitch period cannot exist in isolation: voiced speech is characterized by the imperfect repetition of this basic unit, and so we must now try to extend TACK to periodic signals.

Arguably the most natural route to periodicity is to replace the linear time coordinate in the bias-augmented input of TACK, namely $\mathbf{x}_t = (1, t)^\top$, by a set of *harmonic features* that are periodic simply by construction. This is the classical idea of embedding time on a circle (MacKay, 1998): functions of the embedded coordinates are automatically periodic in the original time variable. The construction proceeds in three steps: define the embedding, apply the arc cosine kernel, and normalize.

5.1.1 Harmonic embedding

Let $\omega_0 = 2\pi/T$ denote the fundamental angular frequency for a period $T > 0$, and let $J \geq 1$ be the *harmonic order*, which controls the richness of the embedding. Define the harmonic feature map

$$\boldsymbol{\varphi}(t) = \begin{pmatrix} \sigma_b \\ \sigma_{c,1} \cos(\omega_0 t) \\ \sigma_{c,1} \sin(\omega_0 t) \\ \vdots \\ \sigma_{c,J} \cos(J\omega_0 t) \\ \sigma_{c,J} \sin(J\omega_0 t) \end{pmatrix} \in \mathbb{R}^{2J+1}, \tag{160}$$

where $\sigma_b > 0$ and $\sigma_{c,j} > 0$ for $j = 1, ..., J$ are scale parameters playing analogous roles to σ_b and σ_c in Section 4.3.2, but act on the embedded space rather than linear time. These scale parameters modulate the harmonic basis (160) and thereby control how much energy the embedding allocates to each harmonic, albeit in a non-obvious way. In any case, where TACK lived in $\mathbb{R}^2$, we now work in $\mathbb{R}^{2J+1}$ with $(J+1)$ independent harmonic scales, which gives the kernel much greater spectral flexibility, while still inheriting (some of) the changepoint behavior from its ancestor.

An interesting property of this particular embedding is that its squared norm is constant in t:

$$\|\varphi(t)\|^2 = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 (\cos^2(j\omega_0 t) + \sin^2(j\omega_0 t)) = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 := R^2. \quad (161)$$

Every point $\varphi(t)$ therefore lies on a sphere of radius R in $\mathbb{R}^{2J+1}$. For the inner product between two embedded points, the cosine addition formula $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta)$ collapses the harmonic pairs:

$$\varphi(t)^\top \varphi(t') = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 \cos(j\omega_0(t-t')). \quad (162)$$

This depends on t and t' only through their difference, and is manifestly periodic with period T. Define θ as the angle between the two embedded points,

$$\theta(t-t') := \arccos\left(\frac{\varphi(t)^\top \varphi(t')}{R^2}\right), \quad (163)$$

which takes values in $[0, \pi]$ and satisfies $\theta(0) = 0$.

5.1.2 Definition

Applying the ACK (137) to $\varphi(t)$ and $\varphi(t')$ gives

$$\mathtt{ACK}(\varphi(t), \varphi(t')) = \frac{1}{\pi} R^{2d} J_d(\theta(t-t')), \quad (164)$$

which depends on t and t' only through their difference since R is constant. The prefactor R^{2d}/π couples the overall differentiability of sample paths (controlled by d) to the output scale. To circumvent this, we apply standard normalization and define the *periodic arc cosine kernel of degree d* as

$$\mathtt{PACK}(t, t') := \sigma_a^2 \frac{J_d(\theta(t-t'))}{J_d(0)}, \quad (165)$$

where $J_d(\theta)$ was defined in (139) and $J_d(0) = \pi(2d-1)!!$ is the value of the angular kernel at zero angle. Positive definiteness is preserved because composing a positive definite kernel with a deterministic feature map yields another positive definite kernel. By construction, the variance $\mathtt{PACK}(t,t) = \sigma_a^2$, so σ_a is the only parameter that controls the overall amplitude scale. Note that for legibility we generally suppress the dependence of $\mathtt{PACK}(t,t')$ on its hyperparameters; we list them in Table 5 for an oversight.

Symbol	Name	Role
σ_a	amplitude	marginal standard deviation
σ_b	bias amplitude	DC component of the embedding
$\sigma_{c,1}, ..., \sigma_{c,J}$	harmonic scales	modulate each harmonic pair
T	period	fundamental period of the kernel
d	degree	smoothness class (RePU order)
J	harmonic order	number of harmonics in embedding

Table 5 | **Hyperparameters of PACK.** Of these, σ_a, σ_b and T are positive scalar, the harmonic scales $\sigma_{c,j}$ are J-dimensional, and d and J are discrete.

How come PACK is stationary even though ACK and TACK were not? This is simply because we assigned equal weight to each cosine-sine pair in (162). It is a practical choice to ease training the kernel, and does not impair learning nonstationary phenomena. The thresholding behavior of the RePU activations is still present, but it is now replicated periodically.[22] This means that PACK can still express very sharp within-cycle structure even though it is now stationary and periodic.

5.1.3 Hyperparameters and the weight simplex

The definition (165) involves $J+1$ scale parameters (σ_b and $\sigma_{c,1}, ..., \sigma_{c,J}$) that only enter through the angle θ in (163). Since θ depends on the normalized inner product $\varphi(t)^\top \varphi(t')/R^2$, only the *ratios* of these scales matter. This motivates introducing the *harmonic weights*

$$w_0 = \frac{\sigma_b^2}{R^2}, \quad w_j = \frac{\sigma_{c,j}^2}{R^2} \quad (j = 1, ..., J), \tag{166}$$

which satisfy the simplex constraint $w_0 + \sum_{j=1}^J w_j = 1$ with $w_j \geq 0$. In terms of these weights, the cosine of the angle (163) becomes

$$\cos\theta(t-t') = w_0 + \sum_{j=1}^J w_j \cos(j\omega_0(t-t')), \tag{167}$$

and the kernel (165) can be evaluated by computing $\theta(t-t')$ from (167) and substituting into J_d. We learn the harmonic weights directly in the simplex space to avoid degeneracies.

The zeroth order harmonic weight w_0 plays an interesting role. It descends directly from the bias amplitude σ_b in the TACK construction, but its behavior is altered nontrivially by the harmonic embedding. When $w_0 = 1$ (all harmonic weights zero), the angle θ is identically zero and the kernel degenerates to a constant σ_a^2, describing a DC signal. Intermediate values

[22] The neural network interpretation of Section 4.3.2 carries over directly. PACK is the $H \to \infty$ limit of a single hidden layer with RePU activations applied to the $(2J+1)$-dimensional harmonic input (160) rather than the 2-dimensional bias-augmented input.

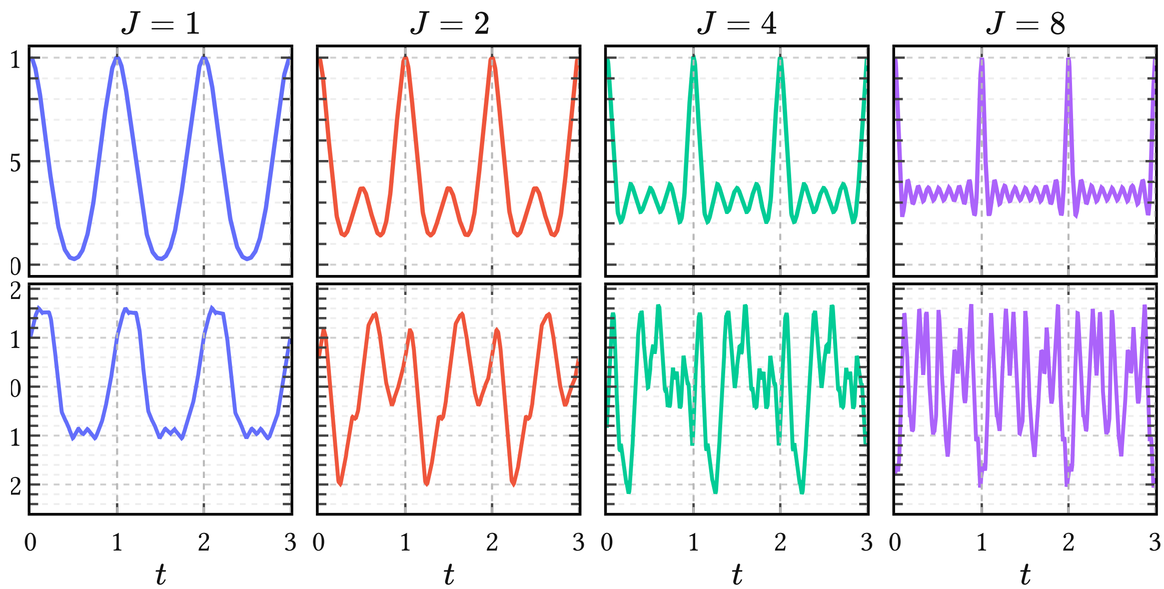


Figure 18 | **Effect of harmonic order J on PACK.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a prior sample. At fixed degree $d = 1$, increasing J decreases the effective within-cycle lengthscale and yields progressively wigglier covariance structure and sample paths. The remaining hyperparameters are fixed at $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and uniform harmonic weights $w_1, \ldots, w_J$ over the remaining mass.

of w_0 let the cosines oscillate about zero in (162) so that θ can reach $\pi/2$ and beyond, where $J_d(\theta)$ is able to produce steep, changepoint-like transitions of the type we need to model sharp glottal closures. If $w_0 = 0$, the autocovariance $\texttt{PACK}(t, 0)$ is allowed to reach zero periodically, which for higher d effectively creates kernels with compact support, although it isn't immediately clear what this implies for modeling speech signals.

At least J has a clear geometric effect. Increasing J inserts higher harmonics directly into the embedding and therefore decreases the effective within-cycle lengthscale of the process. In prior draws this appears as progressively finer oscillatory detail, as seen in Figure 18. Practically, increasing J while keeping all $w_j \sim O(1/J)$ gives rise to smaller effective lengthscales.

5.1.4 Spectral decay and smoothness

Because PACK is both stationary and periodic with period T, its covariance is an even T-periodic function that admits a Fourier cosine expansion

$$\texttt{PACK}(t, t') = a_0 + \sum_{m=1}^{\infty} a_m \cos(m\omega_0(t - t')), \tag{168}$$

with coefficients

$$a_0 = \frac{1}{T}\int_0^T \texttt{PACK}(t, 0)\,\mathrm{d}t, \quad a_m = \frac{2}{T}\int_0^T \texttt{PACK}(t, 0)\cos(m\omega_0 t)\,\mathrm{d}t \quad (m \geq 1). \tag{169}$$

By Bochner's theorem, positive definiteness of PACK is equivalent to $a_m \geq 0$ for all $m \geq 0$, so we can interpret these coefficients as encoding power. The Fourier expansion (168) has a direct interpretation as a *Bochner line spectrum* with spectral density given by

$$\tilde{k}(\xi) = 2\pi a_0\, \delta(\xi) + \pi \sum_{m=1}^{\infty} a_m[\delta(\xi - m\omega_0) + \delta(\xi + m\omega_0)]. \tag{170}$$

The decay of the harmonic weights a_m determines both the spectral character of the prior and the mean square differentiability of GP sample paths. For stationary kernels, this is governed by the behavior of the covariance near the origin: the more derivatives $k(t) = k(t, 0)$ has at $t = 0$, the more mean square derivatives the process admits. A familiar example is the Matérn family, whose sample paths are r times mean square differentiable for every integer $r < \nu$. Now, it can be shown that the sample paths of a degree-d PACK kernel are exactly d times mean square differentiable, independent of J. This implies that the harmonic weights decay in power as

$$a_m = \mathcal{O}\big(m^{-(2d+2)}\big), \tag{171}$$

corresponding to an approximate rolloff in magnitude of $-6(d + 1)$ dB per octave, as illustrated for the $d = 1$ case in Figure 20.

This behavior is also visible in Figure 19: increasing d removes the cusp at the origin and produces progressively smoother kernels with faster spectral decay. Low degrees ($d =$

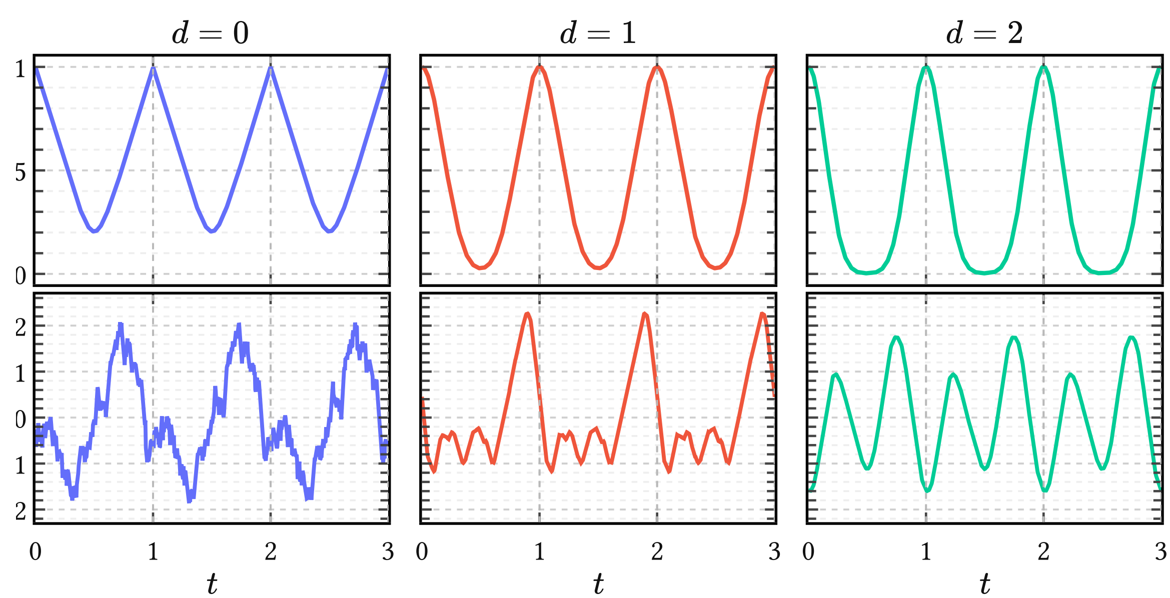


Figure 19 | **Effect of degree d on within-cycle smoothness.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a single prior sample. Holding J and the harmonic weights fixed, increasing d softens the changepoint-like geometry and produces smoother periodized waveforms. The remaining hyperparameters are fixed at $J = 1$, $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and $w_1 = 0.8$.

0 or $d = 1$) therefore yield broad spectra with slowly decaying harmonics, appropriate for modeling sharp glottal closure events.

Classical source models report a spectral tilt of roughly -12 to -18 dB per octave for the glottal flow $u(t)$ itself (Doval et al., 2006; Rosenberg, 1971). Differentiation increases spectral slope by $+6$ dB per octave, so the corresponding glottal flow derivative $u'(t)$ exhibits a tilt of approximately -6 to -12 dB per octave, which corresponds to PACK kernels of degree $d = 0$ and $d = 1$.

5.1.5 Comparison to the standard periodic kernel

The standard periodic kernel in GP literature is the benign periodic squared exponential, which MacKay (1998) obtained by composing the RBF kernel with a circular embedding:

$$k_{\text{per}}(t, t') = \sigma^2 \exp\left(-2 \sin^2\left(\pi \frac{t - t'}{T}\right) / \ell^2\right). \tag{172}$$

Its exact cosine expansion is (Riutort-Mayol et al., 2020, Eq. B.2):

$$k_{\text{per}}(t, t') = \sigma^2 e^{-\frac{1}{\ell^2}} \left[I_0(\ell^{-2}) + 2 \sum_{m=1}^{\infty} I_{m(\ell^{-2})} \cos(m\omega_0(t - t')) \right]. \tag{173}$$

Thus its entire harmonic envelope is controlled by a single lengthscale ℓ. Changing ℓ only broadens or narrows a smooth low-pass spectrum.

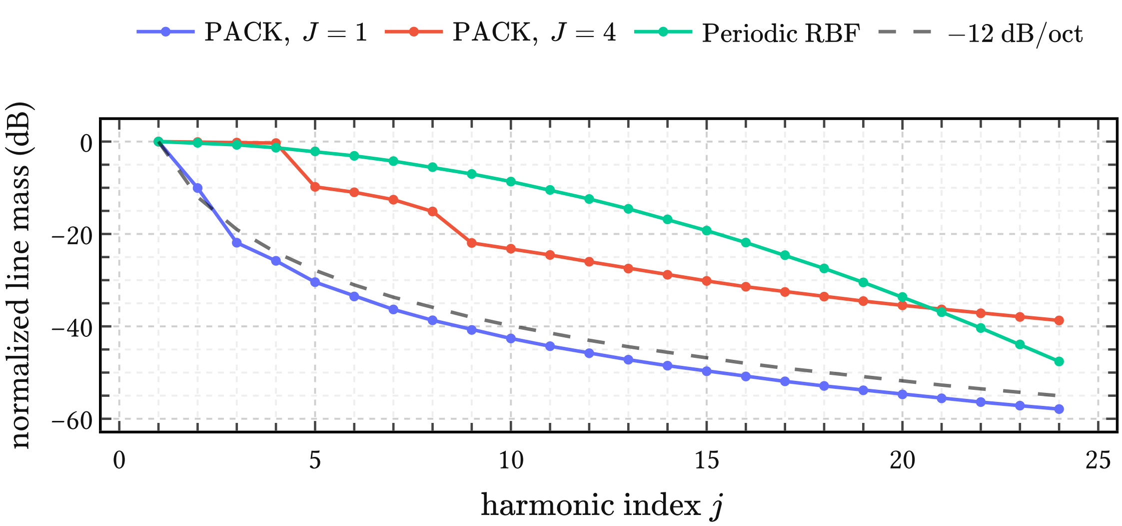


Figure 20 | **PACK versus the periodic squared exponential in the line spectrum.** The curves show the harmonic weights a_m from (170) in dB. PACK with $d = 1$ exhibits the expected -12 dB per octave tail, while the periodic squared exponential traces out a one-parameter low-pass envelope. The PACK curves use $T = 1$, $\sigma_a^2 = 1$, and $w_0 = 0.2$, with uniform weights over the remaining harmonic mass; the periodic squared exponential uses lengthscale $\ell = 0.20$. The dashed reference line indicates -12 dB per octave.

Figure 20 illustrates how PACKs are different in two ways. First, the harmonic weights w_j can redistribute mass freely across harmonics rather than merely broadening a fixed envelope. Second, the coupling between degree d and the asymptotic decay allows the model to learn more realistic, non-exponential spectral rolloff.

5.2 The quasiperiodic arc cosine kernel

Of course, perfect periodicity is too rigid for actual glottal flow; even synthesized glottal flow contains all sorts of drifts and interesting imperfections, which add organic warmth to the synthesized voice. Therefore the next step is to relax PACK into a quasiperiodic variant.

We thus define the *quasiperiodic arc cosine kernel* by

$$\texttt{QPACK}(t,t') := \texttt{env}(t,t') \cdot \texttt{PACK}(t,t') + \texttt{base}(t,t'). \tag{174}$$

This is a composite kernel modeling amplitude modulation of a periodic carrier, plus a second term that adds a small low pass correction. The full parameterization of (174) is listed below in Table 6. In what follows, we choose the envelope $\texttt{env}(t,t')$ and baseline kernel $\texttt{base}(t,t')$ to be stationary kernels. This makes sure that QPACK is still stationary such that it has a well defined Bochner spectrum. We give an example of the latter in Figure 21 and now turn to each of the three components that make up QPACK.

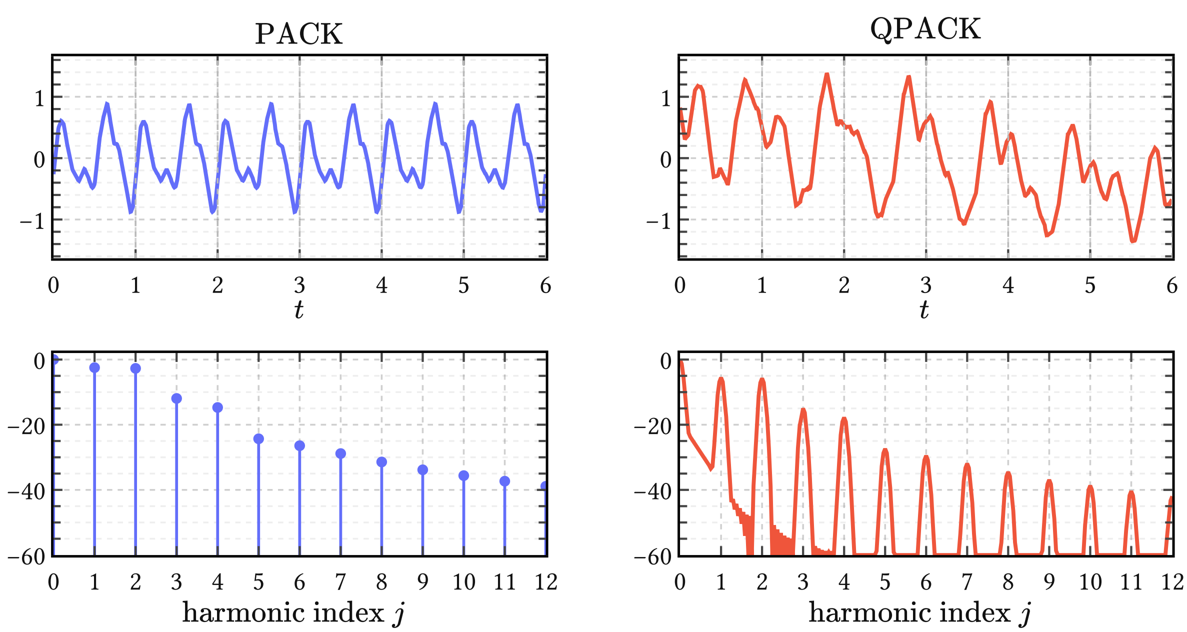


Figure 21 | **From PACK to QPACK in time and frequency.** The top row shows prior samples over six periods; the bottom row shows the corresponding normalized spectra in dB against harmonic index j. The shared carrier hyperparameters are $d=1, J=2, T=1, \sigma_a^2 = 1$, and $w=(0.2,0.4,0.4)$. For QPACK the envelope is a unit-variance RBF with lengthscale $\ell_{\text{env}} = 3$, and the baseline is a rational quadratic kernel with $\sigma^2_{\text{base}} = 0.02$, $\ell_{\text{base}} = 0.5$, and $\alpha_b = 1$.

Carrier kernel The carrier is a PACK of degree d from the previous section. The amplitude modulation by the envelope kernel does not materially influence the spectral rolloff, so the degree d still governs closure sharpness and spectral rolloff, while the weights and harmonic order J model the details of the glottal cycle.

Envelope kernel The envelope kernel, written $\mathtt{env}(t, t')$, modulates this carrier on a much longer timescale ℓ_{env} than the fundamental period T – typically on the order of a dozen periods. We use a unit-variance RBF,

$$\mathtt{env}(t, t') := \exp\left(-\frac{(t-t')^2}{2\ell_{\text{env}}^2}\right), \tag{175}$$

for the envelope because this does not interfere with the differentiability class set by d. In the frequency domain, the product of a slowly varying RBF envelope with the PACK carrier broadens each of the harmonic lines into narrow bands like the ones on display in Figure 21.

Baseline kernel The baseline kernel, written $\mathtt{base}(t, t')$, is a small additive correction. We choose the rational quadratic kernel

$$\mathtt{base}(t, t') := \sigma_{\text{base}}^2 \left(1 + \frac{(t-t')^2}{2\alpha_b \ell_{\text{base}}^2}\right)^{-\alpha_b}, \tag{176}$$

with scale $\sigma_{\text{base}} \ll \sigma_a$, lengthscale $\ell_{\text{base}} \ll \ell_{\text{env}}$, and a general shape parameter $\alpha_b > 0$. Its role is to capture weak residual structure which is not phase-locked enough to belong in the carrier but also not unstructured enough to be dismissed as white noise. This includes open-phase ripple caused by glottal-vocal-tract interaction (Ananthapadmanabha & Fant, 1982), as well as violations of the closure condition that occur frequently in more irregular or pathological voices (Kreiman et al., 2007). In spectral terms the baseline adds a broad low-frequency "pedestal" underneath the broadened carrier bands.

5.3 Gated variants

The PACK and QPACK developed above are stationary: they treat every phase of the glottal cycle with equal fidelity. This is a clean idealization but not one shared by classical GIF practice, where the closed phase is routinely suppressed before estimation. The reason is straightforward. During the closed phase the vocal folds are in contact and glottal flow is nominally zero, so the observed signal is dominated by the vocal tract's free response rather than by the source. Including this region at full weight forces the kernel to spend capacity modeling structure that carries little information about the source waveform.

Classical GIF methods handle this by windowing: the closed phase is masked out by a rectangular, ramped, or Gaussian taper before the filter estimation step (Alku, 2011). The choice of window shape and placement is typically manual and constitutes one of the most persistent

Symbol	Component	Role
σ_a	carrier	PACK amplitude scale; fixed to 1 in the whitened calibration experiments
$w_0, \ldots, w_J$	carrier	harmonic weight simplex inherited from PACK
d	carrier	within-cycle smoothness and spectral slope
J	carrier	harmonic order; effective within-cycle lengthscale control
T	carrier	fundamental period
ℓ_{env}	envelope	timescale of cycle-to-cycle amplitude modulation
σ_{base}	baseline	amplitude of the additive correction
ℓ_{base}	baseline	correlation lengthscale of the additive correction
α_b	baseline	rational quadratic shape parameter

Table 6 | **Hyperparameters of QPACK.** QPACK inherits the full PACK carrier and adds a long-scale envelope and a weak baseline. In the EGIFA calibration experiments the envelope is taken to have unit variance, so only its lengthscale is learned.

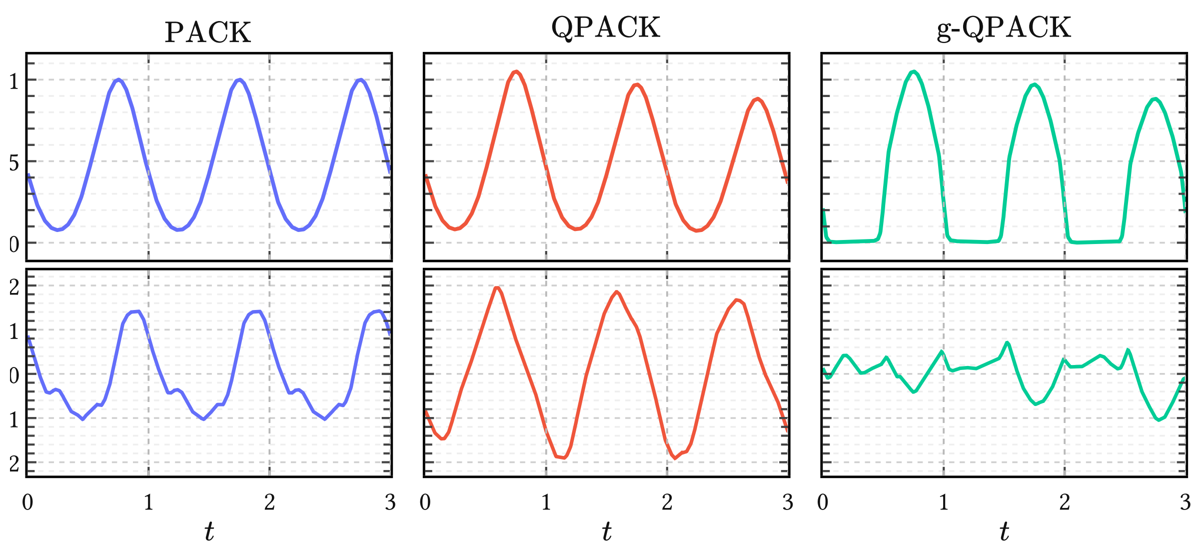


Figure 22 | **From PACK to QPACK to g-QPACK.** The top row shows kernel sections at anchor time $t' = 0.75$ and the bottom row prior samples, all with matched carrier hyperparameters. QPACK relaxes strict periodicity by adding a slow envelope and baseline; g-QPACK then suppresses covariance in the closed phase of the cycle. The shared carrier settings are $d = 1$, $J = 1$, $T = 1$, and $w = (0.2, 0.8)$. Here the QPACK envelope is a unit-variance RBF with $\ell_{\text{env}} = 4$, the baseline is rational quadratic with $\sigma^2_{\text{base}} = 0.05$, $\ell_{\text{base}} = 0.12$, and $\alpha_b = 1$, and the gate sharpness of g-QPACK is $\alpha = 12$.

heuristics in the field.[23] We now fold this idea into the kernel itself by introducing a *periodic gating function* that modulates the covariance from the inside. The result is a family of gated kernels, g-PACK and g-QPACK, in which the degree of phase selectivity is a single learnable parameter optimized alongside the other kernel hyperparameters via evidence maximization.

These gated variants are not an optional refinement. In the calibration experiments at the end of this chapter, they consistently outperform their ungated counterparts, and g-QPACK emerges as the practical model of choice for the remainder of the thesis.

5.3.1 The gating function

Let $\texttt{base}(t, t')$ be any of the periodic or quasiperiodic kernels defined above. We define the *gated kernel*

$$\texttt{gate}(t, t') = g_\alpha(t)\, g_\alpha(t')\, \texttt{base}(t, t'), \tag{177}$$

where $g_\alpha(t) \in (0, 1)$ is a periodic gating function controlled by a *sharpness parameter* $\alpha > 0$. The gate is constructed from a smooth transformation of a sinusoid:

$$g_\alpha(t) = \sigma(-\alpha \sin(2\pi t/T)), \tag{178}$$

with $\sigma(z) = (1 + e^{-z})^{-1}$ the standard logistic sigmoid.[24] This produces a differentiable approximation to a square wave with 50% duty cycle: over one half of the period $g_\alpha(t) \approx 1$, over the other $g_\alpha(t) \approx 0$, and the transitions are smooth. The 50% split is a statement in the *registered phase coordinate* used by the kernel, not in physical time. In physical time the open quotient may vary substantially from cycle to cycle; GOI registration below simply normalizes the detected GCI-to-GOI and GOI-to-GCI subintervals to common half-cycles in phase.

When the time coordinate is registered so that GCIs fall at integer multiples of T and GOIs at half-integer multiples, the gate naturally suppresses the closed phase and preserves the open phase.[25]

5.3.2 Sharpness and the interpolation between regimes

The single parameter α controls the transition width between open and closed regimes. Near the transition points the gate's derivative scales as $g'_\alpha(t) \propto \alpha$, giving a transition width of order $\mathcal{O}(\alpha^{-1})$. The model therefore interpolates continuously between three behaviors:

- **No phase preference** ($\alpha \to 0$): the gate tends to the constant $1/2$ and the gated kernel reduces to a rescaled version of the base kernel. Phase information is ignored.

[23] Appendix G discusses the broader evaluation consequences of such preprocessing choices.

[24] The bare σ here denotes the logistic function and should not be confused with the kernel scale parameters σ_a, σ_b, $\sigma_{c,j}$ introduced in Section 5.1, which always carry subscripts.

[25] This registration is described below in the alignment section. For gated models it is important to normalize not only GCIs to integers but GOIs to half-integers, so that the gate's transition aligns consistently with the open/closed boundary across the corpus.

- **Soft weighting** (moderate α): smooth, approximately linear transitions produce a gradual taper analogous to classical Gaussian or cosine windows.
- **Near-binary masking** (large α): the gate approaches a hard square wave $g_\alpha(t) \to \mathbb{1}_{\{\sin(2\pi t/T)<0\}}$ and the closed phase is effectively excised from the covariance. This recovers the behavior of rectangular closed phase windows.

The point is that the model does not commit a priori to any of these regimes. When α is optimized via the marginal likelihood, the data determine the appropriate degree of phase selectivity. This stands in contrast to classical windowing, where the window type and width must be specified before estimation begins.

5.3.3 Effect on the covariance structure

Multiplying by $g_\alpha(t)g_\alpha(t')$ is equivalent to applying a multiplicative envelope to the feature map of the base kernel:

$$\boldsymbol{\varphi}(t) \mapsto g_\alpha(t)\,\boldsymbol{\varphi}(t). \tag{179}$$

Wherever $g_\alpha(t)$ is small, the effective features are shrunk towards zero and the GP's marginal variance at that point is reduced: $\texttt{gate}(t,t) = g_\alpha(t)^2\,\texttt{base}(t,t)$. Correlations involving the closed phase are suppressed, and the model concentrates its capacity on the open phase where the source is active.

This also means that the gated kernel is no longer stationary, even when the base kernel is. The marginal variance now varies within the period, and the Bochner line spectrum from Section 5.1.4 no longer applies directly. This is a deliberate departure: the spectral elegance of the stationary representation is traded for a direct encoding of the open/closed distinction, which turns out to matter more in practice. In fact, the best performing models in our calibration experiments are all nonstationary precisely because the gate captures the single most important structural feature of the glottal cycle: the difference between the phase where the source is active and the phase where it is not.

5.3.4 g-PACK and g-QPACK

Applying (178) to the two kernel families gives

$$\texttt{g-PACK}(t,t') = g_\alpha(t)\,g_\alpha(t')\,\texttt{PACK}(t,t'), \tag{180}$$

$$\texttt{g-QPACK}(t,t') = g_\alpha(t)\,g_\alpha(t')\,\texttt{QPACK}(t,t'). \tag{181}$$

Both inherit all hyperparameters of their respective base kernel and add the single scalar α. The periodic g-PACK is the obvious baseline, but once gating is admitted it is usually more useful to keep the quasiperiodic envelope and baseline of QPACK than to cling to strict periodicity. In practice, then, the models of real interest are QPACK and especially g-QPACK.

The gate adds one parameter to the kernel, yet its effect is substantial. Where QPACK already handles amplitude modulation and baseline drift, gating captures a qualitative feature that these quasiperiodic modifications cannot: the distinction between the open phase, where the source signal carries information, and the closed phase, where it does not. In classical GIF this distinction is always encoded somewhere, be it through a windowing function, a phase detector, or an explicit closed phase analysis step (Wong et al., 1979). The gated kernel simply absorbs it into the covariance, where it participates in inference on equal footing with the other hyperparameters.

This is also the point at which stationarity is given up on purpose. QPACK still preserves a stationary carrier, but g-QPACK does not: the covariance now changes with phase because the model treats the closed and open parts of the cycle differently by construction. Empirically this trade is worth making. On EGIFA, the gain from gating is larger than the step from strict periodicity to quasiperiodicity alone, and the best calibrated models in this chapter are therefore nonstationary for a very concrete reason: they encode the most important asymmetry of the glottal cycle directly.

The practical effect is visible in held-out fits. Once calibrated on EGIFA, the learned g-QPACK basis interpolates complete voiced groups while de-emphasizing the half-cycle that was registered as closed phase.

5.4 Calibration on EGIFA

The kernels developed above are still only families. To turn them into useful surrogate priors we must *calibrate* their global hyperparameters on data. Here the data are the aligned EGIFA waveforms constructed in Section 5.5: simulator ground-truth glottal flows converted into registered voiced groups and then into standardized DGF waveforms. The calibration problem is therefore not to fit each waveform separately, but to learn a single shared basis that explains a large registered corpus well *a priori*.

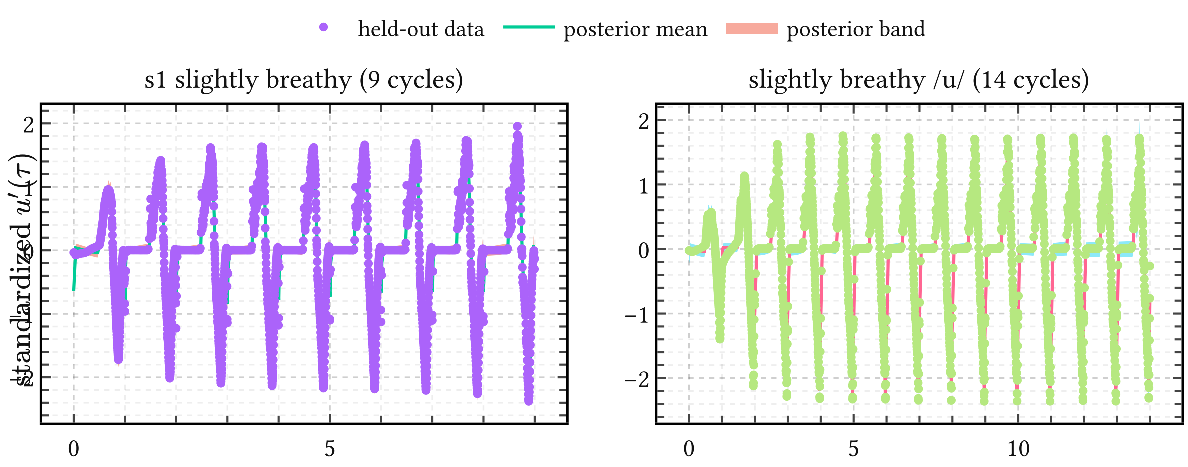


Figure 23 | **Held-out fits from calibrated g-QPACK on EGIFA.** Two retained EGIFA groups are shown together with posterior means and uncertainty bands from the best single held-out g-QPACK run (`pack:2`, $J = 16$, $M = 128$). The shaded half-cycles $[n, n + 1/2]$ indicate the phase registered as closed. After GOI registration, gating leaves these intervals comparatively de-emphasized while retaining flexible interpolation across the informative open phase.

That task is handled by PRISM (Algorithm 1). PRISM learns one global low-rank GP basis from many variable-length waveforms at once, while keeping the kernel interpretation explicit. On the EGIFA corpus used here this matters concretely: after registration there are 5612 voiced groups in total, of which 5000 are used for training and 500 held out for testing, amounting to about 8.3 million effective training samples at a padding width of 4096. An algorithm that stored or optimized a full local variational state for every waveform would quickly become cumbersome. PRISM avoids this by collapsing the local Gaussian states analytically during training, so that minibatching over waveforms updates only the global quantities: inducing locations, kernel hyperparameters, and observation noise.

The resulting picture is straightforward. Alignment turns EGIFA into a common cycle-time corpus; PRISM calibrates one shared basis on that corpus; and only in the next chapter do we *refine* this basis into richer latent generative models. This section is therefore the chapter's empirical endpoint: corpus-level calibration, not yet individualized modeling.

5.4.1 Why PRISM?

PRISM is useful here for three reasons. First, it models *functions* rather than fixed-length vectors: each voiced group may span a different number of glottal cycles and may be sampled on a different registered grid, yet all can still be trained jointly. Second, it yields a probabilistic finite-dimensional representation rather than a deterministic score vector. After calibration, each waveform can be projected to a Gaussian over shared basis amplitudes, which becomes the input to the refinement machinery of Chapter 6. Third, the collapsed variational bound makes large-corpus learning practical: training scales with minibatches of waveforms and does not require synchronized per-waveform variational parameters.

This last point is the decisive one for EGIFA. The aligned corpus is large enough that ordinary full-rank GP learning is out of the question, while methods like PCA would require committing to a common rectangular discretization *before* uncertainty is handled and would return only point embeddings. PRISM instead learns a kernel-induced basis directly on the irregular registered waveforms and retains posterior uncertainty in the shared coefficient space. The full derivation is given in Algorithm 1; here we use it simply as the calibration engine.

5.4.2 Experimental setup

All calibration experiments in this chapter use the aligned EGIFA corpus from Section 5.5. For ungated models we register only GCIs to integers; for gated models we additionally register GOIs to half-integers so that the gate transition occurs at a common phase. Each voiced group is smoothed, differentiated, whitened within group, and then padded to width 4096 for batched learning.

The kernel grid is deliberately modest. We compare carrier degrees 0, 1, and 2 and harmonic orders 16 and 32, always with an RBF envelope, a rational quadratic baseline, and 128 inducing basis functions. For each configuration we run 32 random initializations. Training maximizes the collapsed PRISM objective on the 5000-group training set. Model selection is then done

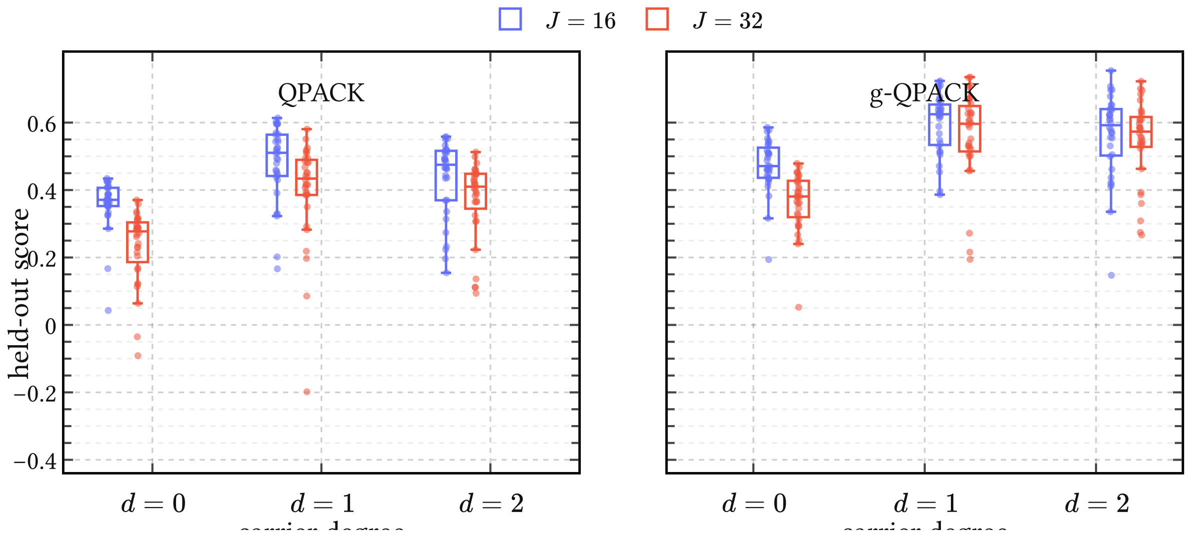


Figure 24 | **Held-out calibration scores on EGIFA.** Each box summarizes 32 independent PRISM initializations for a fixed carrier degree and harmonic order, using 128 inducing basis functions, an RBF envelope, and a rational quadratic baseline. The vertical axis reports mean held-out reduced-rank log evidence per effective sample on 500 test groups. All shown models lie far above their matched white-kernel nulls, omitted for readability. Gating shifts the entire score distribution upward, while harmonic order 16 is consistently more reliable than harmonic order 32 at fixed basis size.

not by the training bound itself but by a held-out reduced-rank log evidence per effective sample on the 500-group test set. As a sanity check, each trained model is also compared to a matched white-kernel null that keeps the same noise scale and inducing structure but removes the learned temporal covariance.

The practical comparison is therefore between QPACK and g-QPACK rather than between every conceivable intermediate model. PACK remains the conceptual starting point, and g-PACK exists as the obvious gated analogue, but once quasiperiodicity and gating are both available it is their combination that matters empirically.

5.4.3 Results

The main empirical conclusion is immediate from Figure 24: gating matters more than the step from strict periodicity to quasiperiodicity alone. Across the entire grid, the gated models score higher than their ungated counterparts, and the gain is not a one-off artifact of a single favorable initialization. This is the empirical reason the chapter's emphasis shifts from stationary PACK and QPACK to the nonstationary g-QPACK construction.

Within the ungated family, the strongest average configuration is `pack:1` with harmonic order 16, which attains a mean held-out score of about 0.297 plus or minus 0.112 across the 32 runs. Within the gated family, `pack:1` with harmonic order 16 is again best on average at about 0.412 plus or minus 0.089. The best *single* ungated run reaches about 0.429, while the best *single* gated run reaches about 0.569 and is obtained by `pack:2` with harmonic order 16. The degree-0 carrier is clearly too crude, whereas degrees 1 and 2 both capture the open-phase geometry well enough to be competitive.

The role of harmonic order is also visible now from the data rather than only from prior samples. At fixed basis size, moving from 16 to 32 usually hurts rather than helps. This is consistent with the interpretation from Figure 18: a larger harmonic order yields a wigglier carrier and therefore asks the finite PRISM basis to resolve sharper within-cycle structure. That extra flexibility is useful only when the data demand it strongly enough, and on EGIFA the more conservative order 16 turns out to be the better compromise.

Visual inspection agrees with the score summary. The held-out fits in Figure 23 show that a well-calibrated g-QPACK model interpolates complete voiced groups cleanly while leaving the registered closed phase comparatively de-emphasized. This is exactly what the score improvement suggests should happen: the model spends more of its capacity on the part of the cycle where the source is informative.

Why not do PCA?

For exploratory work, PCA is of course a useful baseline. But as a calibration engine it is mismatched to the present problem. The aligned EGIFA groups still have varying length and irregular registered grids, so a PCA pipeline must first commit to a common rectangular representation and then discard uncertainty in the resulting scores. PRISM does the reverse: it learns the shared basis directly on the registered functions and only then projects each waveform to a Gaussian in a common coefficient space.

There is also a conceptual difference. PCA learns whatever orthogonal basis best explains empirical variance in the training matrix. PRISM learns a basis induced by a specific kernel family, so calibration remains tied to the qualitative assumptions built into PACK, QPACK, and g-QPACK. That is the real point of the exercise. We do not merely want a convenient compression of EGIFA; we want a computationally tractable way to tune a theoretically motivated prior before carrying it into the downstream BNGIF model.

5.5 Alignment and pitch tracking

Although described here after the kernel constructions, alignment is logically the prerequisite for calibration. The kernels above were written in physical time, but corpus-level learning only becomes meaningful after each waveform is re-expressed in shared cycle time, a dimensionless coordinate in which GCIs occur at integers and, for gated models, GOIs at half-integers. The practical task of this section is therefore to turn the raw EGIFA glottal flows into a collection of clean voiced groups represented as registered time-value pairs.

This is the *alignment* step in the working vocabulary adopted in this chapter: alignment for data registration, calibration for corpus-level kernel learning, and refinement for the downstream individualized models of the next chapter. Its quality directly limits calibration, because a shared kernel can only be learned meaningfully if the waveforms it is asked to explain are placed on a common temporal footing.

The alignment pipeline consists of four steps applied to each recording:
1. detect GCI and GOI events from the glottal flow waveform,
2. compute inter-GCI periods and feed these to a pitch track model,
3. use the model's robust weights to select clean voiced groups,
4. warp each voiced group into shared cycle time by piecewise linear interpolation from the detected landmarks.

Steps 1 and 2 rely on two algorithms, LBGID and t-PRISM, whose full descriptions are deferred to their respective algorithm chapters (Algorithm 4, Algorithm 2). Here we describe only their roles and the rationale for combining them.

5.5.1 Glottal instant detection with LBGID

The EGIFA recordings already provide simulator ground-truth glottal flow, but this does not mean that calibration can start from the raw samples directly. Glottal closure appears not as an acoustic spike but as a soft, context-dependent low-flow regime, and the recordings still contain brief irregular closures, weakly voiced stretches, and silent gaps. The first requirement is therefore a reliable sequence of glottal events.

LBGID (Algorithm 4) is the detector used for this purpose. It constructs adaptive upper and lower envelopes, marks the samples where the flow falls below a small fraction of the local range, splits these into contiguous low-flow regions, and within each region chooses the strongest negative/positive derivative pair in causal order. The first member of that pair is reported as the GCI and the second as the GOI. No specific waveform template is assumed; only the weak physical prior that closure corresponds to low flow bounded by strong opposing transitions is built in.

The result is a sequence of GCI and GOI sample indices for each recording, from which the inter-GCI periods are immediately available.

5.5.2 Pitch track model and voiced group selection

The period sequence extracted by LBGID is noisy. Some periods correspond to genuine voiced cycles; others arise from detection errors in unvoiced segments, unstable phonation, or long silent gaps on which the detector happened to fire. Calibrating kernel hyperparameters on such a mixture would corrupt the learned model.

We address this with a probabilistic pitch track model learned on APLAWD, a speech corpus with synchronized electroglottograph recordings that provide reliable GCI reference marks (Naylor et al., 2007). The model is a GP over log-period trajectories, trained with t-PRISM rather than ordinary PRISM so that gross period errors can be downweighted rather than fitted. Among the APLAWD models tested in `experiments/svi/aplawd`, an RBF kernel with 64 inducing points gave the best average test score at about 2.12 plus or minus 0.01 log-likelihood units per effective data point.[26]

At inference time, t-PRISM returns per-period weights, namely the expected latent precisions of the Student-t augmentation. These are best read as robust *inlier scores*. When a period lies close to the smooth pitch track, its residual is small and the corresponding weight stays near or above one; when the residual is large, the weight is pushed down and that period is effectively treated as an outlier. The operational rule used here is conservative: periods with weight above one are retained, while periods with weight at or below one are excluded from voiced grouping.

Voiced groups are then defined as maximal contiguous runs of high-weight periods. Groups shorter than five periods are discarded as too brief to be useful for calibration.[27] The result is a fully automatic probabilistic sieve. Sudden jumps in inter-GCI distance, isolated detector failures, and weakly voiced regions are not removed by hand; they are simply assigned low weight by the robust pitch model and never enter the aligned corpus.

5.5.3 Registration to shared cycle time

Each surviving voiced group is now warped from physical sample indices to the shared cycle time. The mapping is piecewise linear, with GCI positions as anchor points: the first detected GCI is sent to 0, the next to 1, the next to 2, and so on, so that the shared cycle coordinate advances by one unit per glottal cycle and the period is locally normalized.[28] For gated kernel models (Section 5.3.4), the GOI positions serve as additional anchors and are mapped to half-integers, so that each GOI lands halfway through its registered cycle. This places the open/closed boundary consistently at phase 0.5 within each cycle across the corpus. This is important because the gate function (178) expects its transition to occur at a fixed phase within each period. It should not be read as a claim that the physical open quotient is 50%; rather, the detected pre-GOI and post-GOI subintervals are re-parameterized onto equal half-cycles so that phase selectivity is comparable across speakers and utterances.

The registered flow is then smoothed on a temporal scale of 0.1 ms and differentiated to obtain the DGF.[29] Only after these steps do we whiten each voiced group's derivative and pad the resulting registered time-value pairs for batching.

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[26] The periods are first log-transformed and whitened. The score quoted here is the average test log likelihood per effective data point reported in `experiments/svi/aplawd/README.md`.

[27] In everyday speech, steady-state vowels typically span 3 to 5 pitch periods (Rabiner & Schafer, 2007), so this threshold is a mild requirement.

[28] In the language of functional data analysis, this is *landmark registration*: the GCIs are the landmarks, and the warp is the simplest monotone interpolant consistent with them (Ramsay & Silverman, 1997). A full Bayesian treatment would model the GCI locations as uncertain and sample posterior phase tracks conditioned on them. We opt for deterministic linear interpolation for simplicity.

[29] In practice we smooth the glottal flow with a Gaussian filter whose effective width corresponds to 0.1 ms, differentiate that smoothed signal, and then resample to approximately two samples per smoothing window. At EGIFA's native rate this lands essentially back at 20 kHz, so the step acts primarily as regularization rather than aggressive downsampling.

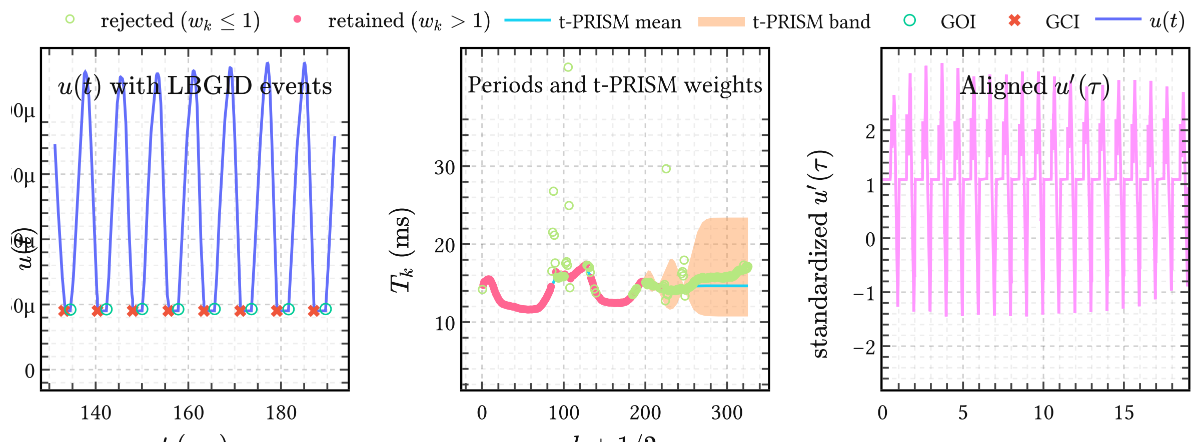

Figure 25 | **The alignment pipeline on an EGIFA recording.** (a) A short raw-flow window with LBGID-detected GCI and GOI events. (b) The corresponding period sequence over the full recording, together with the t-PRISM posterior mean and uncertainty band; periods with weight above one are retained as inliers and contiguous retained runs become voiced groups. (c) One retained voiced group after GOI-aware registration, smoothing, differentiation, and per-group standardization, shown as a waveform in shared cycle time. The lightly shaded half-cycles indicate the phase later suppressed by the gated kernels.

5.5.4 The aligned corpus

The output of the alignment pipeline is a collection of varying-length registered DGF waveforms, one per voiced group, all expressed in shared cycle time and whitened within group. Typical voiced groups span 5 to 30 glottal cycles, so their shared cycle-time extent usually ranges from about 0 to 5 up to about 0 to 30, with the exact extent determined by the pitch track model's confidence weights. These groups are drawn from recordings at different fundamental frequencies, phonation types, and vowel contexts, yet they now live on a common temporal footing where the PACK and QPACK kernels can be applied directly.

This aligned corpus is the input to the calibration step that follows.

Figure 25 makes the three coordinate systems of the chapter explicit. LBGID operates in physical time, the robust pitch model operates on the discrete period sequence, and the calibrated kernels finally see standardized DGF waveforms expressed in shared cycle time. Alignment is the step that makes these three views consistent.

5.6 Summary

We extended the TACK to multiple pitch periods. Our idealization of perfect periodicity allowed us to derive an efficient Karhunen-Loève representation of the corresponding PACK.

We now have a kernel that checks following properties of theoretical GFMs:
- periodicity
- differentiability and spectral case
- closure constraint

- hard/soft GCI support

To add more realism, we relax these hard constraints and learn expand support again using simulations.

6 Surrogate glottal flow models

So far, our aim to learn a good DGF prior which is theoretically grounded but data-driven can be seen as progressive refinement of prior probability, from general to more realistic. But since our data (whence the data-driven comes from), an equal view is that we have been learning *surrogate models* from hi-fi expensive simulations. Since our surrogate models are just quantized (reduced-rank) GPs, our surrogates are simple linear models, and this is why they lend themselves to efficient inference systems. In this chapter we take the final step in refining our surrogate. Our journey was:

1. Define a theoretical density based on the properties of classic GFMs: changepoints lead to the ACK (arc cosine kernel); this was Chapter 4. This was for a single pitch period.
2. We then extended this to multiple pitch periods; first incorporating perfect periodicity, later relaxing this to quasiperiodicity; Chapter 5. Learning hyperparameters from large scale expensive glottal flow simulations happened with PRISM-VFF.
3. And finally, in this chapter, we use the PRISM-VFF embeddings as a starting point for further refinement into more realistic kernels.

Indeed, the kernels learned in step (2.) already scored competitively on EGIFA, but when used as a generative model, were seen not to even look like GF waveforms. We will rectify that in this chapter. First by going “all-out” and learning almost “digital twins” of DGF; then relaxing this into more stationary variants to make sure to have robustness against time warping faults.

6.1 Two levels of learning

We can learn on the level of hyperparameters, which is a nonlinear optimization of the basis functions in the weight-space GP view.

Via PRISM we learn a basis for the problem.

However, to get better models, it is also necessary to learn on the second level, which is that of amplitudes.

6.2 Quantization, again

From LF model we know that inherent dimensionality of GF models is 5.

Can we learn this from data?

See Algorithm 3

This is effectively a deep GP but linearized, and the time warping adds another GP layer.

6.3 Compression

$K = 1$ with outlier rejection. Still a $M = 128$ model.

6.4 Quantization

$K \geq 1$. Specialize.

6.5 Generative models of voiced speech

We can bring together

- time warping (the phase connection)
- learning formant tracks so PRISM can shine again, + combine with tilt and ARPrior to get better AR coeffs
- q-GPLVM quantized qpack models which have natural OQ variations (rather than learning separately). Thus we use QPACK, not g-QPACK kernels such that we can use our time warps.

Can be used to generate large quantities of voiced speech automatically

This is a surrogate model of voiced speech

6.5.1 Generating pitch tracks

Instantaneous phase and pitch A large portion of the nonstationary variation of voiced speech can be described via a monotone *phase function* alone. Let

- t be physical time (in seconds),
- $\tau(t)$ be the cycle index or phase, dimensionless,
- $f_0(t)$ be the instantaneous pitch (Hz).

By definition,

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = f_0(t) \tag{182}$$

This says that phase advances at a rate given by instantaneous pitch. The key is now that we will describe time as a function of phase, not the other way around; this is known as *time warping*: the local pitch warps time.

In the simplest case, that of constant pitch, we have $f_0(t) := 1/T$ such that, integrating (182), we have

$$\tau(t) = \frac{t - t_0}{T} \tag{183}$$

where $t_0 \in \mathbb{R}$ is an arbitrary offset. Thus

$$t(\tau) = t_0 + T\tau. \tag{184}$$

This is simply rescaling and re-offsetting when going from τ space from t, as people would naievely do.

The most general solution of (182) however, is:

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{f_0(t(\tau))} := T(\tau), \tag{185}$$

such that

$$t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u. \tag{186}$$

This defines physical time t as time-warped by local curvature $T(\tau)$, which is the instantaneous period given by the reciprocal of the instantaneous fundamental frequency $f_0(\tau) = 1/T(\tau)$. These can be indexed either in phase τ or physical time t.

Phase warping necessitates having a model with continuous time like GPs. This is not as evident in straight signal-processing approaches.

Note: this construction is like a GP with a single hidden layer.

6.5.2 Generating vowels

6.6 Summary

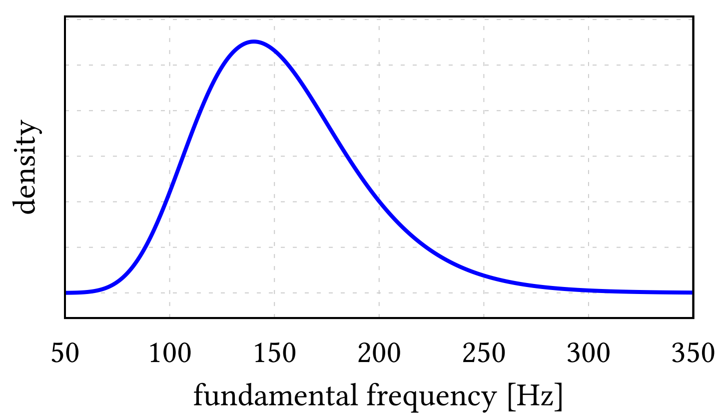

Figure 26 | **Analytic marginal fundamental frequency distribution** implied by the log-period prior. The distribution is lognormal with median 150 Hz, 5-95% range approximately 100-224 Hz (2.5–97.5%: 92–244 Hz).

7 Bayesian nonparametric glottal inverse filtering

7.1 Recap of all the ingredients

- IKLP
- AR prior
- Quasiperiodic arc cosine kernel
  - Learnt from synthetic Paule data
- Latent variable models for refining

7.2 Learning priors from synthetic data

7.2.1 PAULE and CommonVoice

PAULE is a neural model that inverts speech waveforms to the control parameter (CP) trajectories that drive VocalTractLab (VTL). In other words it attempts to reconstruct what the articulators did to arrive at loosely similar sounding speech (like children acquiring and mimicking speech). While (as its authors acknowledge) listening to the resynthesized recordings show that reconstruction quality is quite poor, our aim here is not the latter but rather exploration of human possible glottal flow, operationalized via VTL.

To do this, we forked VTL and patched it to allow extracting glottal flow and used that on the PAULE training set (ie, inverted CP trajectories which are then inputted to VTL to get the GF waveforms). This got us some 21k examples, of which most were "duds", ie, synthesized GFs and speech that are nondescript murmured babblings.

The next step is to use LBGID on this to extract reasonable quality GCI and GOI estimates, then detect voiced regions with our pitch track model (t-PRISM), then take the DGF, killing the very large (unphyisically breathiness) DC in the signal.[30]

A direct indicator of synth quality is just synth length (ie the length of synthesized speech and GF waveforms). We found that short lengths generally sounded very similar like the same muffled grunt, and we filter these out.

We then can make accurate measurements of T and OQ. Our goal here is to cover enough variation of OQ to learn the closed phase and spectral rolloff (since latter is heavily correlated with the former). Due to synth quality we also artificially make the closed phase more perfectly closed, since this is phsyiologically observed and apparantly non-reached (non-modeled) part of GF waveforms by PAUL.

7.2.2 Finding high quality voiced groups

We use t-PRISM for that, and our LBGID algorithm. The main conceptual idea here is that we use *an explicit pitch track model* to do the extraction rather than ad hoc heuristics.

[30] Although we could have also modeled it with PRISM-VFF directly by adding a RBF kernel to soak up low varying baselines. But this would then need to be unlearned by the model during BNGIF inference time.

Tricks we use:

- Normalize data to (0,1): this means we freeze mean to 0 **and** kernel variance to 1.
- Lengthscale in ballpark order: 10. This is simply to cut number of iterations, the model does find lengthscales 15 if initialized from 1, but takes much longer.
- Inverse-ECDF initialization of inducing points. This is important because trajectory lengths are long tailed, and we need to place predictive power where it matters. In 1D we can just sample from the empirical distribution, with interpolation.
- Set ν based on event statistics. Optimizing ν is possible, but more stable optimization. Results depend only very slightly on exact value of ν as long as < 10.

For the rest standard Adam optimizer with standard settings.

Heuristic for setting ν We set ν using a simple tail-probability heuristic derived from the expected frequency of spike events. After normalization, the GP prior has unit variance and the learned observation noise standard deviation is approximately $\sigma \approx 0.15$. Typical spikes have amplitude around 5, corresponding to residuals of size $r \approx \frac{5}{0.15} \approx 30 - 35$ noise standard deviations. We interpret spikes as events that should lie in the heavy tails of the Student-t likelihood rather than being explained by the Gaussian core. Therefore we choose ν such that the tail probability of a Student-t distribution satisfies $P(|X| > r) \approx p_{\text{spike}}$, where p_{spike} is the empirical spike rate (here about 1%). Using the large-r asymptotic tail behaviour $P(|X| > r) \approx C * r^{-\nu}$ and ignoring constants gives the practical rule $\nu \approx -\frac{\log(p_{\text{spike}})}{\log(r)}$. For $r \approx 35$ and $p_{\text{spike}} \approx 0.01$, this yields $\nu \approx 1 - 2$, which corresponds to a Cauchy-like heavy-tailed likelihood. This choice ensures that extreme residuals are downweighted rather than forcing the GP to explain them through excessive curvature.

7.3 EGIFA

7.3.1 About the dataset

(Chien et al., 2017) compare state-of-the art GIF algorithms on a synthetic database.

The good

The bad

The ugly

Problem with their `gci_area` backend can also be seen in Figure 9 in the paper, where the cycle endpoint (GCI) is marked, but seen not to align with the actual event.

Paper finds limited impact of GCI errors on the trad algos. But is this because their baseline truth is already very bad? Do we recover this?

7.3.2 Algorithms used

7.3.3 Problems

See Appendix G for the gauge discussion

In order to apply gauge with confidence (ie to restrict shifting only backwards) we need a confident ground truth GCI detector. Ie if our detector systematically overestimates the pitch period, algorthms have more leeway to get good results For this reason it is important that LBGID be as robust as possible No matter which `gci_backend` used for GCI detection, we use ground truth periods to restrict lag leeway as much as possible.

7.4 Evaluation on EGIFA

Low F1 vowels cannot be resolved witg GIF?

From (Chien et al., 2017): > It has been observed by some researchers that some vowels with a low first formant frequency cannot be adequately analyzed by an inverse filtering algorithm, whereas the vowel /ä/ has a first-formant frequency that is sufficiently high to avoid interference with the primarily low-frequency energy distribution of glottal source

Is this the case for us? Check if close rounded vowels /o/ and /u/ are also harder. This is case for all traditional algos

F0 dependence

Our decoupling and larger window size should also remedy F0 performance. Is that the case? Just for our best model

GCI estimates

Rather than using error injected ground truth estimates, we just use ground truth (LBGID) and actual GCI detectors, which provide for a more realistic error injection

We compare performance per GCI backend

We can also comparee GCI backend quality by testing on APLAWD

Window sizes

All trad algos work with 32 msec, while we do 128 msec

This buys us more data (crucial) but makes us more susceptible to the assumption that the vocal tract filter stays constant during that period. 128 msec is in general too long for that

So we can test if shorter windows work too, but only for our best model on 128 msec

It should be noted that due to short window length of trad algos that they have a major advantage as to more lag shift freedom, and still we beat them

About CCD

CCD comes out really badly in (Chien et al., 2017), but relatively well in our paper Anecdotal evidence(ie, looking at actual inferred DGFs) suggests that CCD often has really good performance relative to other trad algos

The reason for that is probably that CCD is very sensitive to GCI estimation, and (Chien et al., 2017) area_gci estimator failed heavily there.

7.5 Evaluation on VTRFormants

BNGIF is not just a GIF; it's a full grey-box generative model of both voiced and nonvoiced speech, though GIF is only defined reasonably accurately for the voiced case, as the algorithms need and feed upon the correlations in the quasiperiodic glottal to resolve the inherent ambiguity in GIF (ie blind inversion where both filter and source must be inferred).

7.6 Summary

We optimize by projecting to a common basis. Memory becomes $\mathcal{O}(BMr + Ir^2)$ and compute $\mathcal{O}(Mr^2 + Ir^2 + r^3) \approx \mathcal{O}(Mr^2 + Ir^2)$ as $M \gg r$. With these optimizations in place, we can get to BNGIF speed: linear complexity in frame size. We work at native EGIFA fs of 20 kHz. Raw runtimes: in x32 runs at 5x realtime, in x64 runs at 0.5x realtime. So in x32 1 sec of audio is processed in 0.2 sec, and in x64 in 2 sec. The entire EGIFA corpus can be processed in about 3 minutes in x32 (this does not include jax tracing and loading the corpus and postprocessing results, which about double the time).

Working in x32 can sometimes result in nans due to the GIG distribution-associated computations. This happens in 1% in case of `EGIFA/vowel` and 0.1% in case `EGIFA/speech`. In the former case nan-occurence can be reduced to below 0.001% by doing random restarts without any effect on performance. Future work could wrap the numerically sensitive parts in x64 in JAX so both speed and stability are retained.

8 Conclusion

Here is a `test of mono` fonts. `EGIFA_raw_speech_file`.

A person speaks 16,000 words on average each day: glottal folds are used plenty, so disorders arise quickly when used inappropriately. (Andrade-Miranda et al., 2024)

Possible improvements/future directions:
- $p(a)$ not as a delta function in IKLP
- work in spectral domain completely
- model $h(t)$ as a GP conditional on $e(t)$ given: alternate optimization
- interframe correlations as in (Mehta et al., 2012)
- learn from more realistic glottal flow simulations like (Avhad et al., 2022; Schoder et al., 2024; Zhang et al., 2020); perhaps learn first a mapping from glottal flow area to glottal flow from physics and then use real life measured glottal flow area databases like (Andrade-Miranda et al., 2024)

Recent article in Nature Parkison's disease "Speech impairments, on the other hand, are particularly prevalent, affecting up to 90% of people with PD" "Growing evidence indicates that speech and language alterations often precede the defining motor signs and PD diagnosis **by as much as a decade**" (Cao et al., 2025) "Recent research highlights the value of **objective acoustic speech markers** in detecting PD in the initial or even prodromal (e.g., RBD) stages23,25,26. This implicates a potential window for timely interventions, such as monoamine oxidase B inhibitors, anticholinergics and future novel therapies that may confer improved efficacy in early PD stages" (Cao et al., 2025)

> The recommended and widely employed tests are 1) sustained phonation of the vowel /a/ in a single breath, which helps assess breath control, vocal fold function and vocal quality (Cao et al., 2025) > 2) rapid repetition of syllables such as /pa/, /ta/ and /ka/ to evaluate consonant and vowel articulation, articulation rate and regularity, coordination and speech timing;

Important features: vocal fold oscillation irregularity, breathiness and noise, **and vocal tract resonance fluctuations**

Detection rate in clinical lab conditions is already very high But algorithms could be improved And perhaps we could have apps on mobile phone to cut costs even more; this would mean nonideal conditions so better algos needed

Online learning Priors for the expansion coefficients $\mathbf{a}$ can be updated from the posterior from previous frame. Same for the AR prior $\mathbf{a}$ (nameclash here). Likewise, values of $\{T, t_c\}$ are correlated across frames and this can be exploited too.

8.1 Future work

Learning the time warp from data It would be better to ditch LBGID and learn it from data. But scaling would be an issue. (Kazlauskaite et al., 2018).

IKLP extensions To KARMA and mixture of AR priors

q-GPLVM q-GPLVM as used here is a very stable two-step procedure, but it would an interesting research direction to include the GMM objective directly into the Bayesian GPLVM ELBO. This would enable "activation-aware" quantization.

Spectral priors for ARMA Extend spectral priors for MA and ARMA models. The linear divisibility machinery itself is agnostic: it works just as well for numerator polynomials, hence for antiresonances, notches, and radiation factors in ARMA or pole-zero models. What fails here is only the **all-pole interpretation** of roots of A as dips in |H|^2.

Algorithms

Algorithm 1. PRISM

PRISM stands for **process-induced surrogate modeling**. The central challenge motivating PRISM is one of representation: glottal flow waveforms arrive as time series of varying length, not as fixed-dimensional vectors, yet everything downstream — clustering, density modeling, latent variable inference — wants fixed-dimensional inputs with calibrated uncertainty. One could fit a GP posterior to each waveform separately using whatever kernel one happens to like, and store the resulting function. But storing a posterior function per waveform is expensive, and more importantly there is nothing shared: each waveform lives in its own GP, and you cannot meaningfully cluster or compare "GP posteriors" unless they decompose onto a common basis.

In Chapter 5 this shared-basis view is used for *calibration*: one global kernel is fit to a large registered EGIFA corpus. In Chapter 6 the same representation becomes the input to richer latent models. PRISM is therefore best thought of as the adapter stage of the thesis: from variable-length functions to fixed-dimensional Gaussian coordinates.

PRISM resolves this by learning a single shared kernel-induced basis from all waveforms jointly, then projecting each waveform into a Gaussian posterior over a fixed-dimensional vector of basis amplitudes. The result is a collection $\{(\mathbf{m}_i, \mathbf{S}_i)\}_{i=1}^{I}$ of Gaussians in a common $\mathbb{R}^M$, one per waveform, carrying both the best estimate of how that waveform decomposes

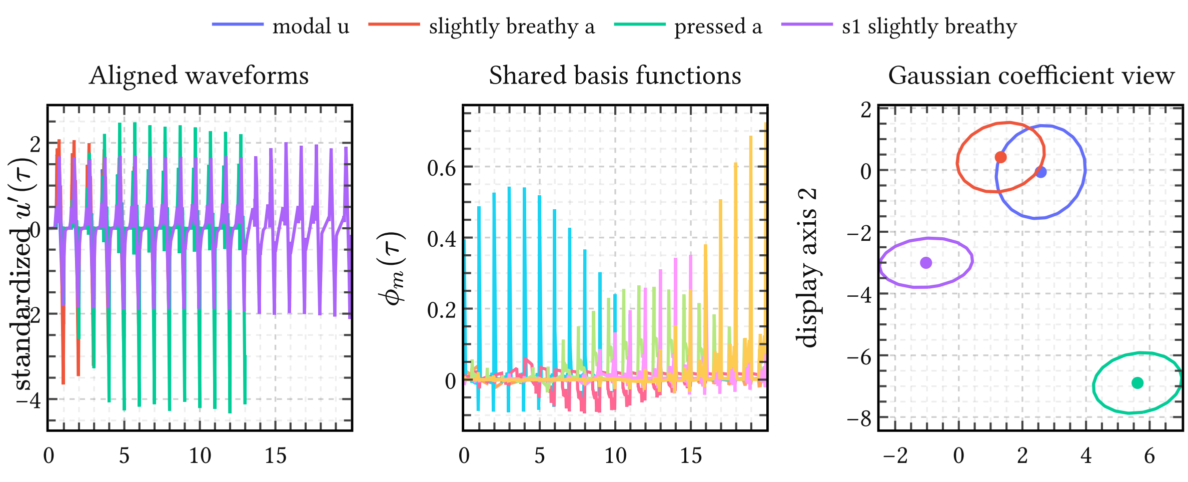

Figure 27 | **PRISM as a shared-basis adapter.** Left: four aligned EGIFA DGF waveforms of different extent in shared cycle time. Middle: six representative entries of the learned shared feature map $\varphi_{m(t)}$ induced by one calibrated g-QPACK basis. Right: the same waveforms after projection to a two-dimensional linear view of the common coefficient space, where each example is represented as a Gaussian rather than a point. The figure makes explicit what PRISM shares across examples: not a common rectangular grid, but a common probabilistic basis.

onto the shared basis and the uncertainty in that decomposition. Figure 27 shows this conversion in the concrete setting of the thesis.

The mechanism that makes this tractable is a collapsed variational bound in the style of (Titsias, 2009). The essential observation is that for Gaussian likelihoods, the per-waveform variational parameters can be eliminated analytically during training, so only global basis parameters need to be optimized. Projection to per-waveform Gaussians then becomes a cheap post-training step requiring no gradient computation. This appendix derives the bound, the projection, and explains why collapse is not merely a computational convenience but the design choice that distinguishes PRISM as a shared-basis learner from a collection of independent GP fits.

1.1. Data and model

We have I independent waveforms. Waveform i consists of a time grid $\mathbf{t}_i = \left(t_{i1}, ..., t_{iN_i}\right)$ and observations $\mathbf{y}_i = \left(y_{i1}, ..., y_{iN_i}\right)$, with N_i varying freely across waveforms. Each is modeled by an independent latent function

$$f_i(.) \sim \texttt{GaussianProcess}(0, k_\theta(., .)), \tag{187}$$

sharing the same kernel hyperparameters θ across all i but with independent draws of f_i. Observations are conditionally independent given f_i:

$$\mathbf{y}_i \mid f_i \sim \texttt{Normal}\left(f_{i(\mathbf{t}_i)}, \sigma^2 \mathbf{I}_{N_i}\right). \tag{188}$$

The shared kernel k_θ is the central object being learned. Its hyperparameters θ, together with the noise variance σ^2, are the global parameters of the model. The per-waveform objects — the f_i themselves, or any variational approximation to them — are local. The tension between learning global parameters efficiently while marginalizing over local ones is precisely what the collapsed bound resolves.

1.2. From kernel to shared basis

Choose M inducing locations $\mathbf{Z} = (z_1, ..., z_M)$ in the time domain, also treated as global learnable parameters. The inducing variables $\mathbf{u}_i = f_{i(\mathbf{Z})} \in \mathbb{R}^M$ are the values of example i's latent function at the shared locations. Since f_i is a GP,

$$\mathbf{u}_i \sim \texttt{Normal}(\mathbf{0}, \mathbf{K}_{ZZ}), \tag{189}$$

where $\mathbf{K}_{ZZ} \in \mathbb{R}^{M \times M}$ with $(\mathbf{K}_{ZZ})_{mn} = k_{\theta(z_m, z_n)}$ is the same for all i. The GP conditional on $\mathbf{u}_i$ is

$$f_{i(\mathbf{t}_i)} \mid \mathbf{u}_i \sim \texttt{Normal}(\mathbf{K}_{iZ}\mathbf{K}_{ZZ}^{-1}\mathbf{u}_i, \mathbf{K}_{ii} - \mathbf{Q}_{ii}), \tag{190}$$

where $\mathbf{K}_{iZ} \in \mathbb{R}^{N_i \times M}$ has (n, m) entry $k_{\theta(t_{in}, z_m)}$, and $\mathbf{Q}_{ii} = \mathbf{K}_{iZ}\mathbf{K}_{ZZ}^{-1}\mathbf{K}_{Zi}$ is the rank-M approximation to $\mathbf{K}_{ii}$.

The whitened parameterization makes the shared structure explicit. Let $\mathbf{L}$ be the Cholesky factor with $\mathbf{K}_{ZZ} = \mathbf{L}\mathbf{L}^\top$, and define whitened amplitudes $\mathbf{a}_i = \mathbf{L}^{-1}\mathbf{u}_i$, so that $\mathbf{a}_i \sim \mathtt{Normal}(\mathbf{0}, \mathbf{I})$ under the prior. Writing $\mathbf{k}_{Z(t)} = \left(k_{\theta(z_1,t)}, ..., k_{\theta(z_M,t)}\right)^\top \in \mathbb{R}^M$ for the cross-covariance vector at time t, the whitened feature map is

$$\varphi(t) = \mathbf{L}^{-1}\mathbf{k}_{Z(t)} \in \mathbb{R}^M, \tag{191}$$

and the mean in (190) can be written as $\mathbf{\Phi}_i \mathbf{a}_i$, where

$$\mathbf{\Phi}_i = \begin{pmatrix} \varphi(t_{i1})^\top \\ \vdots \\ \varphi\left(t_{iN_i}\right)^\top \end{pmatrix} \in \mathbb{R}^{N_i \times M} \tag{192}$$

is the design matrix that maps each observation time to the shared feature space. This is the "prism": every irregular time grid, no matter how long or short, is mapped to a common $\mathbb{R}^M$ through the shared basis. Each waveform thus becomes an instance of Bayesian linear regression,

$$\mathbf{y}_i = \mathbf{\Phi}_i \mathbf{a}_i + \varepsilon_i, \qquad \mathbf{a}_i \sim \mathtt{Normal}(\mathbf{0}, \mathbf{I}), \qquad \varepsilon_i \sim \mathtt{Normal}(\mathbf{0}, \sigma^2 \mathbf{I}). \tag{193}$$

1.3. Collapsed variational bound

The cost of keeping local parameters A standard sparse variational GP (Titsias, 2009) approximates the posterior over $\mathbf{u}_i$ with a Gaussian $q_{i(\mathbf{u}_i)} = \mathtt{Normal}(\mathbf{m}_i, \mathbf{S}_i)$ and maximizes a per-example ELBO

$$\mathcal{L}_i^{\text{SVGP}}(\mathbf{m}_i, \mathbf{S}_i; \mathbf{Z}, \theta, \sigma^2) = \mathbb{E}_{q_i}[\log p(\mathbf{y}_i \mid f_i)] - D_{\text{KL}}(q_i(\mathbf{u}_i) \parallel p(\mathbf{u}_i)). \tag{194}$$

In the present setting, examples are independent GPs sharing $(\mathbf{Z}, \theta)$. Following this route directly, we must store and optimize variational parameters $(\mathbf{m}_i, \mathbf{S}_i)$ for every waveform i — a storage cost of $\mathcal{O}(IM^2)$ in the covariances alone. More fundamentally, the local parameters would need to be maintained and warm-started across minibatches during stochastic optimization, requiring a large synchronized state that makes proper minibatching over waveforms awkward.

Titsias collapse For Gaussian likelihoods, (Titsias, 2009) showed that the optimal $q_{i(\mathbf{u}_i)}$ can be found analytically as a function of $(\mathbf{Z}, \theta, \sigma^2)$ and the observed data $(\mathbf{t}_i, \mathbf{y}_i)$ for that example alone. Substituting this optimal form back into the ELBO yields a bound that depends only on the global parameters, and no per-example variational state needs to be stored or updated.

To state the bound compactly, define the scaled feature matrix and its Gram matrix,

$$\mathbf{A}_i = \frac{\mathbf{\Phi}_i^\top}{\sigma} \in \mathbb{R}^{M \times N_i}, \qquad \mathbf{B}_i = \mathbf{I}_M + \mathbf{A}_i \mathbf{A}_i^\top \in \mathbb{R}^{M \times M}, \tag{195}$$

and the sufficient statistic $\mathbf{v}_i = \mathbf{A}_i \mathbf{y}_i = \mathbf{\Phi}_i^\top \mathbf{y}_i / \sigma \in \mathbb{R}^M$. The collapsed ELBO for example i is

$$\mathcal{L}_i(\mathbf{Z}, \theta, \sigma^2) = \log \texttt{Normal}(\mathbf{y}_i \mid \mathbf{0}, \mathbf{Q}_{ii} + \sigma^2 \mathbf{I}) - \frac{1}{2\sigma^2} \operatorname{Tr}(\mathbf{K}_{ii} - \mathbf{Q}_{ii}). \tag{196}$$

The first term is the log marginal likelihood under the low-rank approximation; the second penalizes the approximation error.

Evaluating the bound Both terms are computed efficiently from $\mathbf{A}_i$, $\mathbf{B}_i$, and $\mathbf{v}_i$, so the cost per waveform scales as $\mathcal{O}(M^2 N_i)$ and the $N_i \times N_i$ matrix $\mathbf{Q}_{ii} + \sigma^2 \mathbf{I}$ is never formed.

For the log marginal term, write $\mathbf{Q}_{ii} + \sigma^2 \mathbf{I} = \sigma^2 (\mathbf{I} + \mathbf{A}_i^\top \mathbf{A}_i)$. By the matrix determinant lemma, $\det\left(\mathbf{I}_{N_i} + \mathbf{A}_i^\top \mathbf{A}_i\right) = \det(\mathbf{B}_i)$, so

$$\log\det(\mathbf{Q}_{ii} + \sigma^2 \mathbf{I}) = N_i \log \sigma^2 + \log\det(\mathbf{B}_i). \tag{197}$$

The Woodbury identity gives $(\mathbf{Q}_{ii} + \sigma^2 \mathbf{I})^{-1} = \sigma^{-2}(\mathbf{I} - \mathbf{A}_i^\top \mathbf{B}_i^{-1} \mathbf{A}_i)$, so the quadratic form is

$$\mathbf{y}_i^\top (\mathbf{Q}_{ii} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_i = \sigma^{-2}(\|\mathbf{y}_i\|^2 - \mathbf{v}_i^\top \mathbf{B}_i^{-1} \mathbf{v}_i). \tag{198}$$

For the trace correction, $\operatorname{Tr}(\mathbf{Q}_{ii}) = \operatorname{Tr}(\mathbf{\Phi}_i \mathbf{\Phi}_i^\top) = \|\mathbf{\Phi}_i\|_F^2$, so

$$-\frac{1}{2\sigma^2} \operatorname{Tr}(\mathbf{K}_{ii} - \mathbf{Q}_{ii}) = -\frac{1}{2\sigma^2} (\operatorname{Tr}(\mathbf{K}_{ii}) - \|\mathbf{\Phi}_i\|_F^2). \tag{199}$$

The full dataset objective is the sum $\mathcal{L}(\mathbf{Z}, \theta, \sigma^2) = \sum_{i=1}^{I} \mathcal{L}_i$. Every quantity here is computable from the global parameters and the data for example i; no per-example state persists between gradient steps.

1.4. Stochastic optimization

Because examples are independent, an unbiased estimate of $\mathcal{L}$ is available from any minibatch $\mathcal{B} \subset \{1, ..., I\}$,

$$\hat{\mathcal{L}} = \frac{I}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \mathcal{L}_i(\mathbf{Z}, \theta, \sigma^2). \tag{200}$$

Gradient ascent on $(\mathbf{Z}, \theta, \sigma^2)$ using $\hat{\mathcal{L}}$ is stochastic variational inference in the sense of (Hoffman et al., 2013): minibatching over independent examples, updating global parameters only. The absence of local parameters to synchronize or warm-start is what makes this natural rather than contrived.

1.5. Projection to the PRISM representation

Once $(\mathbf{Z}, \theta, \sigma^2)$ are trained, the design matrix $\mathbf{\Phi}_i$ is determined for any waveform, and the BLR model (193) is fully specified. The per-example posterior over whitened amplitudes follows immediately from Bayesian linear regression:

$$\mathbf{a}_i \mid \mathbf{y}_i \sim \texttt{Normal}(\mathbf{m}_i, \mathbf{S}_i), \tag{201}$$

$$\mathbf{S}_i = \left(\mathbf{I} + \sigma^{-2}\mathbf{\Phi}_i^\top \mathbf{\Phi}_i\right)^{-1} = \mathbf{B}_i^{-1}, \qquad \mathbf{m}_i = \sigma^{-2}\mathbf{S}_i \mathbf{\Phi}_i^\top \mathbf{y}_i = \mathbf{B}_i^{-1}\mathbf{v}_i. \tag{202}$$

This reuses $\mathbf{B}_i$ and $\mathbf{v}_i$ from the collapsed bound, so the projection costs nothing beyond what training already computed. The PRISM representation is the resulting collection

$$\{(\mathbf{m}_i, \mathbf{S}_i)\}_{i=1}^{I}, \tag{203}$$

a set of Gaussians in a common $\mathbb{R}^M$, one per waveform, encoding amplitude estimates and their uncertainty. Downstream tasks — clustering, density modeling, mixture of factor analyzers — operate entirely in this fixed-dimensional latent space, where the variable-length origin of the data is no longer a complication.

1.6. Summary

PRISM learns a shared, global kernel-induced basis from a collection of independent variable-length time series by maximizing the collapsed objective

$$\max_{\mathbf{Z}, \theta, \sigma^2} \sum_{i=1}^{I} \mathcal{L}_i(\mathbf{Z}, \theta, \sigma^2), \tag{204}$$

without storing or updating per-waveform variational parameters. After training, each waveform is projected to a Gaussian in a fixed-dimensional latent space (201), and downstream modeling proceeds entirely there.

The collapse step is not an implementation detail. Without it, shared basis learning would require maintaining and synchronizing per-waveform Gaussian states across minibatches, at a memory cost that grows with I. With it, the learning problem reduces to optimizing $\mathbf{Z}$, θ, and σ^2, and the per-example representations are assembled on demand from the trained basis alone.

Algorithm 2. t-PRISM

PRISM assumes Gaussian residuals. That is often too brittle in practice. In the alignment pipeline of Chapter 5, for example, the observations sent to t-PRISM are not raw DGF waveforms but inter-GCI period measurements: a single detector failure, silent gap, or irregular closure can create a period observation far away from the local pitch trend. More generally, any irregular functional series may contain isolated observations that should be downweighted rather than fitted at full strength.

t-PRISM replaces the Gaussian likelihood with a Student-t, whose heavier tails allow the model to assign large residuals a modest probability rather than treating them as catastrophic evidence against the current fit. The key is to do this without dismantling the collapsed structure of Algorithm 1. The approach is standard: represent the Student-t as a scale mixture of Gaussians via auxiliary precision variables λ_{in}, derive the ELBO over both the GP amplitudes and the precisions, and then observe that the GP part collapses exactly as before — now with a per-sample weighted noise covariance determined by the current λ_{in} estimates. The λ variables themselves have conjugate closed-form updates. The result is an alternating scheme where each minibatch runs a small inner loop of cheap precision updates, then takes a gradient step on the global parameters, with no large per-waveform state persisting between minibatches.

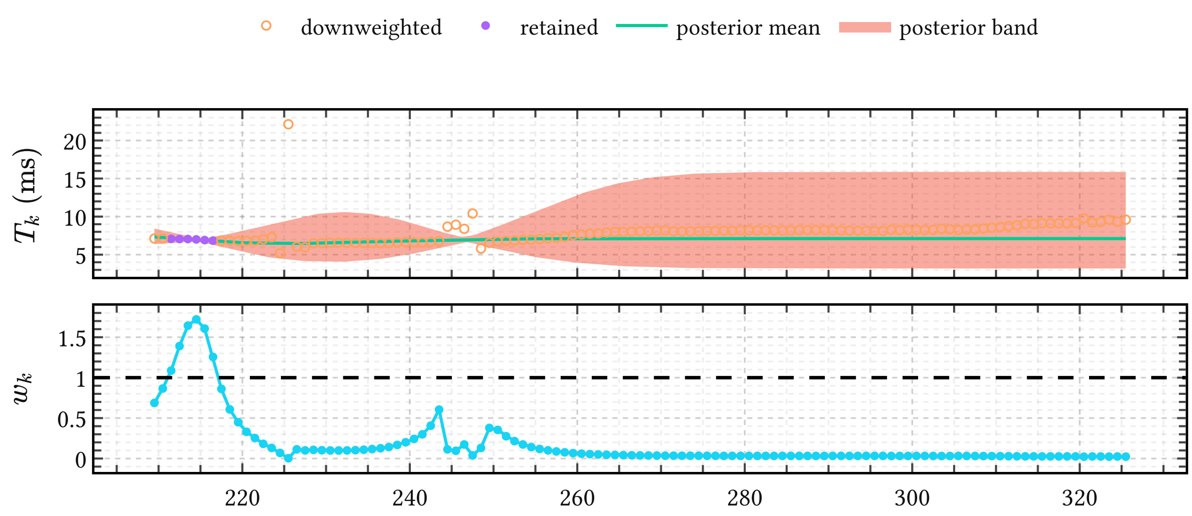

Figure 28 | **t-PRISM on an inter-GCI period track.** The top panel shows a local EGIFA period sequence together with the t-PRISM posterior mean and uncertainty band learned from the APLAWD pitch-track prior. The bottom panel shows the inferred precision weights $w_k = \mathbb{E}[\lambda_k]$ for the same excerpt. When the observed periods deviate sharply from the smooth local trend, their weights collapse below one and they are automatically downweighted rather than fitted.

2.1. Student-t as a scale mixture

The Student-t likelihood with degrees of freedom ν and scale σ^2 is recovered from the Normal-Gamma augmentation

$$\lambda_{in} \sim \texttt{Gamma}(\nu/2, \nu/2), \quad y_{in} \mid f_{in}, \lambda_{in} \sim \texttt{Normal}(f_{in}, \sigma^2/\lambda_{in}). \tag{205}$$

Marginalizing over λ_{in} recovers $\texttt{StudentT}(y_{in} \mid f_{in}, \nu, \sigma^2)$ exactly. Stacking the per-sample precisions into $\boldsymbol{\Lambda}_i = \operatorname{diag}\left(\lambda_{i1}, ..., \lambda_{iN_i}\right)$, the augmented likelihood for waveform i is Gaussian with a diagonal but heteroscedastic noise covariance:

$$\mathbf{y}_i \mid f_i, \boldsymbol{\Lambda}_i \sim \texttt{Normal}\left(f_{i(\mathbf{t}_i)}, \sigma^2 \boldsymbol{\Lambda}_i^{-1}\right). \tag{206}$$

High-precision samples (λ_{in} large) are trusted; low-precision ones (λ_{in} small) are down-weighted. The prior on each λ_{in} is $\texttt{Gamma}(\nu/2, \nu/2)$ with mean one, so in the absence of data the weights default to unity and the model reduces to PRISM.

2.2. t-PRISM ELBO

We introduce a mean-field variational family over the latent precision variables,

$$q_{i(\boldsymbol{\Lambda}_i)} = \prod_{n=1}^{N_i} q_{in}(\lambda_{in}), \quad q_{in}(\lambda_{in}) = \texttt{Gamma}(\alpha_{in}, \beta_{in}), \tag{207}$$

while the GP amplitudes $\mathbf{a}_i$ remain collapsed exactly as in Algorithm 1. The per-waveform ELBO, before collapsing, is

$$\mathcal{L}_i^t = \mathbb{E}_{q_i}[\log p(\mathbf{y}_i \mid f_i, \boldsymbol{\Lambda}_i)] - D_{\mathrm{KL}}\left(q_{i(\mathbf{u}_i)} \parallel p(\mathbf{u}_i)\right) - \sum_{n=1}^{N_i} D_{\mathrm{KL}}(q_{in}(\lambda_{in}) \parallel p(\lambda_{in})). \tag{208}$$

Define the expected precision and expected log-precision under q_{in}:

$$w_{in} = \mathbb{E}[\lambda_{in}] = \alpha_{in}/\beta_{in}, \quad \ell_{in} = \mathbb{E}[\log \lambda_{in}] = \psi(\alpha_{in}) - \log \beta_{in}, \tag{209}$$

where ψ is the digamma function. The expected log-likelihood expands as

$$\mathbb{E}[\log p(\mathbf{y}_i \mid f_i, \boldsymbol{\Lambda}_i)] = -\frac{1}{2} \sum_{n=1}^{N_i} \left(\log(2\pi) + \log \sigma^2 - \ell_{in} + \frac{w_{in}}{\sigma^2}\left((y_{in} - m_{in})^2 + v_{in}\right)\right), \tag{210}$$

where $m_{in} = \mathbb{E}\left[f_{i(t_{in})}\right]$ and $v_{in} = \operatorname{Var}\left(f_{i(t_{in})}\right)$ are the marginal moments of the sparse GP, to be computed from the collapsed GP with the current weights.

Collapsing the GP conditional on $\mathbf{W}_i$

Given fixed weights $\mathbf{W}_i = \operatorname{diag}\left(w_{i1}, ..., w_{iN_i}\right)$, the augmented likelihood (206) is Gaussian with noise covariance $\sigma^2 \mathbf{W}_i^{-1}$. This is exactly the setting of Algorithm 1 with a heteroscedastic but still diagonal noise, and the Titsias collapse applies without modification. Define the weighted design matrix and its Gram matrix,

$$\mathbf{A}_i^W = \mathbf{W}_i^{1/2}\boldsymbol{\Phi}_i/\sigma \in \mathbb{R}^{N_i \times M}, \qquad \mathbf{B}_i^W = \mathbf{I}_M + \left(\mathbf{A}_i^W\right)^\top \mathbf{A}_i^W \in \mathbb{R}^{M \times M}, \tag{211}$$

and the weighted sufficient statistic $\mathbf{v}_i^W = \left(\mathbf{A}_i^W\right)^\top \mathbf{W}_i^{1/2}\mathbf{y}_i \in \mathbb{R}^M$.[31] The collapsed Gaussian contribution to the ELBO is

$$\mathcal{L}_i^{\text{coll}}(\mathbf{W}_i) = \log \texttt{Normal}\left(\mathbf{y}_i \mid \mathbf{0}, \mathbf{Q}_i^W + \sigma^2 \mathbf{W}_i^{-1}\right) - \frac{1}{2\sigma^2}\operatorname{Tr}(\mathbf{W}_i(\mathbf{K}_{ii} - \mathbf{Q}_{ii})), \tag{212}$$

where $\mathbf{Q}_i^W = \boldsymbol{\Phi}_i\left(\mathbf{B}_i^W\right)^{-1}\boldsymbol{\Phi}_i^\top$ and the Woodbury and determinant-lemma identities from Algorithm 1 apply with $\mathbf{A}_i^W$ in place of $\mathbf{A}_i$. The marginal moments needed in (210) are recovered as

$$m_{in} = \varphi(t_{in})^\top \left(\mathbf{B}_i^W\right)^{-1}\mathbf{v}_i^W, \qquad v_{in} = k(t_{in}, t_{in}) - \varphi(t_{in})^\top \left(\mathbf{I} - \left(\mathbf{B}_i^W\right)^{-1}\right)\varphi(t_{in}). \tag{213}$$

Substituting everything, the per-waveform t-PRISM ELBO is

$$\mathcal{L}_i^t(\mathbf{W}_i, \ell_i) = \mathcal{L}_i^{\text{coll}}(\mathbf{W}_i) + \frac{1}{2}\sum_{n=1}^{N_i} \ell_{in} - \sum_{n=1}^{N_i} D_{\text{KL}}(q_{in}(\lambda_{in}) \parallel \texttt{Gamma}(\nu/2, \nu/2)). \tag{214}$$

The middle term arises because the collapsed Gaussian uses $\sigma^2 \mathbf{W}_i^{-1}$ as the noise covariance while the full ELBO requires $\mathbb{E}[\log\det(\boldsymbol{\Lambda}_i)] = \sum_n \ell_{in}$; keeping it ensures the bound is tight with respect to the expected Gaussian normalization. Dropping it would produce a subtly incorrect objective, which is where implementations often silently go wrong.

Local CAVI updates for the precision variables

Differentiating (214) with respect to α_{in} and β_{in}, or equivalently using the conjugacy of Gamma with the Normal-Gamma model, the optimal q_{in} is

$$\alpha_{in}^* = (\nu + 1)/2, \quad \beta_{in}^* = \frac{1}{2}\left(\nu + \left((y_{in} - m_{in})^2 + v_{in}\right)/\sigma^2\right). \tag{215}$$

These are the coordinate ascent updates: the shape is constant and the rate inflates exactly when the current residual $(y_{in} - m_{in})^2 + v_{in}$ is large relative to σ^2. Large residuals drive β_{in} up, which drives $w_{in} = \alpha_{in}^*/\beta_{in}^*$ down, which downweights that sample in the next collapse. This is the robust mechanism: the model learns to distrust samples it cannot explain, without any manual flagging of outliers.

2.3. Training

For a minibatch $\mathcal{B}$ of waveforms, the stochastic objective is

$$\hat{\mathcal{L}}^{\text{t-PRISM}} = I/\ |\mathcal{B}| \sum_{i \in \mathcal{B}} \mathcal{L}_i^t. \tag{216}$$

For each waveform i in the minibatch, the updates are:

[31] In the unweighted case $\mathbf{W}_i = \mathbf{I}$ these reduce to $\mathbf{A}_i, \mathbf{B}_i, \mathbf{v}_i$ from Algorithm 1 exactly.

1. initialize $w_{in} = 1$ for all n (unit weights, equivalent to PRISM),
2. repeat for a small fixed number of steps:
   - run the weighted collapsed GP computation (211) with current $\mathbf{W}_i$, obtaining (m_{in}, v_{in}) via (213),
   - update $(\alpha^*_{in}, \beta^*_{in})$ via (215) and refresh (w_{in}, ℓ_{in}) via (209),
3. evaluate $\mathcal{L}^t_i$ (214).

Then take a gradient step in the global parameters $(\mathbf{Z}, \theta, \sigma^2, \nu)$. The per-example Gamma parameters $(\alpha_{in}, \beta_{in})$ are local to each minibatch step and discarded afterwards; no per-waveform state accumulates across iterations.

2.4. Projection

After training, each waveform is projected to a Gaussian over whitened amplitudes exactly as in Algorithm 1, but using the final converged weights $\mathbf{W}_i$ from the inner loop. The posterior is

$$\mathbf{a}_i \mid \mathbf{y}_i \sim \texttt{Normal}(\mathbf{m}_i, \mathbf{S}_i), \quad \mathbf{S}_i = \left(\mathbf{B}_i^W\right)^{-1}, \quad \mathbf{m}_i = \left(\mathbf{B}_i^W\right)^{-1} \mathbf{v}_i^W, \tag{217}$$

which again reuses the quantities already computed during the inner loop. The PRISM representation $\{(\mathbf{m}_i, \mathbf{S}_i)\}_{i=1}^I$ is therefore the same fixed-dimensional probabilistic object as before, now obtained from a robust fit.

2.5. Summary

t-PRISM extends PRISM by replacing the Gaussian observation model with a Student-t, implemented through per-sample Gamma precision variables λ_{in}. Given these variables, the GP collapse of Algorithm 1 applies without change, with a weighted noise covariance $\sigma^2 \mathbf{W}_i^{-1}$ in place of $\sigma^2 \mathbf{I}$. The λ_{in} variables have closed-form conjugate updates (215) that automatically downweight samples with large residuals, requiring no outer iteration beyond a small fixed-count inner loop per minibatch. Global parameters $(\mathbf{Z}, \theta, \sigma^2, \nu)$ are then updated by a standard gradient step, and per-example states are discarded. In Chapter 5 the weights $w_{in} = \mathbb{E}[\lambda_{in}]$ are used directly as robust inlier scores on period trajectories, which is what turns t-PRISM from a generic robust GP into a voiced-group selector.

Algorithm 3. q-GPLVM

PRISM and PRISM-VFF turn variable-length waveforms into Gaussians in a common coefficient space $\mathbb{R}^M$: each waveform n becomes a posterior $\mathbf{a}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n)$ encoding how it decomposes onto the shared basis. This is useful for fixed-dimensional modeling, but the collection $\{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^N$ is not a generative model. It is still tied to the training set, and scoring a new waveform against it requires keeping all N representatives around.

q-GPLVM compresses this collection into a compact generative prior. The idea is to discover the low-dimensional manifold that the coefficient vectors inhabit using a Bayesian GP latent variable model (Titsias & Lawrence, 2010), then quantize the variational posterior into a small Gaussian mixture, and push each component back to coefficient space through the learned GP map. The result is a mixture of K Gaussians in $\mathbb{R}^M$ that summarizes the training distribution and provides a proper prior for downstream inference without reference to individual training waveforms.

The name refers to the quantization step: the GPLVM posterior, which lives on a continuous manifold, is discretized into a finite mixture. This is the second of two quantizations in the pipeline from waveform to prior. The first was PRISM itself, which quantized a GP into a finite-rank BLR (Algorithm 1). Both exploit the same principle: marginalizing amplitudes in a linear model is analytically tractable and automatically regularizes, in what (Jaynes, 2003, Sec. 17.8.1) calls the "Bayesian safety device."

3.1. Two levels of latent space

The q-GPLVM pipeline involves two distinct reductions, and keeping them separate conceptually avoids confusion.

Level 1: PRISM maps each waveform to a Gaussian in $\mathbb{R}^M$. The feature map $\varphi(\tau)$ ((191) in Algorithm 1 for the spectral variant) defines a linear synthesis model

$$g(\tau) \approx \varphi(\tau)^\top \mathbf{a}, \quad \mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}), \tag{218}$$

and the posterior $\mathbf{a} \mid \mathbf{y} \sim \texttt{Normal}(\mathbf{m}, \mathbf{S})$ is the PRISM representation. This is a linear dimensionality reduction: from a variable-length time series to a fixed M-dimensional vector with calibrated uncertainty.[32]

Level 2: Bayesian GPLVM discovers nonlinear structure among the coefficient vectors by mapping them from a Q-dimensional latent space ($Q \ll M$) through a GP. This is a nonlinear dimensionality reduction: from $\mathbb{R}^M$ to $\mathbb{R}^Q$, with the GP encoding the manifold geometry.[33] q-

[32]In terms of GP approximation, PRISM is a *quantization* of the GP prior to a rank-M BLR. The infinite-dimensional function space is projected onto the span of the learned basis, and the remaining degrees of freedom are absorbed by the observation noise.

[33]The composition Level 2 → Level 1 is a two-layer deep GP: one nonlinear layer (the GPLVM mapping $\mathbb{R}^Q \to \mathbb{R}^M$) followed by one linear layer (the PRISM synthesis $\mathbb{R}^M \to \mathcal{L}^2$). If time warping of the input coordinate τ is also learned nonlinearly (Chapter 5), that adds another nonlinear layer, bringing the total to three (Damianou et al., 2014). Unlike typical deep GPs, we avoid propagating full distributional uncertainty

GPLVM operates entirely at Level 2 and its output is a mixture model in the coefficient space of Level 1, which then lifts to waveform space by simple matrix multiplication.

3.2. The Bayesian GPLVM

We now set up the model that operates on the PRISM coefficient vectors. The theoretical foundation is the Bayesian GP latent variable model of Titsias & Lawrence (2010), which we adapt to our notation and briefly develop below.

Observations and notation

Collect the PRISM posterior means into a data matrix $\mathbf{Y} \in \mathbb{R}^{N \times D}$ where N is the number of training waveforms and $D = M$ is the coefficient dimension. Each row $\mathbf{y}_n = \mathbf{m}_n^\top$ is the PRISM posterior mean for waveform n.[34]

Model

For each output dimension $d = 1, ..., D$, introduce an independent latent function $f_d(\cdot)$ with a shared GP prior

$$f_d(\cdot) \sim \texttt{GaussianProcess}(0, k(\cdot, \cdot)), \tag{219}$$

and an observation model

$$y_{nd} = f_d(\mathbf{x}_n) + \eta_{nd}, \quad \eta_{nd} \sim \texttt{Normal}(0, \beta^{-1}), \tag{220}$$

where $\mathbf{x}_n \in \mathbb{R}^Q$ is the latent coordinate of waveform n and β^{-1} is shared observation noise. The latent prior is isotropic:

$$p(\mathbf{x}_n) = \texttt{Normal}\big(\mathbf{0}, \mathbf{I}_Q\big). \tag{221}$$

The kernel is ARD squared exponential,

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2} \sum_{q=1}^{Q} \alpha_q \big(x_q - x_q'\big)^2\right), \tag{222}$$

with per-dimension inverse lengthscales α_q. When the model drives $\alpha_q \to 0$ during training, dimension q is effectively switched off, providing automatic selection of the intrinsic latent dimensionality (Titsias & Lawrence, 2010).

Variational inference

Integrating out the latent variables $\mathbf{x}_n$ from the marginal likelihood is intractable because they appear nonlinearly inside the kernel matrix. Titsias & Lawrence (2010) resolve this by

through the nonlinear layers and instead use moment matching, which amounts to a local linearization of the forward map.

[34] The PRISM posterior also provides per-waveform covariances $\mathbf{S}_n$ that encode projection uncertainty. These are not used in the GPLVM model itself, but are exploited later during quantization via extreme deconvolution.

introducing P inducing inputs $\mathbf{Z} \in \mathbb{R}^{P \times Q}$ in the latent space and a factorized variational posterior over the latent coordinates,

$$q(\mathbf{X}) = \prod_{n=1}^{N} \texttt{Normal}(\mathbf{x}_n \mid \boldsymbol{\mu}_n, \mathbf{S}_n^x), \tag{223}$$

where $\boldsymbol{\mu}_n \in \mathbb{R}^Q$ and $\mathbf{S}_n^x = \operatorname{diag}\left(S_{n1}^x, ..., S_{nQ}^x\right)$ is a diagonal covariance. The variational bound has the standard form

$$\mathcal{F}(q) = \tilde{\mathcal{F}}(q) - D_{\mathrm{KL}(q(\mathbf{X}) \,\|\, p(\mathbf{X}))}, \tag{224}$$

where $\tilde{\mathcal{F}}(q)$ is the expected log-likelihood under the sparse GP approximation and the KL term is analytic since both $q(\mathbf{X})$ and $p(\mathbf{X})$ are factorized Gaussians.

The variational parameters $\{\boldsymbol{\mu}_n, \mathbf{S}_n^x\}_{n=1}^N$, the inducing inputs $\mathbf{Z}$, and the kernel and noise parameters $\left\{\sigma_f^2, \alpha_1, ..., \alpha_Q, \beta\right\}$ are all optimized jointly by gradient ascent on $\mathcal{F}$. The inducing variables $\mathbf{u}_d = f_d(\mathbf{Z})$ for each output dimension d are eliminated analytically, in exactly the same spirit as the Titsias collapse in Algorithm 1, leaving an objective that depends on the latent variables only through three kernel expectation statistics (Titsias & Lawrence, 2010).

The Ψ statistics

All tractable computation in the Bayesian GPLVM reduces to these expectations of kernel quantities under $q(\mathbf{x}_n)$:

$$\begin{aligned}
\psi_0 &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}[k(\mathbf{x}_n, \mathbf{x}_n)], \\
[\boldsymbol{\Psi}_1]_{np} &= \mathbb{E}_{q(\mathbf{x}_n)}\left[k\left(\mathbf{x}_n, \mathbf{z}_p\right)\right], \\
[\boldsymbol{\Psi}_2]_{pp'} &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}\left[k\left(\mathbf{x}_n, \mathbf{z}_p\right) k\left(\mathbf{x}_n, \mathbf{z}_{p'}\right)\right],
\end{aligned} \tag{225}$$

with $\boldsymbol{\Psi}_1 \in \mathbb{R}^{N \times P}$ and $\boldsymbol{\Psi}_2 \in \mathbb{R}^{P \times P}$. Note that $\boldsymbol{\Psi}_2$ is *not* $\boldsymbol{\Psi}_1^\top \boldsymbol{\Psi}_1$: the integration couples the two kernel evaluations through the shared uncertain input $\mathbf{x}_n$.

For the ARD RBF kernel (222) with $q(\mathbf{x}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \mathbf{S}_n^x)$, all three statistics admit closed forms. The first statistic is trivial: $\psi_0 = N\sigma_f^2$ since $k(\mathbf{x}, \mathbf{x}) = \sigma_f^2$ for any $\mathbf{x}$. For $\boldsymbol{\Psi}_1$, the Gaussian integral yields

$$[\boldsymbol{\Psi}_1]_{np} = \sigma_f^2 \prod_{q=1}^{Q} \left(\alpha_q S_{nq}^x + 1\right)^{-1/2} \exp\left(-\frac{\alpha_q \left(\mu_{nq} - z_{pq}\right)^2}{2\left(\alpha_q S_{nq}^x + 1\right)}\right), \tag{226}$$

which is a kernel evaluation at $\boldsymbol{\mu}_n$ with inflated lengthscales $\alpha_q^{-1} + S_{nq}^x$. When $\mathbf{S}_n^x \to \mathbf{0}$ (no input uncertainty), this recovers the standard kernel evaluation $k\left(\boldsymbol{\mu}_n, \mathbf{z}_p\right)$. $\boldsymbol{\Psi}_2$ has a similar closed form involving the midpoint and distance between inducing pairs $(\mathbf{z}_p, \mathbf{z}_{p'})$ (Titsias & Lawrence, 2010).

Two properties that matter downstream: first, all three statistics decompose as sums over n, so stochastic minibatching is straightforward (Lalchand et al., 2022); second, the same closed forms apply whenever the input distribution is Gaussian with *any* covariance, not only the diagonal $\mathbf{S}_n^x$ used in training. This second property is what makes the push-forward tractable.

3.3. Quantization

After training, the Bayesian GPLVM provides a variational posterior $q(\mathbf{x}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \mathbf{S}_n^x)$ for each training waveform. The aggregated posterior $q(\mathbf{x}) = 1/N \sum_{n=1}^{N} q(\mathbf{x}_n)$ is already a generative model in the latent space, but with N components it encodes the entire training set. The quantization step replaces this N-component mixture by a compact K-component approximation ($K \ll N$):

$$q(\mathbf{x}) \approx \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \quad \sum_{k=1}^{K} \pi_k = 1, \tag{227}$$

where $\mathbf{c}_k \in \mathbb{R}^Q$ and $\mathbf{C}_k \in \mathbb{R}^{Q \times Q}$ are the mean and full covariance of component k. Full covariances are retained because local curvature and anisotropy in the latent space are precisely the structure we want to preserve.

Extreme deconvolution

The variational posteriors $q(\mathbf{x}_n)$ are not point estimates: each carries a covariance $\mathbf{S}_n^x$ that quantifies genuine uncertainty about the latent position. A standard GMM fit to the posterior means $\boldsymbol{\mu}_n$ alone would ignore this uncertainty and overfit to noise in the mean estimates. *Extreme deconvolution* (XDGMM) (Bovy et al., 2011) is a variant of EM for Gaussian mixtures that accounts for known, heterogeneous measurement uncertainties on the inputs. It fits (227) to the collection $\{(\boldsymbol{\mu}_n, \mathbf{S}_n^x)\}_{n=1}^{N}$ by treating each $\boldsymbol{\mu}_n$ as a noisy observation of the true latent coordinate with known covariance $\mathbf{S}_n^x$.

The background component

Training data inevitably contain outliers: waveforms that PRISM projected poorly, or that lie outside the main phonation modes. To prevent these from distorting the learned mixture, we initialize component $k = 0$ as a broad background Gaussian whose mean and covariance are set to the empirical mean and covariance of all $\boldsymbol{\mu}_n$. The remaining $K - 1$ components are initialized from k-means on the posterior means. During EM, the background component absorbs waveforms that do not fit any well-defined cluster, and its weight π_0 is generally observed to become negligible.

The output of the quantization step is a small parametric object $\{\pi_k, \mathbf{c}_k, \mathbf{C}_k\}_{k=1}^{K}$ that stands in for the entire training collection.

3.4. Push-forward

We now have a GMM in the Q-dimensional latent space and a trained GP map from $\mathbb{R}^Q$ to $\mathbb{R}^D$. To obtain a generative model in coefficient space, each GMM component must be pushed through the GP.

Recovered inducing parameters

During training, the inducing variables $\mathbf{u}_d$ were eliminated analytically from the bound. For the push-forward we need them back. From the optimal variational posterior (Titsias, 2009) applied within the GPLVM,

$$\begin{aligned}\hat{\mathbf{S}}_u &= \mathbf{K}_{ZZ}\big(\mathbf{K}_{ZZ} + \beta\boldsymbol{\Psi}_2^{\mathrm{tr}}\big)^{-1}\mathbf{K}_{ZZ},\\ \hat{\mathbf{m}}_d &= \beta\,\mathbf{K}_{ZZ}\big(\mathbf{K}_{ZZ} + \beta\boldsymbol{\Psi}_2^{\mathrm{tr}}\big)^{-1}\big[\boldsymbol{\Psi}_1^{\mathrm{tr}}\big]^{\top}\mathbf{y}_d,\end{aligned} \tag{228}$$

where $\boldsymbol{\Psi}_1^{\mathrm{tr}}$, $\boldsymbol{\Psi}_2^{\mathrm{tr}}$ are the training-time Ψ statistics and $\mathbf{y}_d \in \mathbb{R}^N$ is the d-th column of $\mathbf{Y}$. The posterior covariance $\hat{\mathbf{S}}_u \in \mathbb{R}^{P\times P}$ is shared across all output dimensions; the posterior means are collected into $\widehat{\mathbf{M}} = [\hat{\mathbf{m}}_1, ..., \hat{\mathbf{m}}_D] \in \mathbb{R}^{P\times D}$.

Moment matching

For one latent component $\mathbf{x} \sim \mathtt{Normal}(\mathbf{c}, \mathbf{C})$, the induced distribution of $\mathbf{y} = (f_1(\mathbf{x}), ..., f_D(\mathbf{x}))^{\top}$ under the GP is non-Gaussian due to the nonlinearity of the map. The standard approach is moment matching: choose the Gaussian $\mathtt{Normal}\big(\boldsymbol{\mu}_y, \mathbf{S}_y\big)$ whose first two moments equal those of the true push-forward.

The required expectations are the same Ψ statistics of (225), now evaluated at the GMM component $(\mathbf{c}, \mathbf{C})$ rather than at a variational posterior $(\boldsymbol{\mu}_n, \mathbf{S}_n^x)$. Since the ARD RBF formula (226) remains valid for any Gaussian input, including one with full covariance $\mathbf{C}$, no new derivations are needed.[35] Write

$$\begin{aligned}&\psi_0 = \mathbb{E}_{\mathbf{x}}[k(\mathbf{x}, \mathbf{x})],\\ &\boldsymbol{\psi}_1 \in \mathbb{R}^P \text{ with } [\boldsymbol{\psi}_1]_p = \mathbb{E}_{\mathbf{x}}\big[k\big(\mathbf{x}, \mathbf{z}_p\big)\big],\\ &\boldsymbol{\Psi}_2 \in \mathbb{R}^{P\times P} \text{ with } [\boldsymbol{\Psi}_2]_{pp'} = \mathbb{E}_{\mathbf{x}}\big[k\big(\mathbf{x}, \mathbf{z}_p\big)k\big(\mathbf{x}, \mathbf{z}_{p'}\big)\big],\end{aligned} \tag{229}$$

for the per-component statistics, suppressing the dependence on $(\mathbf{c}, \mathbf{C})$.

3.4.2.1. Mean

The predictive mean for output d at a known point $\mathbf{x}$ is $\hat{\mathbf{m}}_d^{\top}\mathbf{K}_{ZZ}^{-1}\mathbf{k}_Z(\mathbf{x})$, where $\mathbf{k}_Z(\mathbf{x}) \in \mathbb{R}^P$ is the cross-covariance vector to the inducing inputs. Taking the expectation over $\mathbf{x} \sim \mathtt{Normal}(\mathbf{c}, \mathbf{C})$ replaces $\mathbf{k}_Z(\mathbf{x})$ by its mean $\boldsymbol{\psi}_1$:

[35] With a full $\mathbf{C}$, the product over dimensions in (226) generalizes to the matrix expression $|\boldsymbol{\Lambda}^{-1}\mathbf{C} + \mathbf{I}|^{-1/2}\exp\Big(-1/2\big(\mathbf{c} - \mathbf{z}_p\big)^{\top}(\boldsymbol{\Lambda} + \mathbf{C})^{-1}\big(\mathbf{c} - \mathbf{z}_p\big)\Big)$, where $\boldsymbol{\Lambda} = \mathrm{diag}\big(\alpha_1^{-1}, ..., \alpha_Q^{-1}\big)$.

$$\boldsymbol{\mu}_y = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} \boldsymbol{\psi}_1 \in \mathbb{R}^D. \tag{230}$$

3.4.2.2. Covariance

The law of total variance gives

$$\mathbf{S}_y = \underbrace{\mathbb{E}_{\mathbf{x}}[\mathtt{Cov}(\mathbf{y} \mid \mathbf{x})]}_{\text{averaged predictive variance}} + \underbrace{\mathtt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{y} \mid \mathbf{x}])}_{\text{input uncertainty}}. \tag{231}$$

The first term averages the GP predictive variance (which is the same for every output d since all outputs share the kernel) under the input distribution. Since outputs are conditionally independent given $\mathbf{x}$, this term is isotropic:

$$\mathbb{E}_{\mathbf{x}}[\mathtt{Cov}(\mathbf{y} \mid \mathbf{x})] = \gamma\, \mathbf{I}_D, \tag{232}$$

where

$$\gamma = \psi_0 + \beta^{-1} - \mathrm{Tr}(\mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2) + \mathrm{Tr}\Big(\mathbf{K}_{ZZ}^{-1} \hat{\mathbf{S}}_u \mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2\Big). \tag{233}$$

The four contributions are respectively: the prior variance at $\mathbf{x}$, the observation noise, minus the variance explained by the inducing inputs, plus the posterior uncertainty on those inducing inputs.

The second term in (231) captures how the predictive mean varies as $\mathbf{x}$ moves over the component. It introduces cross-output correlations via

$$\mathtt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{y} \mid \mathbf{x}]) = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2 - \boldsymbol{\psi}_1 \boldsymbol{\psi}_1^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}, \tag{234}$$

which has rank at most P.

Combining, the moment-matched covariance is

$$\mathbf{S}_y = \gamma\, \mathbf{I}_D + \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2 - \boldsymbol{\psi}_1 \boldsymbol{\psi}_1^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}. \tag{235}$$

The message is that the push-forward covariance decomposes into an isotropic term (from averaged predictive noise) and a low-rank term (from input uncertainty). Only the latter carries information about the manifold geometry of the latent space.

The mixture weights

The weights π_k of the latent GMM (227) are a prior on latent space. Under the push-forward, they remain the mixture weights in coefficient space:

$$p(\mathbf{y}) \approx \sum_{k=1}^{K} \pi_k\, \mathtt{Normal}(\mathbf{y} \mid \boldsymbol{\mu}_k, \mathbf{S}_k), \tag{236}$$

where $(\boldsymbol{\mu}_k, \mathbf{S}_k)$ is the moment-matched push-forward of the k-th component via (230) and (235). The weights transform only under Bayesian conditioning on observed data, which happens downstream in IKLP, not here.

Composing with PRISM

The mixture (236) lives in the PRISM coefficient space $\mathbb{R}^D$. To obtain a prior over waveforms $\mathbf{g} \in \mathbb{R}^W$ evaluated on a grid of W time points, recall that PRISM synthesizes via the linear map

$$\mathbf{g} = \mathbf{\Phi a} + \varepsilon, \quad \varepsilon \sim \texttt{Normal}(\mathbf{0}, \sigma^2 \mathbf{I}_W), \tag{237}$$

where $\mathbf{\Phi} \in \mathbb{R}^{W \times M}$ is the design matrix from (192) in Algorithm 1. Because this step is linear, pushing a Gaussian in coefficient space through it preserves Gaussianity:

$$\mathbf{g} \mid k \sim \texttt{Normal}(\mathbf{\Phi} \boldsymbol{\mu}_k, \quad \mathbf{\Phi S}_k \mathbf{\Phi}^\top + \sigma^2 \mathbf{I}_W), \tag{238}$$

and the full prior over waveforms is

$$p(\mathbf{g}) \approx \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{g} \mid \mathbf{\Phi} \boldsymbol{\mu}_k, \mathbf{\Phi S}_k \mathbf{\Phi}^\top + \sigma^2 \mathbf{I}_W). \tag{239}$$

Low-rank structure

The covariance $\mathbf{\Phi S}_k \mathbf{\Phi}^\top$ has rank at most $\min(M, \operatorname{rank}(\mathbf{S}_k))$. From (235), $\mathbf{S}_k$ is a rank-P correction of a scalar multiple of the identity, so the effective rank of the nontrivial part is at most P, which is typically comparable to Q. Each component of (239) is therefore a low-rank Gaussian prior on waveforms plus a diagonal nugget from the PRISM and GPLVM observation noise terms. The cascaded rank reduction is worth emphasizing: PRISM projects from the infinite-dimensional function space to rank M; the GPLVM further reduces to an effective rank of at most Q (selected automatically by the ARD mechanism); quantization preserves this by working within the Q-dimensional latent space; and the push-forward inherits the rank bound.[36]

For DGF waveforms, the ARD mechanism typically selects Q in the range 3 to 8. Each mixture component then defines a waveform prior supported on a subspace of dimension at most Q out of a potentially much larger W. This is genuine dimensionality reduction, and the entire computation required to produce it is amortized at training time.

3.5. Summary

q-GPLVM is a three-stage compression pipeline that converts a collection of PRISM coefficient posteriors into a compact mixture prior over waveforms.

[36]More specialized analyses of pushing Gaussians through nonlinear maps, including higher-order corrections beyond moment matching, are given by (Zhang & Shin, 2021) and (Zhang & Shin, 2022).

1. A Bayesian GPLVM (Titsias & Lawrence, 2010) discovers the low-dimensional manifold structure of the PRISM coefficient vectors, producing a variational posterior $q(\mathbf{x}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \mathbf{S}_n^x)$ for each training waveform in a Q-dimensional latent space with Q selected automatically by ARD.
2. Extreme deconvolution (Bovy et al., 2011) quantizes the aggregated posterior into a K-component GMM in this latent space, respecting the per-waveform uncertainties via an errors-in-variables formulation. A broad background component absorbs outliers.
3. Moment matching pushes each GMM component through the trained GP map using the same Ψ statistics that appear in the GPLVM bound, giving a Gaussian in coefficient space ((230), (235)). Because PRISM synthesis is linear, the result extends to a mixture of low-rank Gaussians over waveforms (239).

The entire computation is amortized: once trained, the pipeline outputs a fixed set of K component parameters $\{\pi_k, \boldsymbol{\mu}_k, \mathbf{S}_k\}_{k=1}^K$ in coefficient space. These are the Gaussian clusters consumed by Chapter 6, where the means are folded into second-moment matrices and a stationary-to-nonstationary interpolation produces the kernel bank for IKLP.

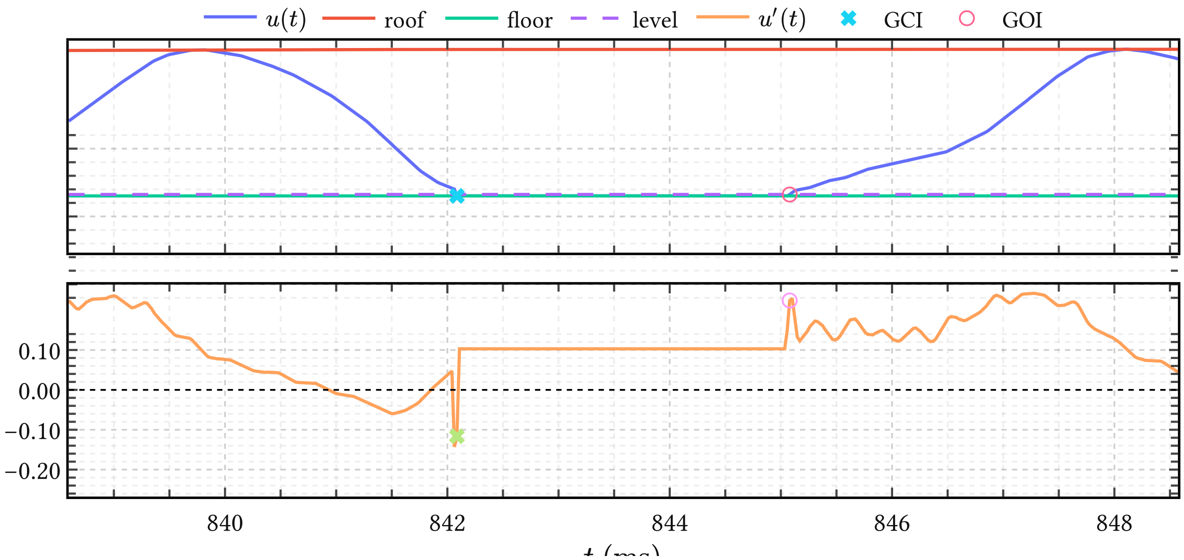

Figure 29 | **LBGID on one detected low-flow region.** The top panel shows the glottal flow together with the adaptive roof, floor, and relative level threshold. The shaded interval is one candidate low-flow region. The bottom panel shows the derivative inside the same window, with the selected negative-going GCI and later positive-going GOI marked explicitly. The detector therefore first localizes the closed phase by amplitude and only then resolves the instant pair by dynamics.

Algorithm 4. LBGID

LBGID stands for level-based glottal instant detection.

Within the BNGIF framework, glottal closure and opening instants serve primarily as *alignment anchors* for the closed-phase analysis used downstream in IKLP. The task is therefore not to reconstruct vocal fold biomechanics or to match a stylized glottal model, but to extract consistent temporal landmarks from the airflow signals $u(t)$ produced by physics-based simulators such as VocalTractLab. This is how the algorithm is used in Chapter 5, where the detected events define the registration map to shared cycle time. Later, in Chapter 7, the same events support the closed-phase treatment inside the final inverse-filtering model.

This changes the detection problem in an important way. In a synthesizer, glottal closure and opening are not predefined symbolic events; they are emergent consequences of biomechanical dynamics, and the closed phase appears not as a hard discontinuity but as a soft, context-dependent region of suppressed airflow embedded in an otherwise continuous signal. Traditional glottal instant detectors assume quasi-periodicity, predefined waveform shapes, or spectral heuristics derived from acoustic signals. None of these assumptions transfer cleanly to synthetic airflow streams.

The guiding intuition behind LBGID is deliberately simple: closure corresponds to *low airflow*, and glottal instants correspond to strong opposing changes in airflow dynamics flanking those low-flow regions. The algorithm therefore first identifies where the flow is small and then extracts the most energetic dynamic transitions within those regions.

4.1. Adaptive envelope normalization

A central difficulty with airflow signals is that absolute amplitude is rarely interpretable on its own. Slow baseline drift, variation in subglottal pressure, and simulator-specific scaling can all alter amplitude without changing the underlying temporal structure. Rather than attempting to remove these effects explicitly, LBGID constructs a local coordinate system by tracking adaptive upper and lower envelopes, obtained by detecting peaks and troughs and interpolating between them. These define a smoothly varying local amplitude range,

$$A(t) = \mathrm{roof}(t) - \mathrm{floor}(t), \tag{240}$$

and all subsequent decisions are made relative to it. The signal is interpreted relationally: what matters is not the absolute airflow value at any point, but where it sits within its current local dynamic range.

4.2. Level-based closed phase detection

With the local amplitude frame in hand, candidate closed phases are identified via a relative threshold,

$$\mathrm{level}(t) = \mathrm{floor}(t) + \alpha(\mathrm{roof}(t) - \mathrm{floor}(t)), \tag{241}$$

for a small constant α, and points satisfying $u(t) \leq \mathrm{level}(t)$ are treated as belonging to a candidate closed phase. This yields contiguous low-flow segments that naturally partition the signal into bounded candidate intervals. The deliberate choice to isolate the closure regime first — before looking for any dynamics — is what distinguishes LBGID from direct peak or derivative detectors: the search for instants is local and physically constrained from the outset.

4.3. Dynamic pairing inside the closed phase

Within each low-flow segment, LBGID searches for a pair of opposing dynamic events. The derivative $u'(t)$ is computed by finite differences. Rather than identifying a single extremum, the algorithm evaluates all pairs of candidate points (i, j) within the segment by the signed product

$$E(i, j) = u'(i)\, u'(j). \tag{242}$$

Pairs with $E(i, j) < 0$ correspond to opposite-sign slopes, i.e., one point is on an ascending transition and the other on a descending one. Among these, the pair maximizing $|E(i, j)|$ is selected as the most energetically opposed dynamic interaction within the closed phase, and its two members are reported as the glottal closure instant (GCI) and glottal opening instant (GOI). Because the search is causal, the selected GCI is always the negative-going event and the selected GOI the later positive-going one, matching the ordering used in the registration pipeline of Chapter 5.

No assumption is made about the waveform morphology between them.

4.4. Closing remarks

LBGID performs three transformations in sequence: convert absolute airflow into a local amplitude frame via envelope tracking, identify the physical regime where closure must occur via the level threshold, and detect the onset and offset of that closure as the most energetically opposed derivative pair within the candidate segment. The result is a pair of instants that emerge from the airflow's own amplitude and dynamic structure rather than from any externally imposed waveform model.

This makes LBGID well-suited to physics-based simulation environments, where the closed phase is gradual and the absence of a sharp GCI spike would cause direct peak detectors to fail. In the BNGIF context it provides a stable alignment mechanism that is robust to the variation in phonatory effort, voice quality, and fundamental frequency present in the VocalTractLab data.

Bibliography

Airaksinen, M., Raitio, T., Story, B., & Alku, P. (2014). Quasi Closed Phase Glottal Inverse Filtering Analysis With Weighted Linear Prediction. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *22*(3), 596–607. https://doi.org/10.1109/TASLP.2013.2294585

Alku, P. (1992). *Glottal Wave Analysis with Pitch Synchronous Iterative Adaptive Inverse Filtering. 11.*

Alku, P. (2011). Glottal Inverse Filtering Analysis of Human Voice Production — A Review of Estimation and Parameterization Methods of the Glottal Excitation and Their Applications. *Sadhana*, *36*(5), 623–650. https://doi.org/10.1007/s12046-011-0041-5

Alku, P., Bäckström, T., & Vilkman, E. (2002). Normalized Amplitude Quotient for Parametrization of the Glottal Flow. *The Journal of the Acoustical Society of America*, *112*(2), 701–710. https://doi.org/10.1121/1.1490365

Alzamendi, G. A., & Schlotthauer, G. (2017). Modeling and Joint Estimation of Glottal Source and Vocal Tract Filter by State-Space Methods. *Biomedical Signal Processing and Control*, *37*, 5–15. https://doi.org/10.1016/j.bspc.2016.12.022

Ananthapadmanabha, T. V., & Fant, G. (1982). Calculation of True Glottal Flow and Its Components. *Speech Communication*, *1*(3), 167–184. https://doi.org/10.1016/0167-6393(82)90015-2

Andrade-Miranda, G., Chatzipapas, K., Arias-Londoño, J. D., & Godino-Llorente, J. I. (2024, December). *GIRAFE: Glottal Imaging Dataset for Advanced Segmentation, Analysis, and Facilitative Playbacks Evaluation* (Issue arXiv:2412.15054). arXiv. https://doi.org/10.48550/arXiv.2412.15054

Antsaklis, P. J., & Michel, A. N. (2006). *Linear Systems*. Springer Science & Business Media.

Atal, B. S., & Hanauer, S. L. (1971). Speech Analysis and Synthesis by Linear Prediction of the Speech Wave. *The Journal of the Acoustical Society of America*, *50*(2B), 637–655.

Auvinen, H., Raitio, T., Airaksinen, M., Siltanen, S., Story, B. H., & Alku, P. (2014). Automatic Glottal Inverse Filtering with the Markov Chain Monte Carlo Method. *Computer Speech & Language*, *28*(5), 1139–1155. https://doi.org/10.1016/j.csl.2013.09.004

Avhad, A., Li, Z., Wilson, A., Sayce, L., Chang, S., Rousseau, B., & Luo, H. (2022). Subject-Specific Computational Fluid-Structure Interaction Modeling of Rabbit Vocal Fold Vibration. *Fluids*, *7*(3), 97. https://doi.org/10.3390/fluids7030097

Becker, T., Jessen, M., & Grigoras, C. (2008,). Forensic Speaker Verification Using Formant Features and Gaussian Mixture Models. *Ninth Annual Conference of the International Speech Communication Association.*

Bleyer, I. R., Lybeck, L., Auvinen, H., Airaksinen, M., Alku, P., & Siltanen, S. (2017). Alternating Minimisation for Glottal Inverse Filtering. *Inverse Problems*, *33*(6), 65005. https://doi.org/10.1088/1361-6420/aa6eb8

Boersma, P. (2001). *Praat, a System for Doing Phonetics by Computer*. *5*(9), 341–345.

Bonastre, J.-F., Kahn, J., Rossato, S., & Ajili, M. (2015). Forensic Speaker Recognition: Mirages and Reality. *S. Fuchs/d*, 255.

Bovy, J., Hogg, D. W., & Roweis, S. T. (2011). Extreme Deconvolution: Inferring Complete Distribution Functions from Noisy, Heterogeneous and Incomplete Observations. *The Annals of Applied Statistics*, *5*(2B). https://doi.org/10.1214/10-AOAS439

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., & Wanderman-Milne, S. (2020). JAX: Composable Transformations of Python+ NumPy Programs, 2018. *URL Http://github. Com/google/jax*, *4*, 16.

Bretthorst, G. L. (1988). *Bayesian Spectrum Analysis and Parameter Estimation*.

Calvetti, D., & Somersalo, E. (2018). Inverse Problems: From Regularization to Bayesian Inference. *Wires Computational Statistics*, *10*(3), e1427. https://doi.org/10.1002/wics.1427

Cao, F., Vogel, A. P., Gharahkhani, P., & Renteria, M. E. (2025). Speech and Language Biomarkers for Parkinson's Disease Prediction, Early Diagnosis and Progression. *Npj Parkinson's Disease*, *11*(1), 57. https://doi.org/10.1038/s41531-025-00913-4

Chang-Sheng Yang, & Kasuya, H. (1998). Automatic Estimation of Formant and Voice Source Parameters Using a Subspace Based Algorithm. *Proceedings of the 1998 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP '98 (Cat. No.98CH36181)*, *2*, 941–944. https://doi.org/10.1109/ICASSP.1998.675421

Chen, C. J. (2016). *Elements of Human Voice*. WORLD SCIENTIFIC. https://doi.org/10.1142/9891

Chien, Y.-R., Mehta, D. D., Guenason, J., Zanartu, M., & Quatieri, T. F. (2017). Evaluation of Glottal Inverse Filtering Algorithms Using a Physiologically Based Articulatory Speech Synthesizer. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *25*(8), 1718–1730. https://doi.org/10.1109/TASLP.2017.2714839

Childers, D. G., & Hu, H. T. (1994). Speech Synthesis by Glottal Excited Linear Prediction. *The Journal of the Acoustical Society of America*, *96*(4), 2026–2036. https://doi.org/10.1121/1.411319

Cho, Y., & Saul, L. (2009). Kernel Methods for Deep Learning. *Advances in Neural Information Processing Systems*, *22*.

Cox, R. T. (1946). Probability, Frequency and Reasonable Expectation. *Physics*.

Cuevas, A., López, S., Mandic, D., & Tobar, F. (2021). Bayesian Autoregressive Spectral Estimation. *2021 IEEE Latin American Conference on Computational Intelligence (LA-CCI)*, 1–7. https://doi.org/10.1109/LA-CCI48322.2021.9769834

Damianou, A. C., Titsias, M. K., & Lawrence, N. D. (2014, September). *Variational Inference for Uncertainty on the Inputs of Gaussian Process Models* (Issue arXiv:1409.2287). arXiv.

Degottex, G. (2010). *Glottal Source and Vocal-Tract Separation* [Doctoral dissertation].

Degottex, G., Kane, J., Drugman, T., Raitio, T., & Scherer, S. (2014). COVAREP — A Collaborative Voice Analysis Repository for Speech Technologies. *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 960–964. https://doi.org/10.1109/ICASSP.2014.6853739

Deller, J. R. (1983). On the Time Domain Properties of the Two-Pole Model of the Glottal Waveform and Implications for LPC. *Speech Communication*, *2*(1), 57–63.

Diaconis, P., & Freedman, D. (1999). Iterated Random Functions. *SIAM Review*, *41*(1), 45–76.

Doval, B., D'Alessandro, C., & Henrich, N. (2006). The Spectrum of Glottal Flow Models. *Acta Acustica United with Acustica*, *92*(6), 1026–1046.

Drugman, T., Alku, P., Alwan, A., & Yegnanarayana, B. (2019a, December). *Glottal Source Processing: From Analysis to Applications* (Issue arXiv:1912.12604). arXiv.

Drugman, T., Bozkurt, B., & Dutoit, T. (2011). Causal–Anticausal Decomposition of Speech Using Complex Cepstrum for Glottal Source Estimation. *Speech Communication*, *53*(6), 855–866. https://doi.org/10.1016/j.specom.2011.02.004

Drugman, T., Bozkurt, B., & Dutoit, T. (2019b). A Comparative Study of Glottal Source Estimation Techniques. *Arxiv:2001.00840 [Cs, Eess]*.

Ekström, A. G., Moran, S., Sundberg, J., & Lameira, A. (2023). *PREQUEL: Supervised Phonetic Approaches to Analyses of Great Ape Quasi-Vowels*.

Fant, G. (1960). *Acoustic Theory of Speech Production*. Mouton.

Fant, G. (1979). Glottal Source and Excitation Analysis. *STL-QPSR*, *1*(1979), 85–107.

Fant, G. (1995). The LF-model Revisited. Transformations and Frequency Domain Analysis. *Speech Trans. Lab. Q. Rep., Royal Inst. Of Tech. Stockholm*, *2*(3), 40.

Fant, G., Liljencrants, J., & Lin, Q.-g. (1985). A Four-Parameter Model of Glottal Flow. *STL-QPSR*, *4*(1985), 1–13.

Flanagan, J. L. (1965). *Speech Analysis Synthesis and Perception*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-00849-2

Frigola, R. (2015). *Bayesian Time Series Learning with Gaussian Processes* [Doctoral dissertation].

Fu, Q., & Murphy, P. (2006). Robust Glottal Source Estimation Based on Joint Source-Filter Model Optimization. *IEEE Transactions on Audio, Speech and Language Processing*, *14*(2), 492–501. https://doi.org/10.1109/TSA.2005.857807

Fujisaki, H., & Ljungqvist, M. (1986). Proposal and Evaluation of Models for the Glottal Source Waveform. *ICASSP '86. IEEE International Conference on Acoustics, Speech, And Signal Processing*, *11*, 1605–1608. https://doi.org/10.1109/ICASSP.1986.1169239

Fulop, S. A. (2011). *Speech Spectrum Analysis*. Springer.

Ghosal, S., & van der Vaart, A. (2017). *Fundamentals of Nonparametric Bayesian Inference* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781139029834

Gibson, J. (2018). Entropy Power, Autoregressive Models, and Mutual Information. *Entropy*, *20*(10), 750. https://doi.org/10.3390/e20100750

Glorot, X., & Bengio, Y. (2010). Understanding the Difficulty of Training Deep Feedforward Neural Networks. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, 249–256.

Gobl, C. (2017). Reshaping the Transformed LF Model: Generating the Glottal Source from the Waveshape Parameter Rd. *Interspeech 2017*, 3008–3012. https://doi.org/10.21437/Interspeech.2017-1140

Harrison, P. (2013). *Making Accurate Formant Measurements: An Empirical Investigation of the Influence of the Measurement Tool, Analysis Settings and Speaker on Formant Measurements* [Doctoral dissertation].

Herbst, C. T., Lohscheller, J., Švec, J. G., Henrich, N., Weissengruber, G., & Fitch, W. T. (2014). Glottal Opening and Closing Events Investigated by Electroglottography and Super-High-Speed Video Recordings. *Journal of Experimental Biology*, *217*(6), 955–963. https://doi.org/10.1242/jeb.093203

Hoffman, M. D., Blei, D. M., Wang, C., & Paisley, J. (2013). Stochastic Variational Inference. *The Journal of Machine Learning Research*, *14*(1), 1303–1347.

Hornik, K. (1993). Some New Results on Neural Network Approximation. *Neural Networks*, *6*(8), 1069–1072.

Jaynes, E. T. (1984). Prior Information and Ambiguity in Inverse Problems. *Inverse Problems. An International Journal on the Theory and Practice of Inverse Problems, Inverse Methods and Computerized Inversion of Data*, *14*, 151–166.

Jaynes, E. T. (1991). Notes on Present Status and Future Prospects. In W. T. Grandy & L. H. Schick (Eds.), *Maximum Entropy and Bayesian Methods: Maximum Entropy and Bayesian Methods* (pp. 1–13). Springer Netherlands. https://doi.org/10.1007/978-94-011-3460-6_1

Jaynes, E. T. (2003). *Probability Theory: The Logic of Science*. Cambridge University Press.

Jopling, R. L. (2009). *Voice Initiation and Voice Offset Patterns in Normal Females: Investigated by High Speed Digital Imaging* [Doctoral dissertation].

Kadiri, S. R., Alku, P., & Yegnanarayana, B. (2021). Extraction and Utilization of Excitation Information of Speech: A Review. *Proceedings of the IEEE*, *109*(12), 1920–1941. https://doi.org/10.1109/JPROC.2021.3126493

Kameoka, H., Ono, N., & Sagayama, S. (2010). Speech Spectrum Modeling for Joint Estimation of Spectral Envelope and Fundamental Frequency. *IEEE Transactions on Audio, Speech, And Language Processing*, *18*(6), 1507–1516. https://doi.org/10.1109/TASL.2009.2036287

Kazlauskaite, I., Ek, C. H., & Campbell, N. D. (2018). Gaussian Process Latent Variable Alignment Learning. *Arxiv Preprint Arxiv:1803.02603*.

Kent, R. D., & Vorperian, H. K. (2018). Static Measurements of Vowel Formant Frequencies and Bandwidths: A Review. *Journal of Communication Disorders*, *74*, 74–97. https://doi.org/10.1016/j.jcomdis.2018.05.004

Klatt, D. H. (1986). Representation of the First Formant in Speech Recognition and in Models of the Auditory Periphery. *Canadian Acoustics*, *14*(3bis), 5–7.

Klatt, D. H., & Klatt, L. C. (1990). Analysis, Synthesis, and Perception of Voice Quality Variations among Female and Male Talkers. *The Journal of the Acoustical Society of America*, *87*(2), 820–857. https://doi.org/10.1121/1.398894

Knuth, K. H., & Skilling, J. (2012). Foundations of Inference. *Axioms*, *1*(1), 38–73. https://doi.org/10.3390/axioms1010038

Kominek, J., & Black, A. W. (2004,). The CMU Arctic Speech Databases. *Fifth ISCA Workshop on Speech Synthesis*.

Kreiman, J., Gerratt, B. R., & Antoñanzas-Barroso, N. (2007). Measures of the Glottal Source Spectrum. *Journal of Speech, Language, And Hearing Research*, *50*(3), 595–610. https://doi.org/10.1044/1092-4388(2007/042)

Lalchand, V., Ravuri, A., & Lawrence, N. D. (2022, April). *Generalised Gaussian Process Latent Variable Models (GPLVM) with Stochastic Variational Inference* (Issue arXiv:2202.12979). arXiv. https://doi.org/10.48550/arXiv.2202.12979

Landau, H. J. (1983). The Inverse Problem for the Vocal Tract and the Moment Problem. *SIAM Journal on Mathematical Analysis*, *14*(5), 1019–1035. https://doi.org/10.1137/0514082

Le Besnerais, G., & Giovannelli, J.-F. (2010). Inverse Filtering and Other Linear Methods. In J. Idier (Ed.), *Bayesian Approach to Inverse Problems: Bayesian Approach to Inverse Problems* (pp. 80–116). ISTE. https://doi.org/10.1002/9780470611197.ch4

Lighthill, M. J. (1958). *An Introduction to Fourier Analysis and Generalised Functions*. Cambridge University Press.

Lindeberg, T. (2013). *Scale-Space Theory in Computer Vision* (Vol. 256). Springer Science & Business Media.

Little, M. A. (2011). Mathematical Foundations of Nonlinear, Non-Gaussian, and Time-Varying Digital Speech Signal Processing. In C. M. Travieso-González & J. B. Alonso-Hernández (Eds.), *Advances in Nonlinear Speech Processing: Vol. 7015. Advances in Nonlinear Speech Processing* (pp. 9–16). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-25020-0_2

Little, M. A., McSharry, P. E., Roberts, S. J., Costello, D. A., & Moroz, I. M. (2007). Exploiting Nonlinear Recurrence and Fractal Scaling Properties for Voice Disorder Detection. *Biomedical Engineering Online*, *6*(1), 23. https://doi.org/10.1186/1475-925X-6-23

Ma, A., Lau, K. K., & Thyagarajan, D. (2020). Voice Changes in Parkinson's Disease: What Are They Telling Us?. *Journal of Clinical Neuroscience*, *72*, 1–7. https://doi.org/10.1016/j.jocn.2019.12.029

MacKay, D. J. (1998). Introduction to Gaussian Processes. *NATO ASI Series F Computer and Systems Sciences*, *168*, 133–166.

MacKay, D. J. C. (2005). *Information Theory, Inference, and Learning Algorithms*. https://doi.org/10.1198/jasa.2005.s54

Markel, J. D., & Gray, A. H. (1976). *Linear Prediction of Speech: 12* (Vol. 12). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-66286-7

Maurer, D. (2016). *Acoustics of the Vowel*. Peter Lang.

Mehta, D. D., Rudoy, D., & Wolfe, P. J. (2012). Kalman-Based Autoregressive Moving Average Modeling and Inference for Formant and Antiformant Tracking. *The Journal of the Acoustical Society of America*, *132*(3), 1732–1746. https://doi.org/10.1121/1.4739462

Milenkovic, P. (1986). Glottal Inverse Filtering by Joint Estimation of an AR System with a Linear Input Model. *IEEE Transactions on Acoustics, Speech, And Signal Processing*, *34*(1), 28–42. https://doi.org/10.1109/TASSP.1986.1164778

Miller, R. L. (1959). Nature of the Vocal Cord Wave. *The Journal of the Acoustical Society of America*, *31*(6), 667–677. https://doi.org/10.1121/1.1907771

Monahan, J. F. (1984). A Note on Enforcing Stationarity in Autoregressive-Moving Average Models. *Biometrika*, *71*(2), 403–404. https://doi.org/10.1093/biomet/71.2.403

Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press.

Naylor, P. A., Kounoudes, A., Gudnason, J., & Brookes, M. (2007). Estimation of Glottal Closure Instants in Voiced Speech Using the DYPSA Algorithm. *IEEE Transactions on Audio, Speech and Language Processing*, *15*(1), 34–43. https://doi.org/10.1109/TASL.2006.876878

Neal, R. M. (1996). Priors for Infinite Networks. In *Bayesian Learning for Neural Networks: Bayesian Learning for Neural Networks* (pp. 29–53). Springer.

O Cinneide, A. (2012). *Phase-Distortion-Robust Voice-Source Analysis.* https://doi.org/10.21427/D7FW3B

Orbanz, P., & Teh, Y. W. (2010). Bayesian Nonparametric Models. *Encyclopedia of Machine Learning, 1.*

Orlikoff, R. F., Golla, M. E., & Deliyski, D. D. (2012). Analysis of Longitudinal Phase Differences in Vocal-Fold Vibration Using Synchronous High-Speed Videoendoscopy and Electroglottography. *Journal of Voice : Official Journal of the Voice Foundation*, *26*(6), 816.e13–816.e20. https://doi.org/10.1016/j.jvoice.2012.04.009

Pandey, G., & Dukkipati, A. (2014). *To Go Deep or Wide in Learning?.*

Perrotin, O., Feugère, L., & d'Alessandro, C. (2021). Perceptual Equivalence of the Liljencrants–Fant and Linear-Filter Glottal Flow Models. *The Journal of the Acoustical Society of America*, *150*(2), 1273–1285. https://doi.org/10.1121/10.0005879

Peterson, G. E., & Barney, H. L. (1952). Control Methods Used in a Study of the Vowels. *The Journal of the Acoustical Society of America*, *24*(2), 175–184.

Plumpe, M., Quatieri, T., & Reynolds, D. (1999). Modeling of the Glottal Flow Derivative Waveform with Application to Speaker Identification. *IEEE Transactions on Speech and Audio Processing*, *7*(5), 569–586. https://doi.org/10.1109/89.784109

Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (1992). *Numerical Recipes in C (2nd Ed.): The Art of Scientific Computing*. Cambridge University Press.

Rabiner, L. R., & Schafer, R. W. (2007). *Introduction to Digital Speech Processing* (Vol. 1). https://doi.org/10.1561/2000000001

Rabiner, L. R., Juang, B.-H., & Rutledge, J. C. (1993). *Fundamentals of Speech Recognition* (Vol. 14). PTR Prentice Hall Englewood Cliffs.

Rahimi, A., & Recht, B. (2008). Weighted Sums of Random Kitchen Sinks: Replacing Minimization with Randomization in Learning. In D. Koller, D. Schuurmans, Y. Bengio, & L. Bottou (Eds.), *Advances in Neural Information Processing Systems: Vol. 21. Advances in Neural Information Processing Systems.*

Ramsay, J. O., & Silverman, B. W. (1997). *Functional data analysis.* Springer.

Rao, A., & Ghosh, P. K. (2018). Glottal Inverse Filtering Using Probabilistic Weighted Linear Prediction. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *27*(1), 114–124.

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

Riutort-Mayol, G., Bürkner, P.-C., Andersen, M. R., Solin, A., & Vehtari, A. (2020). Practical Hilbert Space Approximate Bayesian Gaussian Processes for Probabilistic Programming. *Arxiv:2004.11408 [Stat]*.

Rosenberg, A. E. (1971). Effect of Glottal Pulse Shape on the Quality of Natural Vowels. *The Journal of the Acoustical Society of America*, *49*(2B), 583–590. https://doi.org/10.1121/1.1912389

Sawada, H., Kameoka, H., Araki, S., & Ueda, N. (2012). Efficient Algorithms for Multichannel Extensions of Itakura-Saito Nonnegative Matrix Factorization. *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 261–264. https://doi.org/10.1109/ICASSP.2012.6287867

Schleusing, O. (2012). *Multi-Parametric Source-Filter Separation of Speech and Prosodic Voice Restoration* [Technical report].

Schnupp, J., Nelken, I., & King, A. (2011). *Auditory Neuroscience: Making Sense of Sound*. MIT press.

Schoder, S., Falk, S., Wurzinger, A., Lodermeyer, A., Becker, S., & Kniesburges, S. (2024). A Benchmark Case for Aeroacoustic Simulations Involving Fluid-Structure-Acoustic Interaction Transferred from the Process of Human Phonation. *Acta Acustica*, *8*, 13. https://doi.org/10.1051/aacus/2024005

Schroeder, M. R. (1999). *Computer Speech: Recognition, Compression, Synthesis*. Springer-Verlag. https://doi.org/10.1007/978-3-662-03861-1

Simpson, F., Lalchand, V., & Rasmussen, C. E. (2020,). Marginalised Spectral Mixture Kernels with Nested Sampling. *Third Symposium on Advances in Approximate Bayesian Inference*.

Sivia, D. S., & Skilling, J. (2006). *Data Analysis: A Bayesian Tutorial* (2nd ed). Oxford University Press.

Skilling, J. (2005). Bayesics. *AIP Conference Proceedings*, *803*, 3–24.

Stevens, K. N. (2000). *Acoustic Phonetics*. MIT press.

Stoica, P., & Moses, R. L. (2005). *Spectral Analysis of Signals*. Pearson/Prentice Hall.

Svec, J. G., Schutte, H. K., Chen, C. J., & Titze, I. R. (2021). Integrative Insights into the Myoelastic-Aerodynamic Theory and Acoustics of Phonation. Scientific Tribute to Donald G. Miller. *Journal of Voice*. https://doi.org/10.1016/j.jvoice.2021.01.023

Sørbye, S. H., & Rue, H. (2017). Penalised Complexity Priors for Stationary Autoregressive Processes. *Journal of Time Series Analysis*, *38*(6), 923–935. https://doi.org/10.1111/jtsa.12242

Taylor, A. M., & Reby, D. (2010). The Contribution of Source–Filter Theory to Mammal Vocal Communication Research. *Journal of Zoology*, *280*(3), 221–236. https://doi.org/10.1111/j.1469-7998.2009.00661.x

Titsias, M. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics*, 567–574.

Titsias, M., & Lawrence, N. D. (2010). Bayesian Gaussian Process Latent Variable Model. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, 844–851.

Titze, I. R. (1995). *Workshop on Acoustic Voice Analysis: Summary Statement*. National Center for Voice and Speech.

Titze, I. R. (2000). Principles of Voice Production (Second Printing). *Iowa City, IA: National Center for Voice and Speech.*

Titze, I. R., Baken, R. J., Bozeman, K. W., Granqvist, S., Henrich, N., Herbst, C. T., Howard, D. M., Hunter, E. J., Kaelin, D., Kent, R. D., Kreiman, J., Kob, M., Löfqvist, A., McCoy, S., Miller, D. G., Noé, H., Scherer, R. C., Smith, J. R., Story, B. H., … Wolfe, J. (2015). Toward a Consensus on Symbolic Notation of Harmonics, Resonances, and Formants in Vocalization. *The Journal of the Acoustical Society of America*, *137*(5), 3005–3007. https://doi.org/10.1121/1.4919349

Turner, R. E., & Sahani, M. (2014). Time-Frequency Analysis as Probabilistic Inference. *IEEE Transactions on Signal Processing*, *62*(23), 6171–6183. https://doi.org/10.1109/TSP.2014.2362100

Van Soom, M., & de Boer, B. (2019). A New Approach to the Formant Measuring Problem. *Proceedings*, *33*(1), 29. https://doi.org/10.3390/proceedings2019033029

Van Soom, M., & de Boer, B. (2020). Detrending the Waveforms of Steady-State Vowels. *Entropy*, *22*(3), 331. https://doi.org/10.3390/e22030331

Van Soom, M., & de Boer, B. (2021). A Weakly Informative Prior for Resonance Frequencies. *Physical Sciences Forum*, *3*(1), 2. https://doi.org/10.3390/psf2021003002

Van Soom, M., & De Boer, B. (2022,). Softly Constrained Agent-Based Models. *Joint Conference on Language Evolution.*

Veldhuis, R. (1998). A Computationally Efficient Alternative for the Liljencrants-Fant Model and Its Perceptual Evaluation. *The Journal of the Acoustical Society of America*, *103*(1), 566–571. https://doi.org/10.1121/1.421103

Verdolini, K., & Titze, I. R. (1995). The Application of Laboratory Formulas to Clinical Voice Management. *American Journal of Speech-Language Pathology*, *4*(2), 62–69. https://doi.org/10.1044/1058-0360.0402.62

Viterbi, A. (1967). Error Bounds for Convolutional Codes and an Asymptotically Optimum Decoding Algorithm. *IEEE Transactions on Information Theory*, *13*(2), 260–269. https://doi.org/10.1109/TIT.1967.1054010

Wang, R., Zhang, Y., Ou, Z., & Hasegawa-Johnson, M. (2016). Use of Particle Filtering and MCMC for Inference in Probabilistic Acoustic Tube Model. *2016 IEEE Statistical Signal Processing Workshop (SSP)*, 1–5. https://doi.org/10.1109/SSP.2016.7551748

Whalen, D. H., Chen, W.-R., Shadle, C. H., & Fulop, S. A. (2022). Formants Are Easy to Measure; Resonances, Not so Much: Lessons from Klatt (1986). *The Journal of the Acoustical Society of America*, *152*(2), 933–941. https://doi.org/10.1121/10.0013410

Williams, C. K. I. (1998). Computation with Infinite Neural Networks. *Neural Computation*, *10*(5), 1203–1216. https://doi.org/10.1162/089976698300017412

Wong, D., Markel, J., & Gray, A. (1979). Least Squares Glottal Inverse Filtering from the Acoustic Speech Waveform. *IEEE Transactions on Acoustics, Speech, And Signal Processing*, *27*(4), 350–355. https://doi.org/10.1109/TASSP.1979.1163260

Xuan, J., Lu, J., & Zhang, G. (2021, September). *Bayesian Transfer Learning: An Overview of Probabilistic Graphical Models for Transfer Learning* (Issue arXiv:2109.13233). arXiv.

Yoshii, K., & Goto, M. (2013). Infinite Kernel Linear Prediction for Joint Estimation of Spectral Envelope and Fundamental Frequency. *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, 463–467. https://doi.org/10.1109/ICASSP.2013.6637690

Zhang, B., & Shin, Y. C. (2021). An Adaptive Gaussian Mixture Method for Nonlinear Uncertainty Propagation in Neural Networks. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2021.06.007

Zhang, B., & Shin, Y. C. (2022). A Gaussian Mixture Filter with Adaptive Refinement for Nonlinear State Estimation. *Signal Processing*, *201*, 108677. https://doi.org/10.1016/j.sigpro.2022.108677

Zhang, Y., Zheng, X., & Xue, Q. (2020). A Deep Neural Network Based Glottal Flow Model for Predicting Fluid-Structure Interactions during Voice Production. *Applied Sciences*, *10*(2), 705. https://doi.org/10.3390/app10020705

Appendices

Appendix A. Regularization and priors

As its name suggests, GIF belongs to the class of *inverse problems* (Le Besnerais & Giovannelli, 2010). Inverse problems confront us when we have observations of some effects [noisy oservations of $s(t)$ in our case] of which we want to infer the causes $u(t)$ and $h(t)$ within the context of some model $s(t) = u(t) * h(t) * r(t)$.

In the applied sciences it is very often the case for inverse problems that the observed data are quite consistent with a broad and sometimes colorful class of possible solutions rather than a unique and logically deductible solution. From the viewpoint of pure mathematics therefore such inverse problems are often deemed 'ill-posed', but perhaps a more productive viewpoint[37] is that such problems are simply underdetermined.

Underdetermined problems are ubiquitous in a wide range of applications, from medical diagnosis (what could be causing the patient's persistent cough) to astronomy (what is perturbing the orbit of Uranus). Bayesian theory by itself does not require any particular structure on the "feasible set" of plausible solutions and as such underdetermined problems have no special status within it, nor pose any special challenge to it (Jaynes, 2003). In practice however, we would like to have a more well determined solution to underdetermined problems as our computing budgets (and patience) are limited. For example, it would most certainly help if we can already discard those solutions that lack to some degree a set of required properties we established beforehand. This is where *regularization* and *priors* come into play.

In general, *regularization* is the process of constraining the solution space of some problem to obtain a more stable and perhaps unique solution; in other words, to convert an underdetermined problem into something more 'well-determined' and, hopefully, more amenable to computational optimization. For example, ridge regression is a widely used regularization technique to numerically stabilize linear regression in high dimensions.[38]

In the Bayesian framework regularization is imposed simply through careful assignment of the probability distributions describing a specific problem − of which the bulk of the effort is traditionally concentrated on the prior probabilities, or *priors* (Sivia & Skilling, 2006). Assuming our prior information about the problem at hand is correct and the assignment of the priors accurately reflect that information (and we got away with the approximations made in our inference methods) we will have automatically implemented just the right degree of regularization for our problem; indeed, specially catered to the one data set we

[37] And closer to the historical origin of the epithet "mal posée" (Jaynes, 1984).

[38] Ridge regression (also known as ℓ^2 regularization) can be derived from Bayesian theory as standard linear regression with Gaussian priors on the amplitudes (Murphy, 2022, Sec. 11.3). This principled view allows us to go much further than numerical stabilization. We will show in the next chapters that ridge regression can (approximately) express intricate pieces of prior information such as definite signal polarity, the differentiability class to which a function belongs, and power constraints on waveforms.

happened to observe, because we express everything in terms of posterior probabilities that are conditioned on the observed data.

For simple inverse problems where intuition can see the answer clearly, Bayesian regularization leads to answers that comply with common sense (MacKay, 2005). This is simply because Bayesian probability theory is a formalized theory of plausible reasoning itself (Skilling, 2005), which humans undertake every day. But regularization in the Bayesian framework can go beyond our intuition: it will of course continue to work for complex problems that require considerable analysis, and this consistency makes it a valuable tool for attacking inverse problems (Calvetti & Somersalo, 2018).

Appendix B. Parametric, nonparametric, semiparametric?

We clarify some of the confusing terminology associated with Bayesian nonparametric models.

Parametric models are simply models with a fixed and prespecified amount of unknown parameters R, independent of the size of the dataset N.

Examples are linear regression and neural networks. In BNGIF the filter priors π_{PZ} and π_{AP} priors are parametric as they have a fixed amount of parameters R depending on their order K, similar to linear prediction (LP) methods. And of course the π_{LF} prior with $R = 4$ is parametric as well.

Nonparametric models in a Bayesian context are models for which the effective number of parameters $\tilde{R}$ is capable of adapting to N (Orbanz & Teh, 2010).

They are also referred to as infinite-dimensional parametric models, which is just a shorthand way of saying that they are parametric models which have been mathematically constructed to behave consistently under inference as $R \to \infty$. That is, given the data of size N, a nonparametric model roughly behaves as an ordinary parametric model with $\tilde{R} < \infty$ parameters, with the exception that $\tilde{R}$ is determined automatically from the data rather than specified beforehand.

Examples are Dirichlet process mixture models, for which the effective number of mixture components does not need to be known beforehand, and, of course, GPs, which are infinite-dimensional distributions over functions. In BNGIF the π_{GP} and π_{VS} priors are nonparametric because they are GPs.[39]

The name 'nonparametric' is a bit of a misnomer because nonparametric models almost always have a set of Q hyperparameters which typically determine either the scales present in the data or the degree to which sparsity or smoothness is encouraged. In theory, inference of these Q hyperparameters is no different from inference with a parametric model of order $R = Q + \tilde{R}$ in which $\tilde{R}$ parameters have been marginalized over analytically.

Semiparametric models in a Bayesian context are models with both parametric and nonparametric components for which the latter is considered a nuisance parameter (Ghosal & van der Vaart, 2017, p. 368). This means that we are mainly interested in conclusions about the parametric component of the model, which require marginalizing over the nonparametric component.

In BNGIF the parametric and nonparametric components are very neatly divided into models for the filter $h(t)$ and the source $u(t)$, respectively, reflecting the qualitative difference between the information available from acoustic phonetics about the two. One might be

[39] Strictly, these are not fully nonparametric because they are only reduced-rank GPs and not full-rank GPs. Their capacity to adapt their complexity to growing amounts of data N is limited by the number of basisfunctions M; nonparametric behavior is recovered in the limit $M \to \infty$.

tempted to call BNGIF semiparametric for this reason if it weren't for the fact that the nonparametric component describing $u(t)$ is actually the main object of interest for the GIF problem. If anything the nuisance bit for GIF is in the parametric component describing $h(t)$, not the nonparametric component for $u(t)$.

Appendix C. JAX implementation of the Liljencrants-Fant model

```python
import jax
import jax.numpy as jnp
import jaxopt


def _get_initial_bracket(T0, tol, UPPER_FACTOR=1e3):
    lower = (1 / T0) * tol
    upper = (1 / T0) * UPPER_FACTOR
    return lower, upper


def _bisect_exponential_rate(
    f, tol, initial_bracket, maxiter, fail_val=jnp.nan, **kwargs
):
    lower, upper = initial_bracket
    bisec = jaxopt.Bisection(
        optimality_fun=f,
        lower=lower,
        upper=upper,
        maxiter=maxiter,
        tol=tol,
        check_bracket=False,
    )
    result = bisec.run(**kwargs)
    success = result.state.error < tol
    exponential_rate = result.params
    return jax.lax.cond(success, lambda: exponential_rate, lambda: fail_val)


def _bisect_epsilon(p, tol, initial_bracket, maxiter):
    def f(epsilon, p):
        LHS = epsilon * p["Ta"]
        RHS = 1 - jnp.exp(-epsilon * (p["T0"] - p["Te"]))
        return LHS - RHS

    return _bisect_exponential_rate(f, tol, initial_bracket, maxiter, p=p)


def _bisect_a(p, epsilon, tol, initial_bracket, maxiter):
    def f(a, p, epsilon):
        factor = 1 / (a**2 + (jnp.pi / p["Tp"]) ** 2)
        term1 = (
            jnp.exp(-a * p["Te"])
            * (jnp.pi / p["Tp"])
            / jnp.sin(jnp.pi * p["Te"] / p["Tp"])
        )
        term2 = a
        term3 = (
            jnp.pi / p["Tp"] * (1 / jnp.tan(jnp.pi * p["Te"] / p["Tp"]))
        )  # cotg() is 1/tan()
```

```
        LHS = factor * (term1 + term2 - term3)

        RHS = (p["T0"] - p["Te"]) / (
            jnp.exp(epsilon * (p["T0"] - p["Te"])) - 1
        ) - 1 / epsilon

        return LHS - RHS

    # Assume that the bisection will only fail when `a -> 0`
    return _bisect_exponential_rate(
        f, tol, initial_bracket, maxiter, fail_val=0.0, p=p, epsilon=epsilon
    )


def _dgf(t, p, epsilon, a, **kwargs):
    """Implement the LF model per Doval et al. (2006) Section A1.4."""

    def rise(t):
        return (
            -jnp.exp(a * (t - p["Te"]))
            * jnp.sin(jnp.pi * t / p["Tp"])
            / jnp.sin(jnp.pi * p["Te"] / p["Tp"])
        )

    def decrease(t):
        return (
            -1
            / (epsilon * p["Ta"])
            * (
                jnp.exp(-epsilon * (t - p["Te"]))
                - jnp.exp(-epsilon * (p["T0"] - p["Te"]))
            )
        )

    return jnp.piecewise(
        t,
        [t < 0, (0 <= t) & (t < p["Te"]), (p["Te"] <= t) & (t <= p["T0"])],
        [0.0, rise, decrease, 0.0],
    )


def _nans_like(a):
    return jnp.full_like(a, jnp.nan)


def dgf(t, p, offset=0.0, tol=1e-6, initial_bracket=None, maxiter=100):
    """Calculate the glottal flow derivative (DGF) using the LF (Liljencrants-Fant) model

    This JAX implementation can be jitted and is differentiable with respect to `p`.
```

```
    Jitting this function will gain several a speedup of several orders of magnitude.

    The LF model [1,2] is nonzero only for values of `t` in `[offset, offset + T0]`, where
    `T0 = p['T0']`. If the LF parameters in `p` are inconsistent, such that the
    implicit equations cannot be solved, the function's output `u` will be all `nan`s.

    Args:
        `t` (array): Time points at which to evaluate the DGF `u(t)`
        `p` (dict): Dictionary with the LF parameters. It must have the following keys:
            `['T0', 'Te', 'Tp', 'Ta']`. `Ee` is always assumed to be 1.
        `offset` (scalar): The DGF waveform starts at this offset and is nonzero in
            `[offset, offset + T0]`.
        `tol`, `initial_bracket`, `maxiter` (scalar): Parameters controlling the bisection
            routine `_bisect_exponential_rate()` to solve the two equations implicit in
            the LF model.

    Returns:
        `u` (DeviceArray): The glottal flow derivative evaluated at times `t` if the LF
            parameters in `p` are consistent, otherwise an array of `nan`s shaped as `t`.

    References:
        [1] G. Fant, J. Liljencrants, and Q. Lin, "A four-parameter model of glottal flow",
            STL-QPSR, vol. 4, no. 1985, pp. 1–13, 1985.
        [2] Doval, Boris, Christophe d'Alessandro, and Nathalie Henrich. "The spectrum of
            glottal flow models." Acta acustica united with acustica 92.6 (2006): 1026-1046.
    """
    if initial_bracket is None:
        initial_bracket = _get_initial_bracket(p["T0"], tol)

    epsilon = _bisect_epsilon(
        p, tol=tol, initial_bracket=initial_bracket, maxiter=maxiter
    )
    a = _bisect_a(
        p, epsilon=epsilon, tol=tol, initial_bracket=initial_bracket, maxiter=maxiter
    )
    u = _dgf(t - offset, p=p, epsilon=epsilon, a=a)
    inconsistent_parameters = jnp.any(jnp.isnan(u))
```

```
    return jax.lax.cond(inconsistent_parameters, lambda: _nans_like(u), lambda: u)


def consistent_lf_params(p):
    """Check whether the implicit LF equations can be solved given the LF parameters in `p`"""
    return ~jnp.any(jnp.isnan(dgf(0.0, p)))


def _select_keys(q, *keys):
    return {k: q[k] for k in keys if k in q}


def convert_lf_params(p, s, join=True):
    if s == "R -> generic":
        # Perrotin et al. (2021) Eq. (A1)
        Ra, Rk, Rg = p["Ra"], p["Rk"], p["Rg"]
        Oq = (1 + Rk) / (2 * Rg)
        am = 1 / (1 + Rk)
        Qa = Ra / (1 - Oq)  # Doval (2006) p. 5
        q = dict(Oq=Oq, am=am, Qa=Qa)
    elif s == "generic -> R":
        Oq, am, Qa = p["Oq"], p["am"], p["Qa"]
        Rk = 1 / am - 1
        Rg = (1 + Rk) / (2 * Oq)
        Ra = Qa * (1 - Oq)
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "T -> R":
        # Fant (1994)
        T0, Te, Tp, Ta = p["T0"], p["Te"], p["Tp"], p["Ta"]
        Ra = Ta / T0
        Rk = (Te - Tp) / Tp
        Rg = T0 / (2 * Tp)
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "R -> T":
        T0, Ra, Rk, Rg = p["T0"], p["Ra"], p["Rk"], p["Rg"]
        Ta = T0 * Ra
        Tp = T0 / (2 * Rg)
        Te = Tp * (1 + Rk)
        q = dict(Te=Te, Tp=Tp, Ta=Ta)
    elif s == "generic -> T":
        q = convert_lf_params(convert_lf_params(p, "generic -> R"), "R -> T")
        q = _select_keys(q, "Te", "Tp", "Ta")
    elif s == "T -> generic":
        q = convert_lf_params(convert_lf_params(p, "T -> R"), "R -> generic")
        q = _select_keys(q, "Oq", "am", "Qa")
    elif s == "Rd -> R":
        # Perrotin et al. (2021) Eq. (A1)
        # See also Fant (1994) for the meaning of these dimensionless parameters
```

```python
        Rd = p["Rd"]
        Ra = (-1 + 4.8 * Rd) / 100
        Rk = (22.4 + 11.8 * Rd) / 100
        Rg = Rk * (0.5 + 1.2 * Rk) / (0.44 * Rd - 4 * Ra * (0.5 + 1.2 * Rk))
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "Rd -> T":
        q = convert_lf_params(convert_lf_params(p, "Rd -> R"), "R -> T")
        q = _select_keys(q, "Te", "Tp", "Ta")
    else:
        raise ValueError(f"Unknown conversion: {s}")
    return {**p, **q} if join else q


def fant_params(s):
    """Generate typical LF model parameters for a male or female vowel

    Note that the unit of time is `msec`. Values are taken from [1, p. 121].

      [1] Fant, Gunnar. "The LF-model revisited. Transformations and frequency domain analysis."
          Speech Trans. Lab. Q. Rep., Royal Inst. of Tech. Stockholm 2.3 (1995): 40.
    """
    if s == "female vowel":
        return {"T0": 5.0, "Te": 3.25, "Tp": 2.5, "Ta": 0.3978873577297384}
    elif s == "male vowel":
        return {
            "T0": 8.333333333333334,
            "Te": 4.513888888888889,
            "Tp": 3.4722222222222228,
            "Ta": 0.22736420441699334,
        }
    else:
        raise ValueError(s)
```

Appendix D. More comments on the nonparametric glottal flow model

Some aspects of the nonparametric GFM derived in Chapter 4 are discussed in more detail here.

Infinite precision? In the GP view, increasing H increases the *Monte Carlo resolution* with which the finite neural network model samples the arc cosine kernel — the deviation from the limiting kernel vanishes as $\mathcal{O}\left(1/\sqrt{H}\right)$ by (154). The nonparametric limit replaces explicit random changepoints with their continuous Gaussian measure, and the prior over $u'(t)$ becomes a GP whose covariance is the arc cosine kernel restricted to the open phase.

There is a caveat though. We gained full rank — the prior now has support over an infinite-dimensional function space — but we lost the ability to represent true discontinuities. Arc cosine GP sample paths are continuous; they cannot jump.[40] This is the analogue of the amplitude prior constraining the ellipsoid in the finite-H case (footnote 17 of Chapter 4): there we traded a large discrete family of possible changepoint configurations for a continuous Gaussian prior that concentrates mass inside a smooth ellipsoid; here we trade a distribution over Poisson-style jump processes for a Gaussian prior that concentrates mass on smooth function classes. The best the arc cosine GP can do is approximate a sharp GCI with a steep ramp.

Whether this is a problem in practice is an empirical question. The closed-phase GCI is physiologically sharp but not infinitely so, and the DGF data from VocalTractLab are sampled at finite rate, so the distinction between "true jump" and "ramp steep enough that it spans one sample" may not be detectable.

Why this matters for the choice of modeled signal The continuity constraint has a direct implication for which signal we should model with the arc cosine GP. We model $u'(t)$ — the DGF — because the GCI appears as a sharp negative peak there, which is easier to detect and align than the integral $u(t)$. But if the arc cosine GP is continuous, then $u'(t)$ is smooth, which means $u(t)$ is differentiable — and real glottal flow does not have a differentiable closure.

One way around this is to model $u(t)$ directly with the arc cosine GP instead of $u'(t)$. Then the GP sample paths are continuous (which $u(t)$ should be) and discontinuities can be pushed into the *derivative* implicitly via the radiation factor: the combined vocal tract and radiation impulse response is modeled as the AR filter, and the combined source signal becomes $u(t) * r(t)$ rather than $u'(t)$ alone. This is, at bottom, what classical LPC does — it posits $s(t) = e(t) * h(t)$ and allows $h(t)$ to absorb the radiation effect $r(t)$, since differentiation corresponds to a zero in the transfer function, not a pole, so $h(t)$ can remain all-pole. From this viewpoint the present approach generalizes LPC in exactly that way.

[40] Specifically, sample paths of a GP with a kernel of degree d are d-times continuously differentiable with probability one (Rasmussen & Williams, 2006). For $d = 0$ this gives continuous but nowhere-differentiable paths; for $d \geq 1$, paths are smoother still. In either case there are no jumps.

The alternative of obtaining $u'(t)$ implicitly — from the spectral domain or from the AR pole structure — is also possible and may allow the sharp-closure structure to be recovered without committing the GP to model it directly.

Why not go infinite depth? Increasing depth has a different character from increasing width. For a deep GP with Gaussian kernels at each layer, letting the depth diverge drives the process toward one that is independent of its inputs (Diaconis & Freedman, 1999) — a known pathology in practice, where deep GPs tend to "forget" their inputs and produce nearly constant sample paths. Finite but large depth mitigates this, but one might note that infinite width carries a vaguely analogous cost: in the GP limit, the basis functions are frozen at their prior expectation and no longer adapt to the data as drawn features. What is learned is entirely encoded in the kernel hyperparameters. This makes kernel learning — and in particular the multiple-kernel structure of IKLP — the mechanism through which the model does acquire data-driven spectral features, rather than learning them through random feature adaptation.

Why not spline models? Piecewise spline models are well-studied and effective in low-dimensional nonparametric regression. The issue here is resolution dependence. As GP priors, spline kernels produce posterior means that are splines with knots fixed at the observed input locations (MacKay, 1998, p. 6): add more data, add more knots. The arc cosine kernel, by contrast, learns the effective number of hingepoints from the data through its hyperparameters, and this number can remain $\mathcal{O}(1)$ even as the number of observations grows. For glottal flow this matters: a DGF waveform sampled at 16 kHz and one sampled at 44 kHz should both require $\mathcal{O}(1)$ effective changepoints to describe the open phase, not a number proportional to the sampling rate.

Why not Lévy processes? These encode Poisson-style jumps, $\mathcal{O}(1)$ in number, which is exactly the right prior for a signal with a small number of hard discontinuities. The problem is inference: the posterior over a Lévy process is indexed by the jump locations, and marginalizing those out requires MCMC or particle methods that scale with the number of jumps. The GP approach achieves fast closed-form posterior inference precisely because everything — changepoints, amplitudes, timing — has been absorbed into the kernel. Trading exact jumps for steep smooth ramps is the price of that tractability.

Appendix E. On derivative time scales

The EGIFA and CommonVoice datasets present us with ground-truth $u(t)$ but we need $u'(t)$ for learning. Naive finite difference of $u(t)$,

$$u'_{\Delta}(t) = \frac{u(t+\Delta) - u(t)}{\Delta}, \qquad \Delta = \frac{1}{f_s}, \tag{243}$$

often yields spikes in $u'(t)$ which effectively defeat kernel modeling because the increased observation noise will prevent learning fine structure. Luckily, we are saved by the physiology of the speech production process.

Glottal closure is not an instantaneous discontinuity but a short transition extending over a finite time interval. High-speed imaging and synchronized EGG measurements show that opening and closing events occupy $O(0.1)$ ms durations and may exhibit small temporal offsets depending on how closure propagates along the folds (e.g. zipper-like anterior-posterior contact), as described by Herbst et al. (2014) and Orlikoff et al. (2012). Consequently, a derivative operator that reacts to arbitrarily small sample-scale fluctuations does not reflect the underlying physiology; it instead amplifies numerical artefacts or simulator-specific microstructure. A meaningful derivative therefore requires an explicit temporal scale reflecting the duration over which closure dynamics remain physically interpretable.

Since we already model $u(t)$ by Gaussian processes,[41] we are implicitly asserting that first and second moments alone carry most of the important information. Therefore, we propose to use a *Gaussian-smoothed derivative* to measure the slope of a *locally averaged signal*

$$u'_{\sigma}(t) := \partial_t (G_\sigma * u)(t), \qquad G_\sigma(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{t^2}{\sigma^2}\right), \tag{244}$$

rather than differentiating raw samples directly.[42] This type of differentiation results in another GP, since differentiation and convolution (smearing) are closed operators in GP space.

Selecting σ such that the effective smoothing width corresponds to approximately 0.1 ms (about four to five samples at $f_s = 44.1$ kHz) aligns the operator with the expected duration of glottal closure transitions. At smaller scales the derivative is dominated by high-frequency noise and discretization effects; at larger scales the transition itself becomes blurred. The resulting derivative can therefore be interpreted as a slope measured at the physically relevant temporal resolution rather than as a raw numerical difference, and this removes the occurence of large spikes in $u'(t)$ almost completely; an example is shown in Figure 30.

[41] More precisely, we model its derivative $u'(t)$ as a GP, but the integration in (8) preserves GP-ness.

[42] Gaussian smoothing also turns out to be the unique linear kernel that preserves causality in scale-space, meaning that increasing scale removes fine structure without introducing spurious extrema (Lindeberg, 2013).

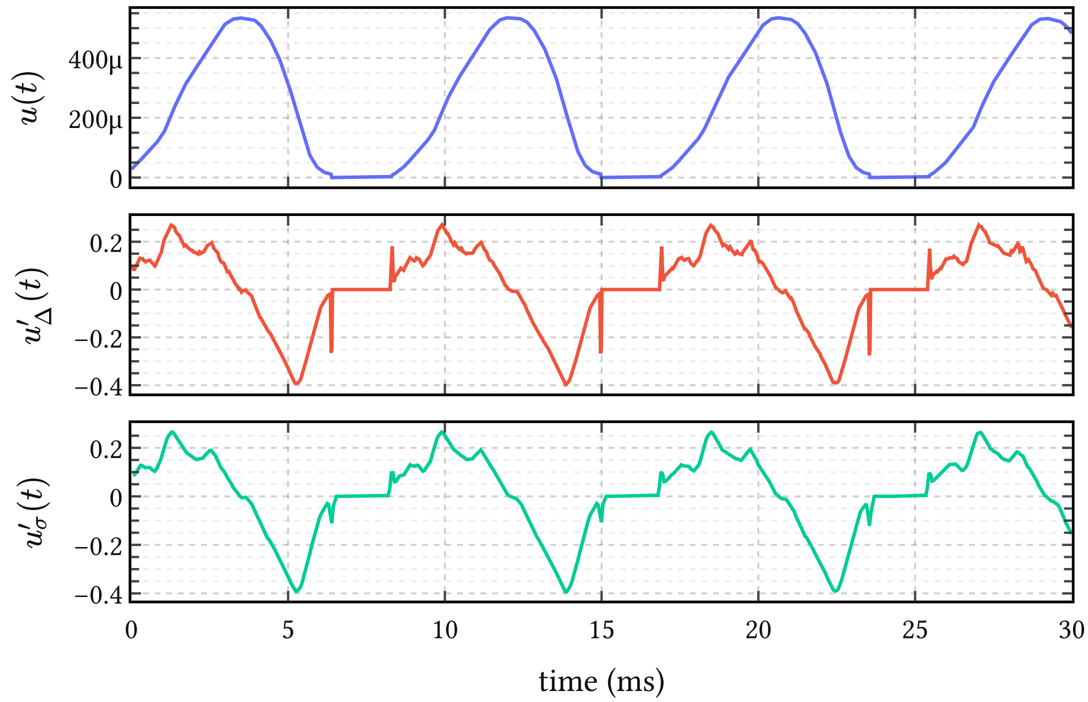

Figure 30 | **Smooth differentiation dampens spikes in $u'(t)$.** The top panel shows the ground truth glottal flow $u(t)$ extracted from a random EGIFA file. The middle panel contains its derivative $u'_\Delta(t)$ obtained by finite differences (243), and the bottom panel the Gaussian-smoothed derivative $u'_\sigma(t)$ proposed in (244). It can be seen that raw differentiation amplifies sample-scale fluctuations into large spikes, whereas the smoothed derivative retains the closure structure at the intended temporal resolution.

Appendix F. Predicting period-to-period jitter

Classical local jitter is defined as the expected absolute change in consecutive periods, normalized by the mean period,

$$\text{jitter} := \frac{\langle |T_{k+1} - T_k| \rangle}{\langle T_k \rangle}. \tag{245}$$

In the present framework, jitter is not introduced as an independent noise source. Instead, it emerges deterministically from a smooth stochastic model of phase warping. All variability originates from curvature of the time warp $t(\tau)$, induced by a Gaussian process prior on the log-period

$$g(\tau) = \log T(\tau). \tag{246}$$

Sampling the process at integer phases $\tau = k$ yields a stationary Gaussian sequence

$$g_k := g(k), \tag{247}$$

with mean μ, marginal variance σ^2, and lag-one correlation

$$\rho(\ell) := \frac{k_{\ell(1)}}{k_{\ell(0)}}, \tag{248}$$

which depends only on the kernel family and the lengthscale ℓ measured in cycles.

For the kernels considered here, the lag-one correlation is given by[43]

The observed periods are lognormal variables,

$$T_k = \exp(g_k), \tag{249}$$

and the pair (T_k, T_{k+1}) is bivariate lognormal. Because g_k and g_{k+1} have equal variance, the sum and difference variables

$$S = \frac{g_{k+1} + g_k}{2}, \quad D = g_{k+1} - g_k \tag{250}$$

are independent Gaussians, with

$$S \sim \text{Normal}\left(\mu, \sigma^2 \frac{1+\rho}{2}\right), \quad D \sim \text{Normal}(0, 2\sigma^2(1-\rho)). \tag{251}$$

Using the identity

[43] Matérn 1/2: $\rho(\ell) = \exp\left(-\frac{1}{\ell}\right)$. Matérn 3/2: $\rho(\ell) = \left(1 + \frac{\sqrt{3}}{\ell}\right) \exp\left(-\frac{\sqrt{3}}{\ell}\right)$. Matérn 5/2: $\rho(\ell) = \left(1 + \frac{\sqrt{5}}{\ell} + \frac{5}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}}{\ell}\right)$. Squared exponential: $\rho(\ell) = \exp\left(-\frac{1}{2\ell^2}\right)$.

$$|\exp(a) - \exp(b)| = 2\exp\left(\frac{a+b}{2}\right)\ |\sinh\left(\frac{a-b}{2}\right)|, \tag{252}$$

the expected absolute period difference factorizes as

$$\langle|T_{k+1} - T_k|\rangle = 2\langle\exp(S)\rangle\left\langle|\sinh\left(\frac{D}{2}\right)|\right\rangle. \tag{253}$$

The first factor evaluates to

$$\langle\exp(S)\rangle = \exp\left(\mu + \sigma^2\frac{1+\rho}{4}\right), \tag{254}$$

while the second admits a closed-form expression in terms of the Gaussian error function. After normalization by $\langle T_k\rangle = \exp\left(\mu + \frac{\sigma^2}{2}\right)$, all dependence on μ cancels, yielding the exact prediction

$$\langle\text{jitter} \mid \ell\rangle = 4\Phi\left(\sigma\sqrt{\frac{1-\rho(\ell)}{2}}\right) - 2, \tag{255}$$

where Φ denotes the standard normal cumulative distribution function.

This expression constitutes a quantitative theory of jitter. Given the marginal log-period variability σ and the kernel-dependent correlation $\rho(\ell)$, it predicts the expected period-to-period jitter without further assumptions.

For small arguments, the Gaussian CDF admits the expansion

$$\Phi(x) \approx \frac{1}{2} + \frac{x}{\sqrt{2\pi}}, \tag{256}$$

which yields the asymptotic scaling law

$$\langle\text{jitter} \mid \ell\rangle \approx \left(2\frac{\sigma}{\sqrt{\pi}}\right)\sqrt{1-\rho(\ell)}, \tag{257}$$

recovering the intuitive result that jitter vanishes as $\rho(\ell) \to 1$, i.e. as the phase warp becomes locally affine.

To make the prediction concrete, we fix a representative adult pitch distribution with median period $T_0 \approx 6.7$ ms (150 Hz) and log-period standard deviation $\sigma = 0.25$. The table below reports the predicted mean jitter for several kernel families and lengthscales.

Normal adult jitter values around 0.5% are only attained for lengthscales of several tens of cycles under smooth kernels. Rougher kernels require substantially larger ℓ to suppress cycle-to-cycle variation. In this sense, ℓ admits a direct physiological interpretation as a cycle-to-cycle smoothness parameter governing the temporal regularity of voicing.

ℓ (cycles)	$\nu = 1/2$	$\nu = 3/2$	$\nu = 5/2$	$\nu = \infty$
5	12.0%	6.2%	5.0%	4.0%
10	8.7%	3.3%	2.6%	2.0%
20	6.2%	1.7%	1.3%	1.0%
40	4.4%	0.85%	0.64%	0.50%
80	3.1%	0.43%	0.32%	0.25%

Table 7 | **Predicted mean period-to-period jitter** as a function of lengthscale ℓ for Matérn kernels with $\nu \in \left\{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\right\}$ and the squared exponential kernel ($\nu = \infty$). Predictions use $\sigma = 0.25$ and are obtained from the exact closed-form expression

$$\langle \text{jitter} \mid \ell \rangle = 4\Phi\left(\sigma\sqrt{\frac{1-\rho(\ell)}{2}}\right) - 2. \tag{258}$$

This framework yields falsifiable predictions linking observed jitter statistics to latent smoothness of the underlying pitch trajectory, which can subsequently be tested by fitting the model to data.

Appendix G. Gauge invariance in glottal inverse filtering evaluation

Every evaluation metric carries an implicit assumption about what constitutes a meaningful difference between two signals. Spectral metrics make this assumption explicit: envelope-based measures deliberately ignore phase, ignore absolute timing, and ignore excitation fine structure. The question is whether source-domain evaluation — comparing an estimated DGF waveform against a ground-truth one — should be held to the same standard, or whether it is legitimate to compare raw waveforms sample by sample.

This chapter argues that it is not legitimate, and that doing so systematically penalizes algorithms for choices that are physically unobservable — choices about what physicists call the *gauge*. The argument is not a matter of taste. It follows directly from the structure of the source-filter model and from the physics of acoustic propagation. The practical consequence is a specific evaluation procedure, aligned at frame level with an affine correction, which we use throughout the experimental chapters.

G.1. The source-filter gauge group

The speech production model is convolutional:

$$y(t) = (u'(t) * h(t)) + \varepsilon(t), \tag{259}$$

where $u'(t)$ is the glottal flow derivative, $h(t)$ is the vocal tract impulse response including radiation, and $\varepsilon(t)$ is noise. The inverse filtering problem asks: given $y(t)$, recover $u'(t)$ and $h(t)$.

The trouble is that the factorization is not unique. Two transformations leave $y(t)$ invariant.

The first is *scale*: for any nonzero constant α, the pair $\left(\alpha u'(t), \frac{h(t)}{\alpha}\right)$ produces exactly the same observed signal as $(u'(t), h(t))$. Absolute source amplitude is therefore not recoverable from acoustics alone; it is determined by convention.

The second is *time-translation*: for any constant delay τ, the pair $(u'(t-\tau), h(t+\tau))$ again produces the same convolution. The two transformations together form a gauge group acting on (u', h): the observable signal depends only on the equivalence class

$$[(u', h)] = \{(\alpha\, u'(t-\tau),; h(t+\tau)/\alpha) \mid \alpha \in \mathbb{R} \setminus \{0\}, \tau \in \mathbb{R}\}. \tag{260}$$

Inverse filtering does not recover a unique (u', h) pair. It selects one representative from this class.

The implication for evaluation is immediate: different algorithms can legitimately choose different representatives while producing identical acoustic reconstructions. An evaluation that compares representatives directly — rather than comparing the equivalence classes they represent — is measuring gauge choice, not signal quality. This is not a hypothetical concern. The closed-phase LP-based algorithms (CPCA, SLP, WLP) anchor the filter by minimizing LP residual energy in the closed phase, which drives the gauge toward a particular delay convention. Complex cepstrum decomposition (CCD) (Drugman et al., 2011) does something entirely different: it separates the maximum-phase anticausal component of the cepstrum, which selects a different gauge altogether. The two families are not wrong in different ways; they are right in different gauges. Evaluating them against the same raw waveform reference therefore introduces a systematic bias.

G.2. Physical dispersion reinforces the ambiguity

The gauge ambiguity from the source-filter model is compounded by a physical one. The vocal tract is a resonant system with a frequency-dependent phase response. Its group delay

$$\tau_{g(\omega)} = -\frac{\mathrm{d}}{\mathrm{d}\omega} \arg H\left(e^{j\omega}\right) \tag{261}$$

is not constant across frequency: narrow formants store and release energy over extended time windows, smearing the temporal relationship between the excitation event and the observed pressure waveform. This smearing varies across vowels — close rounded vowels like /o/ and /u/ with their narrow first-formant bandwidths are empirically the most challenging cases for GIF (Chien et al., 2017) — and there is no single "correct" delay that compensates it.

The standard evaluation benchmark of Chien et al. (2017) accounts for this with a fixed 0.65 ms delay applied uniformly to all ground-truth waveforms, motivated by an assumed 22 cm radiation distance. This is a reasonable first-order correction, but it is a single number applied to a phenomenon that varies with vowel, pitch, voice quality, and fundamental frequency. More importantly, it corrects for the mean delay of one physical effect while leaving the gauge ambiguity from the source-filter decomposition entirely unaddressed. Applying a fixed shift does not constitute gauge-fixing; it shifts the entire evaluation into a particular gauge that may match the CPCA convention reasonably well but will systematically misalign with CCD estimates.

G.3. Per-cycle alignment is the wrong granularity

The same benchmark evaluates performance in a pitch-synchronous, per-cycle fashion: ground-truth and estimated waveforms are aligned within each glottal cycle separately, using glottal closure instants extracted from the synthesizer. This procedure is motivated by the desire to isolate the pulse shape from pitch-period-to-pitch-period variability.

The problem is that it introduces a reference frame — the glottal cycle — that is not implied by the generative model. The source-filter interaction in (259) occurs at frame scale: the filter $h(t)$ is estimated from a windowed segment spanning many cycles, and the gauge it selects is therefore a frame-level property, not a cycle-level one. Different LPC-based algorithms apply their weighting functions (the closed-phase window in CPCA, the Gaussian weights around glottal closure in SLP, the piecewise-linear weights in WLP) at frame scale, and the effective delay they induce drifts slowly across frames as the filter estimate adapts. Cycle-level alignment obscures this drift by correcting it separately at each cycle, which has the unintended effect of partially correcting for LPC-induced gauge variation on behalf of the algorithm.

Frame-level alignment is the granularity that matches the statistical assumptions of the model and the practical structure of the algorithms.

G.4. Affine alignment at frame level

The physically motivated evaluation procedure aligns each estimated frame to the corresponding ground-truth frame by fitting an affine map and a nonnegative integer lag jointly,

$$(a^*, b^*, \tau^*) = \underset{a,b,\tau}{\operatorname{argmin}} \| \mathbf{y}_{\mathrm{est}[\tau:]} \cdot a + b - \mathbf{y}_{\mathrm{true}[:n-\tau]} \|^2, \tag{262}$$

where $\tau \in \{0, 1, ..., \tau_{\max}\}$ is a nonnegative integer lag measured in samples.[44] The scalar a absorbs scale and polarity; the scalar b absorbs any mean offset introduced by pre-emphasis or measurement convention. The lag τ handles the frame-level delay that varies across algorithms and physical conditions.

The aligned estimate is

$$\tilde{\mathbf{y}}_{\mathrm{est}} = a^* \mathbf{y}_{\mathrm{est}[\tau^*:]} + b^*, \tag{263}$$

defined on the overlap $[: n - \tau^*]$. Performance is then measured by centered cosine similarity between $\tilde{\mathbf{y}}_{\mathrm{est}}$ and $\mathbf{y}_{\mathrm{true}[:n-\tau^*]}$:

$$\text{score} = \frac{\langle \tilde{\mathbf{y}}_{\mathrm{est}} - \overline{\tilde{\mathbf{y}}_{\mathrm{est}}}, \mathbf{y}_{\mathrm{true}} - \overline{\mathbf{y}}_{\mathrm{true}} \rangle}{\| \tilde{\mathbf{y}}_{\mathrm{est}} - \overline{\tilde{\mathbf{y}}_{\mathrm{est}}} \| \| \mathbf{y}_{\mathrm{true}} - \overline{\mathbf{y}}_{\mathrm{true}} \|}. \tag{264}$$

Centering before taking the inner product removes the b degree of freedom redundantly, making the score invariant to the bias correction. Cosine similarity is a monotone function of RMSE after affine alignment and is bounded in $[-1, 1]$, which makes it convenient for visualization and for averaging across frames of different lengths.

The procedure is summarized in Table 8, alongside the corresponding treatment in the EGIFA benchmark (Chien et al., 2017) for comparison.

G.5. Summary

The source-filter model has a gauge group: scale and time-translation transformations that leave the observed signal invariant and therefore make the corresponding source properties unobservable from acoustics alone. Different inverse filtering algorithms choose different representatives from this group — LP-based methods anchor on closed-phase energy, CCD

[44] The lag is constrained to be nonnegative and to shift the estimate earlier in time, reflecting the physical direction of propagation delay: the acoustic observation $y(t)$ always lags the source, so a positive lag on the estimate corresponds to shifting it left to match the ground-truth timing. Allowing both directions would conflate physical delay correction with arbitrary time-warping.

Gauge source	Origin	EGIFA treatment	This work
Scale / polarity	Source-filter non-identifiability	Orthogonal projection per cycle	Affine fit per frame (262)
Absolute delay	Source-filter non-identifiability	Fixed 0.65 ms shift	Lag search per frame (262)
Dispersive group delay	Vocal tract resonances	Fixed 0.65 ms shift (partial)	Lag search per frame
Algorithmic gauge	LP vs. cepstral estimation	Not corrected	Corrected uniformly
Mean offset	Pre-emphasis / measurement	Not corrected	Affine fit (b term)

Table 8 | **Gauge sources in GIF evaluation** and their treatment in the EGIFA benchmark of Chien et al. (2017) versus this work. The EGIFA fixed delay of 0.65 ms addresses one physical effect; it does not address the gauge ambiguity from the source-filter decomposition or the algorithm-dependent gauge drift. Affine alignment at frame level handles all sources uniformly.

anchors on cepstral phase — and any evaluation that compares representatives directly rather than correcting for this choice conflates algorithmic style with algorithmic error.

The same argument applies to the fixed-delay and per-cycle alignment choices in the EGIFA benchmark (Chien et al., 2017). A fixed 0.65 ms delay corrects for one specific physical effect at one specific propagation distance; it does not address the gauge freedom of the model, and it does not adapt to the algorithm being evaluated. Per-cycle alignment introduces a reference frame that the generative model does not privilege and that partially absorbs algorithm-induced gauge drift.

Frame-level affine alignment with a lag search, followed by centered cosine similarity, addresses all of these uniformly. It is the evaluation procedure used throughout the experimental chapters of this thesis.

Appendix H. Related contributions

Van Soom & de Boer (2019) Measuring formant frequency $\mathbf{F}$ and bandwidth $\mathbf{B}$ from steady-state vowel waveforms, taking into account the low-frequency effects of the DGF $u'(t)$ during the open phase of the pitch period. Inference is implemented using a maximum-a-posteriori (MAP) approximation. In BNGIF these low-frequency effects are taken into account automatically.

Van Soom & de Boer (2020) Same as the above, but now using robust nested sampling inference instead of a MAP approximation. A new heuristic derivation of the model function from source-filter theory is added. In BNGIF variational inference is the preferred inference method.

Van Soom & De Boer (2022) Theoretical derivation of a new weakly informative prior for resonance frequencies. Shown to have superior performance in predicting steady-state vowel waveforms compared to the usual priors in inferring the pole frequency $\mathbf{x}$ and bandwidth $\mathbf{y}$, while being roughly equally (un)informative.